

PRODUCT DESIGN

Mid-Semester Exam Notes

AE ZG531 / AEL ZG531 / DE ZG541 / DM ZG541 / MM ZG541

Sessions 1 to 8 — Complete Coverage

EC-2 Mid-Semester Test • Closed Book • 30% Weightage

Duration 2 Hours • 90 Marks

Compiled from lecture slides, class transcripts,
past question papers and the instructor's own answer keys

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How to Use These Notes — Read This First

The instructor's exam briefing (Session 8, verbatim guidance)

In the final class before the exam, Prof. Mujumdar spent roughly twenty minutes explaining exactly how the paper is set and how she wants it answered. Everything below comes from that briefing. It is the single most valuable thing in this document — the marks are won or lost on **presentation format**, not on volume of writing.

Paper structure

Item	What was said
Scope	Sessions 1 to 8 only. "I cannot go beyond these 8 sessions' content." Anything from Session 9 onwards is out.
Nature	Closed book. Nothing permitted in the exam hall.
Weightage	30% — "don't look at this 90 marks, look at it as 30% weightage." Each question is only 5-6% of your grade.
Questions	5 or 6 questions , each 15-18 marks (may be 12 in some sets). Four paper sets exist (India regular, Gulf regular, India makeup, Gulf makeup) — content identical, mark split may vary slightly.
Time budget	120 minutes ÷ 6 questions = 20 minutes per question . If a question overruns 20 minutes, move on and come back.
Character	"Mainly theoretical with very less numerical." Expect at most one or two calculation questions.
Products used	Everyday / general products — not industry-specific ones, because the class spans many industries.
Difficulty	"There is no tough level of toughness... there is no wrong answer in this question paper." Application and method matter more than a single right number.

How she wants answers written

Do not write essays

This was repeated four times. Her instruction: "*Take help of flow charts, take help of example, take help of matrix, take help of tree, and then explain it.*" The reason is practical — roughly 500 answer scripts, 6 days, 2-3 TAs. Crisp, structured answers get read properly; essays do not.

Every answer should contain at least one of:

Block diagram	Phases, process flows, system schematics
Flow chart	Step sequences (Pugh steps, AHP process, concept testing)
Matrix / table	Comparisons, HOQ, Pugh, weighted decision matrix, needs-metrics
Graph	Kano curves, S-curve, benchmarking chart, trade-off map
Tree	Function tree, objective tree, FAST diagram, AHP hierarchy
Example	A named real product illustrating the concept

Warning she gave explicitly: "Just 4 or 5 bullet points is not accepted at this level. **How you are applying it is more important.**" So bullets alone will not carry a 15-mark answer — bullets *plus* a diagram *plus* a worked example will.

Specific instructions per topic

Topic	Her exact instruction
AHP	Never more than 3 alternatives and maximum 4 criteria . The consistency check may be skipped — "I have given instructions to my TA that consistency check is eliminated." Work up to normalising matrix [C] and choosing the best option.
Kano model	"If a question on Kano model comes, it is mandatory to draw this " — the satisfaction vs achievement graph with the three curves.
Concurrent vs traditional	Minimum 5-6 differentiating aspects are compulsory. She suggests framing via PQCDSM (Productivity, Quality, Cost, Delivery, Safety, Morale/Maintainability). Plus the two timeline diagrams.
Functional decomposition	If "black box function structure" is asked, you must show the three flows — Material, Energy, Signal (information) — crossing the boundary, and the decomposition into subfunctions.
Pugh / concept screening	Build the actual matrix. Pick a datum, mark +/0/−, sum $\Sigma+$, $\Sigma-$, $\Sigma 0$, rank. "Highest positive, eliminate negatives." Different students may pick different winners — both accepted.
Ratings in any matrix	"Whatever rating you feel and you mention is acceptable" — but you must state your assumption .

The five content blocks she named

She wrote these on screen as the shape of the syllabus, and said questions will be framed one per block:

#	Block	Sessions	Core tools you must be able to produce
1	Planning (introduction & pre-planning)	1, 2	Generic PD phases diagram, stage-gate/spiral, organisation types, concurrent engineering, product planning process
2	Customer need identification & specification	3, 4	5-step need process, Kano diagram, needs-metrics matrix, benchmarking chart, House of Quality
3	Concept generation	5, 6A	Black box, function tree, FAST, SOP, morphological chart, TRIZ contradiction, brainstorming/6-3-5
4	Concept evaluation, scoring & testing	6B, 7	Pugh matrix, weighted decision matrix, objective tree, AHP, concept testing steps
5	Product architecture	8	Modular vs integral, modularity taxonomy, architecture design process, delayed differentiation

Exam-day tactic

Read the question for ~2 minutes and assemble the answer mentally before writing — her advice. Then: **(1)** one or two lines of definition, **(2)** the diagram/matrix, **(3)** bullet points of explanation keyed to the diagram, **(4)** one named real-world example. That whole shape fits in 20 minutes and is exactly what the answer key rewards.

Legend used in these notes

- PROF** Content taken from the instructor's own answer key or worked solution.
- PYQ** This question appeared in a past question paper.
- MUST DRAW** A diagram is compulsory or near-compulsory for this answer.
- HIGH** High probability of appearing, based on past papers and class emphasis.

Contents

Session 1	Introduction to Product Design and Development
Session 2	Product Development Process, Organisation and Planning
Session 3	Identifying Customer Requirements
Session 4	Developing Product Specifications and QFD
Session 5	Establishing Product Function and Concept Generation
Session 6	Concept Development — Morphological, TRIZ, Axiomatic; Concept Selection Basis
Session 7	Concept Selection and Testing — Pugh, Weighted Matrix, AHP
Session 8	Product Architecture
Appendix A	Past Question Papers — Full Text
Appendix B	Instructor's Own Answer Key (EC-2 Make-up) — Reproduced
Appendix C	Instructor's Handwritten AHP Solution
Appendix D	Last-Minute Revision Sheet

Session 1 — Introduction to Product Design and Development

1.1 Product and Product Development

A **product** is something sold by an enterprise to its customers. **Product development** is the set of activities beginning with the perception of a market opportunity and ending in the production, sale and delivery of a product.

Economic success of firms depends on three abilities

1. Ability to **identify the needs** of customers
2. Ability to **launch the product quickly** so that it meets customer requirements
3. Ability to **make and sell the product at lower cost**

The three functions that intersect in product development

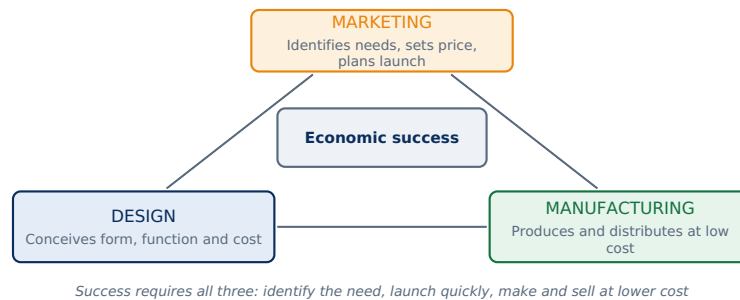


Fig 1.1 — Marketing, Design and Manufacturing intersect in product development

These three abilities map onto the three functions that intersect in product development: **Marketing — Design — Manufacturing**. Draw this as a three-circle or triangle diagram; it is a one-mark freebie in almost any Session 1 question.

Five performance characteristics of product development

Characteristic	What it measures
Product quality	How good is the product? Does it satisfy customer needs? Is it robust and reliable? Reflected in market share and price customers will pay.
Product cost	Manufacturing cost — capital equipment, tooling, incremental cost of each unit. Determines profit at a given sales volume and price.
Development time	How quickly the team completed the effort. Determines responsiveness to competition and to technology change; determines how soon returns begin.
Development cost	How much the firm had to spend to develop the product. Usually a significant fraction of the investment required for profit.
Development capability	Are the team and firm better able to develop future products as a result of this project? An asset the firm carries forward.

Five characteristics of product development performance



Fig 1.2 — The five characteristics of product development performance

Five challenges of product development

Trade-offs	An aircraft can be made lighter but that increases cost. Recognising, understanding and managing such trade-offs is the hardest part of design.
Dynamics	Technologies improve, customer preferences shift, competitors launch, the macroeconomic environment changes — all while you are developing.
Details	A modest product involves thousands of decisions, each with cost implications.
Time pressure	Decisions must usually be taken quickly and without complete information.
Economics	Developing, producing and marketing a product requires large investment; to earn a return, the resulting product must be appealing and inexpensive to produce.

1.2 What is Design?

- To design is to **pull together (synthesise)** something new, or to arrange existing things in a new way.
- Design establishes and defines solutions to, and pertinent structures for, problems not solved before — or new solutions to problems

previously solved differently.

- Good design requires **both analysis and synthesis**.

The Four C's of Design

Creativity • Complexity • Choice • Compromise

A frequently asked one-liner. Remember all four — they are easy marks.

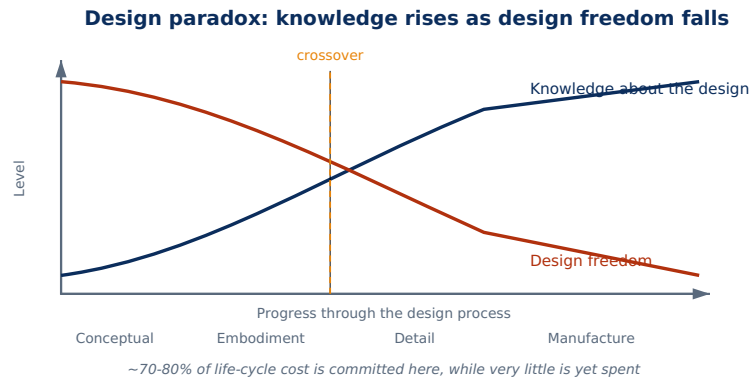


Fig 1.3 — The design paradox: knowledge rises as design freedom falls

1.3 Types of Design HIGH

Type	Definition	Example from class
Original design (innovative)	Generates an original concept for solving a specific need. Few successful original designs occur over time; they disrupt markets by replacing old equipment and manufacturing methods.	Refrigeration/AC moved from ozone-depleting CFCs (R-12, R-11) and HCFCs (R-22, R-502) to zero-ODP HFCs (R-410A, R-134a), forcing change in most components and processes. Fitbit revolutionising wearable health tech.
Adaptive design	Adapting an existing system to a new problem .	Domestic AC was dominated by reciprocating compressors; rotary compressors (half the size/weight, tolerant of harsh climates) replaced most of them.
Re-design (variant)	Extension of product platforms or improvement in an existing design — variation of size, geometry, material properties, control parameters.	A firm making motors up to 15 kW extending the range to 100 kW. Aircraft engine fan case made in composite instead of metal.
Selection design	Selection of off-the-shelf components having the needed properties.	Bearings of specific load and speed. Standard fasteners (bolts and nuts) connecting aircraft engine fan cases.
Industrial design	Products designed for the market, focusing primarily on form, aesthetics and ergonomics for mass usage.	Chairs, shoes.

1.4 Problem-Solving Methodology MUST DRAW

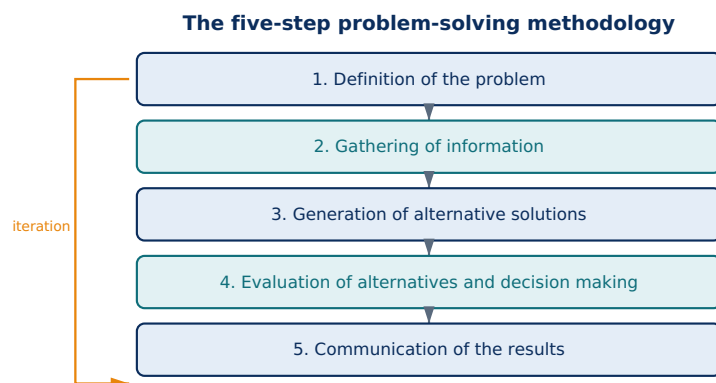


Fig 1.4 — The five-step problem-solving methodology

1. **Definition of the problem**
2. **Gathering of information**
3. **Generation of alternative solutions**
4. **Evaluation of alternatives and decision making**
5. **Communication of the results**

The accompanying curve is the classic one: as the design progresses, **knowledge about the design increases** while **design freedom decreases**. This is exactly why early-phase decisions commit most of the cost.

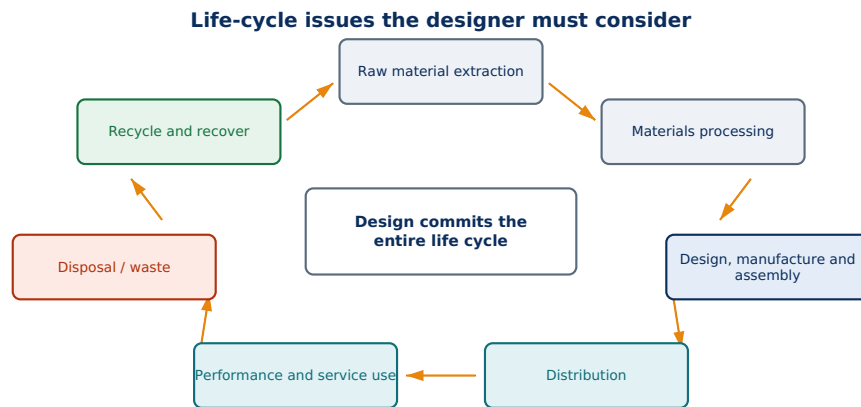
1.5 Considerations of a Good Design

Three groups of considerations:

(1) Achievement of performance requirements

- **Primary performance requirements** — the core function the product must deliver.
- **Complementary performance requirements** — useful life, reliability, ease of use, robustness, safety, maintenance.
- **Environmental requirements for performance** — the service conditions under which the product must operate, and how the product behaves with regard to maintaining a safe and clean environment (green design, sustainable design).
- **Aesthetic requirements** — form, colour, texture, perceived quality.

(2) Life-cycle issues



Good design answers for every loop: extraction, manufacture, distribution, use, retirement and recovery

Fig 1.5 — Life-cycle issues the designer must consider

(3) Regulatory and social issues

Compliance with standards and regulatory bodies — ASME, ASTM, FAA, OSHA, Consumer Product Safety Commission, EPA and their equivalents. Social issues include liability, ethics, and public acceptance.

1.6 The Design Process — Asimow's Morphology of Design **MUST DRAW**

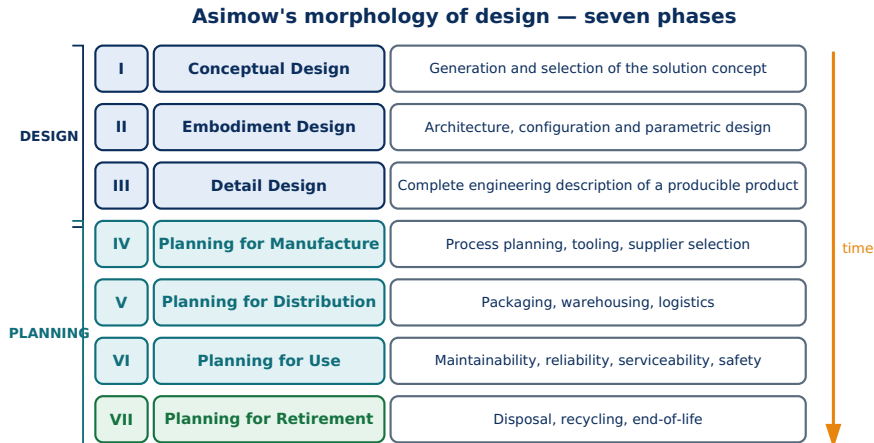


Fig 1.6 — Asimow's morphology of design: seven phases

I	Conceptual Design	Generation and selection of the solution concept
II	Embodiment Design	Architecture, configuration and parametric design
III	Detail Design	Complete engineering description of a tested and producible product
IV	Planning for Manufacture	Process planning, tooling, supplier selection
V	Planning for Distribution	Packaging, warehousing, logistics
VI	Planning for Use	Maintainability, reliability, serviceability, safety in use
VII	Planning for Retirement of the Product	Disposal, recycling, end-of-life

Design Review

- A **retrospective study** of the design, conducted by a review team.
- Held **three to six times** in the life of a project.
- Covers both **technical and business** aspects.
- Central question: "Are we building it right, and are we building the right product?" (verification vs validation).

1.7 Product Development Process Types HIGH

A **product development process** is a sequence of steps or activities that an enterprise employs to conceive, design and commercialise a product. Three types:

(a) Stage-Gate (Generic) PD Process

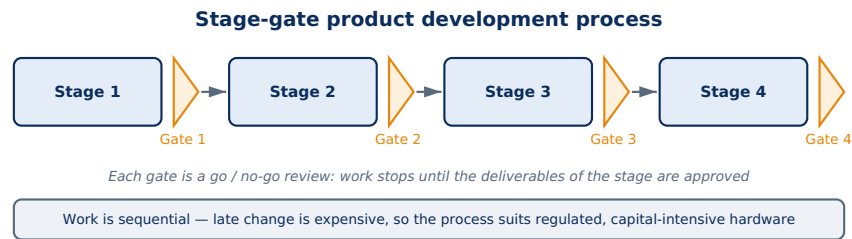


Fig 1.7 — Stage-gate product development process

(b) Spiral PD Process

Spiral model = {(1st stage-gate process) + 1st version of working product} + {(2nd stage-gate process) + 2nd version} + ... + {(nth stage-gate process) + nth version}

The spiral model emphasises **iterative development** and allows budget adjustments based on feedback and insights gained during each iteration. It sits between the pure **Waterfall** model and the **Agile** model.

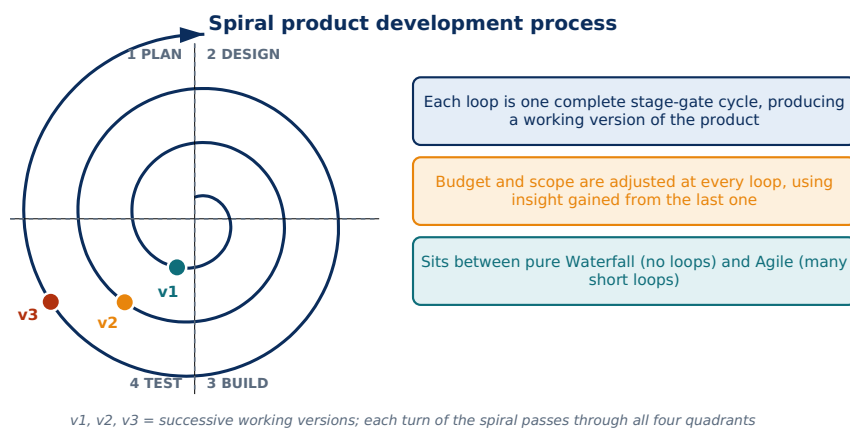


Fig 1.8 — Spiral product development process

(c) Hybrid PD Process

A waterfall skeleton with agile sprints running inside each phase. The formal gates are retained where they are unavoidable — tooling release, certification, production sign-off — while the work inside a phase proceeds in short iterative cycles. This is the practical choice for complex electromechanical products, where the hardware needs gates but the software needs sprints.

Stage gates retained for hardware milestones + sprint cycles inside each stage for software and rapid-build elements

1.8 Factors for Success

- The **quality, performance and price** of the product
- The **cost to manufacture** the product over its life cycle
- The **cost of product development**
- The **time needed to bring the product to market**

Profit = Product Price (↑) – Product Cost (↓)

1.9 Variations in the Generic PD Process HIGH

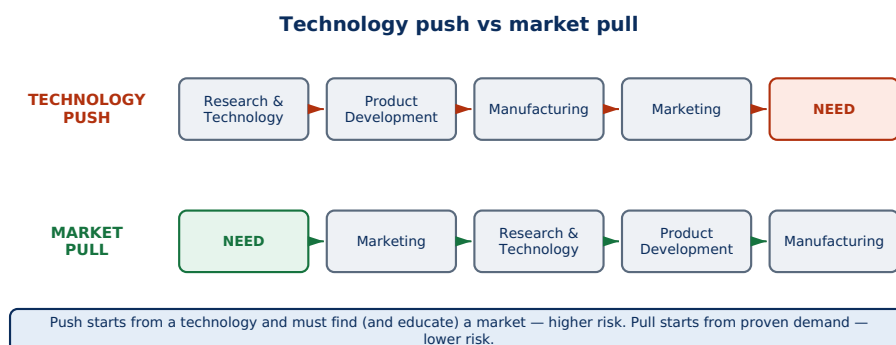


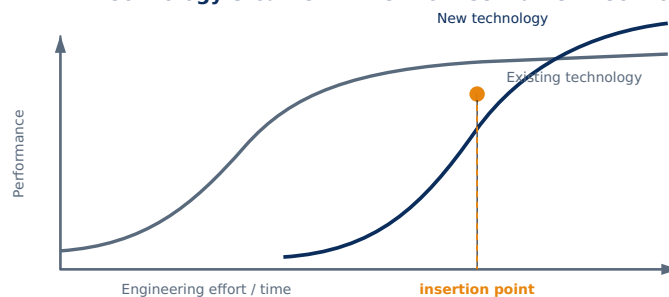
Fig 1.9 — Technology push vs market pull — the two sequences

Process type	Distinct features	Examples
Generic (market pull)	Process begins with a market opportunity; selects technologies to meet customer needs.	Sporting goods, furniture, tools
Technology push	Firm begins with a new proprietary technology and looks for an appropriate market.	Gore-Tex rainwear, Tyvek envelopes
Platform products	Assumes the new product will be built around an established technological subsystem.	Consumer electronics, computers, printers
Process-intensive	The production process constrains the product properties; product cannot be separated from process.	Snack foods, breakfast cereals, chemicals, semiconductors
Customised	New products are slight variations of existing configurations.	Switches, motors, batteries, containers
High-risk	Technical or market uncertainties create high risk of failure; risks identified early and addressed.	Pharmaceuticals, space systems
Quick-build	Rapid modelling and prototyping enable many design-build-test cycles.	Software, mobile phone apps
Complex systems	System decomposed into several subsystems and many components, developed in parallel by many teams.	Aircraft, jet engines, automobiles

Technology Push vs Market Pull — comparison

Dimension	Technology Push	Market Pull
Starting point	New proprietary technology exists; a market must be found	Market need identified; technology selected to satisfy it
Risk	High — market may not exist	Lower — demand already demonstrated
Investment	Very high (pharma, space)	Low to moderate
Marketing effort	Market must be created and educated ; reshaping takes time and money	Channels and awareness already exist; marketing is easier
Time to revenue	Long	Short
Example	Paracetamol/Dolo before the 2000s; Gore-Tex	Incremental variants in automobiles and consumer goods

Technology S-curve — when to insert a new technology



Insert when the new curve's performance crosses the incumbent's — not before, not after

Fig 1.10 — Technology S-curve: when to insert a new technology

1.10 Session 1 Summary Diagram

The generic product development process — six phases

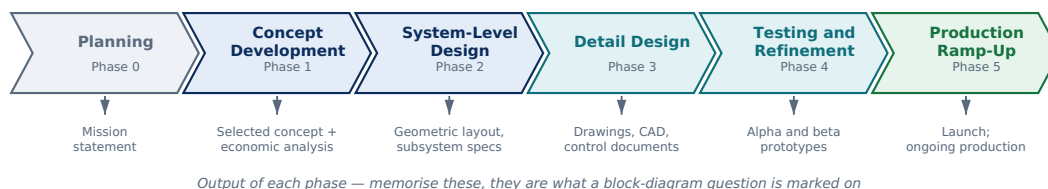


Fig 1.11 — The generic product development process and the output of each phase

Key takeaways to memorise for Session 1

- Three abilities driving economic success; Marketing-Design-Manufacturing triangle
- Five PD characteristics; five PD challenges
- Four C's of design
- Five types of design **with one example each**
- Five-step problem-solving methodology
- Asimow's seven phases
- Stage-gate vs spiral vs hybrid
- Eight process variations with examples; technology push vs market pull

1.11 Expected Questions — Session 1

Q1. Explain the key stages of product development with a neat block diagram. "Iterative product development has [15] become increasingly dominant in technology firms over the past decade." — True or False? Justify your answer with examples. PYQ MUST DRAW

Part A — block diagram (draw first). Six phases of the generic PD process:

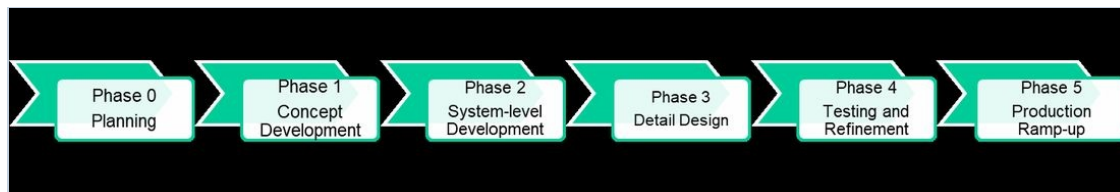


Fig 1.12 — The six-phase generic PD process — reproduce this exact chevron diagram

Phase	Key activities and output
Phase 0 — Planning	Corporate strategy, technology assessment, market objectives. Output: mission statement (target market, business goals, assumptions and constraints).
Phase 1 — Concept Development	Identify customer needs, establish target specifications, generate and select concepts, test concept, set final specifications, plan downstream development. Output: selected concept + economic analysis.
Phase 2 — System-Level Design	Product architecture, decomposition into subsystems and components, preliminary layout, define final assembly scheme. Output: geometric layout, functional specification of subsystems, process flow diagram.
Phase 3 — Detail Design	Complete specification of geometry, materials and tolerances of all unique parts; identification of standard parts; process plans and tooling design. Output: control documentation / drawings and files.
Phase 4 — Testing and Refinement	Construction and evaluation of alpha prototypes (production-intent parts, non-production process) and beta prototypes (production parts and process). Reliability, life, performance and regulatory testing.
Phase 5 — Production Ramp-Up	Product made using the intended production system; workforce trained; residual problems resolved. Transition to ongoing production and launch.

Part B — True or False. Answer: **TRUE**, with justification.

- **Why it is true:** product life cycles have shortened, customer preferences shift quickly, and digital/connected products can be updated after sale. Iteration lets the firm start earning earlier and correct direction using real feedback rather than forecasts.
- **Mechanism:** the spiral model — each loop is a complete stage-gate cycle producing a working version. Budget and scope are adjusted each loop based on insight gained.
- **Examples:** smartphone operating systems shipped and improved through over-the-air updates; electric vehicles where powertrain software is upgraded post-delivery; SaaS and app development where release cadence is weekly.
- **Balanced view (add this — it earns marks):** iteration is **not** dominant everywhere. Process-intensive and safety-critical products — aircraft structures, pharmaceuticals, nuclear systems, medical implants — remain largely stage-gate/waterfall because certification, tooling cost and liability make late change prohibitively expensive. So the correct statement is: iterative development dominates **technology firms and software-rich products**, while regulated hardware retains stage-gate discipline. Hybrid processes are the practical middle ground.

Q2. Classify the various types of design. Illustrate each with a suitable engineering example, and explain which type dominates in a mature industry and why. [15]

- Draw a simple **five-branch tree**: Design → Original / Adaptive / Re-design / Selection / Industrial.
- Use the table in section 1.3 — definition plus the class example for each.
- **Which dominates in a mature industry: re-design (variant) and selection design.** In a mature industry the technology is stable, platforms are established, and the economics favour extending existing platforms (motor range extension, new model variants) and buying standard components rather than inventing. Original design is rare and episodic — it occurs when an external forcing function appears, such as the CFC→HFC refrigerant transition or an emissions regulation change.
- Close with the knowledge-vs-design-freedom curve to show why mature industries avoid original design: the cost of change is high once platforms and tooling are committed.

Q3. Compare stage-gate, spiral and hybrid product development processes. For each, state one product category [15] where it is the appropriate choice and justify.

Aspect	Stage-Gate	Spiral	Hybrid
Structure	Linear stages separated by go/no-go gates	Repeated stage-gate loops, each producing a working version	Waterfall skeleton with agile sprints inside phases
Feedback	At gates only	Every iteration	Continuous within phase, formal at gates
Change cost	High — late change is expensive	Low — change absorbed in next loop	Moderate
Budget control	Fixed up front	Adjusted per iteration	Phase-level control
Risk handling	Risk retired at gates	Risk retired progressively by building	Both
Best for	Regulated, capital-intensive hardware: aircraft structures, pharmaceuticals — certification demands frozen configurations and documented gate evidence	Software-rich and rapidly evolving products: mobile apps, infotainment, connected devices — the market rewards early release and learning	Complex electromechanical systems: automotive ECUs, appliances — hardware needs gates for tooling release, software needs sprints

Add the three diagrams (Figs 1.8–1.10). Close with the spiral equation.

Q4. Distinguish between technology push and market pull products. Why does technology push carry higher risk? [15] Support with examples.

- Draw the two sequence diagrams (Fig 1.9): *Technology push*: Research & Technology → Product Development → Manufacturing → Marketing → Need. *Market pull*: Need → Marketing → Research & Technology → Product Development → Manufacturing.
- Use the comparison table in section 1.9.
- **Why higher risk**: the need is unproven. The firm must invest to develop the technology **and** to create and educate a market, so the outlay is larger and the payback is later. If the market does not materialise, the entire investment is stranded. Market pull starts from demonstrated demand, so commercial risk is largely limited to execution.
- **Examples**: push — Gore-Tex membrane, which needed applications to be found; paracetamol, unknown to the mass Indian consumer before branding as Dolo. Pull — a new SUV variant responding to identified segment demand.
- Add the S-curve (Fig 1.10) to explain *when* to insert a new technology — at the inflection where the new curve's performance crosses the incumbent's.

Q5. What constitutes a good design? Discuss the considerations a designer must satisfy, and explain the role of design reviews in ensuring them. [15]

- Structure the answer as the **three groups** of section 1.5, drawn as a three-branch tree.
- Under group 1, subdivide into primary performance, complementary performance, environmental, aesthetic.
- Under group 2, draw the life-cycle loop (material extraction → manufacture → distribution → use → retirement → recycle).
- Under group 3, name specific bodies: ASME, ASTM, FAA, OSHA, EPA, CPSC.
- **Design review role**: a retrospective, team-based study held 3–6 times in a project, covering technical and business aspects. It is the formal mechanism that asks *are we building it right* (verification against specification) and *are we building the right product* (validation against customer need). Reviews catch the trade-offs and details that individual designers miss, and they are the gates at which life-cycle and regulatory considerations are formally checked.

Q6. Explain the challenges of product development. Using the knowledge–design–freedom relationship, explain why decisions made in the early design phase have a disproportionate effect on product cost. [15]

- List and briefly explain the five challenges (table in section 1.2), each with a one-line example.
- Draw the curve from Fig 1.5: x-axis = design progress/time; two curves — **knowledge about the design rising** and **design freedom falling**. They cross early.
- **The argument**: in the conceptual and embodiment phases the team has maximum freedom but minimum knowledge; by the detail-design and manufacturing phases it has maximum knowledge but almost no freedom. Because architecture, material and process choices are locked in early, roughly 70–80% of life-cycle cost is **committed** during design while only a small fraction is actually **spent** there.
- **Consequence**: the cheapest way to reduce product cost is to invest more effort in the early phases — which is precisely the argument for concurrent engineering, DFMA and front-loaded customer need identification.
- Use the design-freedom curve (Fig 1.3) to reinforce.

Session 2 — PD Process, Organisation and Planning

2.1 Product Development Organisations

Organisations are formed by establishing links among individuals. Two basic classification schemes: grouping by **function** (area of specialisation) or grouping by **project/product**.

Three product development organisation structures

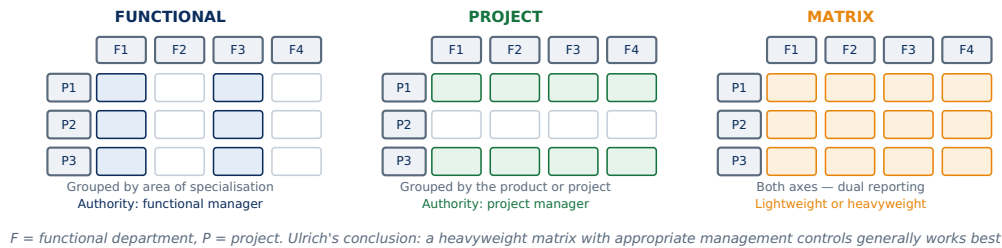


Fig 2.1 — Functional, project and matrix organisation structures

Functional vs Project organisation HIGH

#	Functional Organisation	Project Organisation
1	Employees grouped by expertise , which helps performance	Employees grouped by the project/product , which helps them see the bigger picture
2	More accountability in roles and responsibilities	More ownership
3	Clear career path	Product-centric and varied → versatile and flexible
4	Well-engaged department	Finding peers in the team may be challenging
5	Feel safe and comfortable	Sense of urgency, driven by tight schedule and team setup
6	More job security → loyal to organisation	Sense of insecurity — project closes once done → less loyal
7	Monotonous work → lose interest	Exposure to entire PD gives learning opportunities
8	Cross-functional / inter-departmental effectiveness is poor	Team is inter-disciplinary → more exposure
9	Change management is challenging	Change management is easy to perform
10	Delayed decision making, sometimes unsatisfactory/conflicting	Faster decision making
11	Challenges compiling communication and feedback	Communication and feedback are good
12	Resources usually flexible → cost managed well	Resources sometimes specialised → cost can be high

Matrix organisation

Advantages

1. Readiness, high efficiency and faster to market
2. Suited to large and multi-project organisations — resources can be managed well
3. Effective management of highly skilled experts and capable resources
4. Engaged team → well-integrated organisation
5. Learn and grow — opportunities to work on multiple projects

Disadvantages

1. Role confusion when reporting to multiple supervisors/managers
2. Project Manager vs Functional Manager conflict over priority and resource deployment — affecting the employee
3. More managers → increased overhead cost
4. KPI and career development vs project deliverables tension

How to choose the organisation — four factors

1. **Cross-functional integration** needed
2. **Cutting-edge functional expertise** required
3. **Resource utilisation** strategy
4. **Product development speed**

Ulrich's conclusion, quoted in the Session 1 summary: *generally a **heavyweight matrix organisation** with appropriate management controls works best.* Say this in any organisation question.

Globally distributed PD teams — six reasons

- (1) Access to information — regional and global
- (2) Local skills / availability of technical expertise
- (3) Location of manufacturing plants
- (4) Collaboration with local suppliers
- (5) Low cost (labour and operation)
- (6) Outsourcing to increase PD capacity.

2.2 Concurrent Engineering PROF HIGH

The **conventional (sequential / "over the wall") way** has (1) distinct functions and (2) separate departments with no inter-linking. Each function completes its work and throws it over the wall to the next.

Key elements of concurrent engineering

1. **Cross-functional team** → multi-disciplinary collaboration, increased quality
2. **Parallel design** → reduced cycle time and cost
3. **Vendor partnering**
4. **Revenue Sharing Partner (RSP)** and **Risk and Revenue Sharing Partner (RRSP)** — giving access to new technologies, improved manufacturing, lower development costs, knowledge exchange

Industry case given in class: Melrose / GKN Aerospace — contracts lasting 50 years (life-of-programme), 19 RRSP contracts.

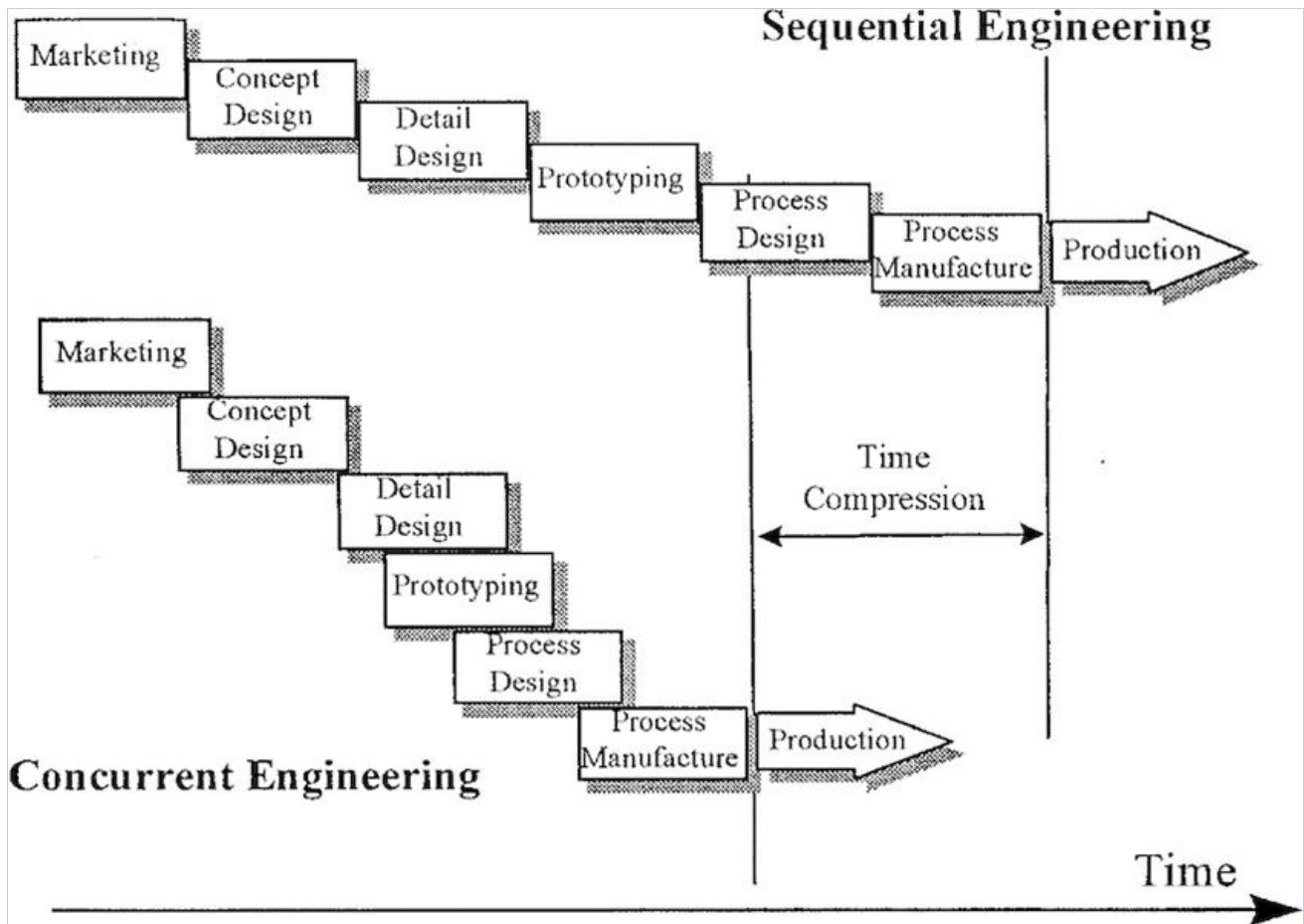


Fig 2.2 — From the instructor's own answer key. Sequential engineering (top) vs concurrent engineering (bottom), showing time compression. Reproduce this exact diagram — she said drawing it takes 3-4 minutes and is expected.

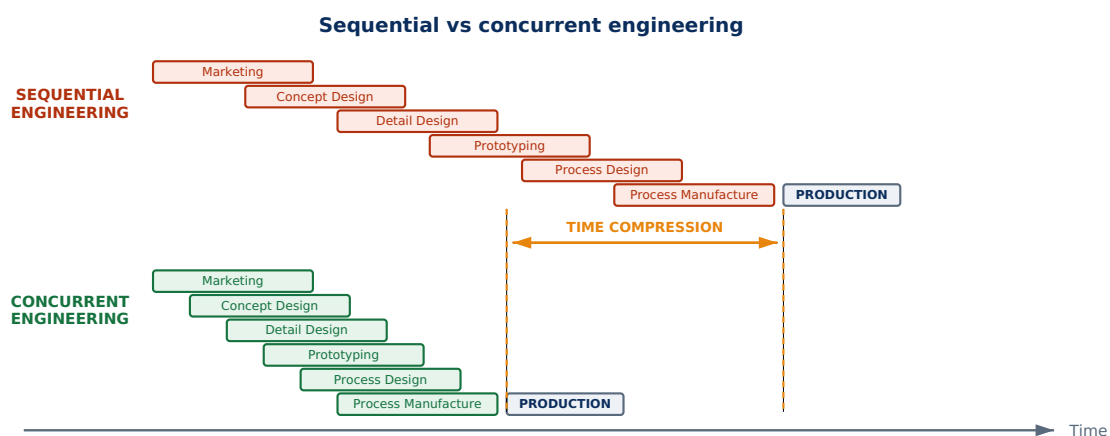


Fig 2.3 — Sequential vs concurrent engineering, showing time compression

2.3 Product Planning

The **product planning process** takes place **before** a product development project is formally approved. It considers both the current product line and the potential portfolio of projects the organisation might pursue, and determines **what subset** of possible development projects will be pursued over **what time period**.

Questions product planning must answer

1. What product development projects will be undertaken?
2. What mix of fundamentally new products, platforms and derivative products should be pursued?

- How do the various projects relate to each other as a portfolio?
- What will be the timing and sequence of the projects?

Four types of PD projects HIGH

Type	Description
New product platforms	New product based on a new, common platform. Requires major development effort.
Derivatives of existing platforms	Extend an existing platform to address familiar markets with a new product or feature.
Incremental improvements	Modify some features of existing products to keep the line current and competitive (cost, serviceability).
Fundamentally new products	Radically different product or production technologies; may address new and unfamiliar markets.

2.4 The Five-Step Product Planning Process MUST DRAW

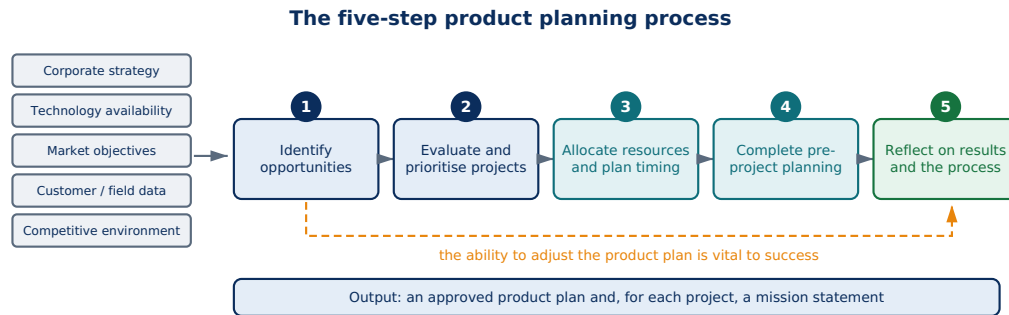


Fig 2.4 — The five-step product planning process

Step	Activity	Content
1	Identify opportunities	Collect ideas for new products from all sources into an opportunity funnel
2	Evaluate and prioritise projects	Four perspectives: competitive strategy, market segmentation, technological trajectories, product platforms
3	Allocate resources and plan timing	Aggregate planning; balance budget vs promising projects vs limited resources
4	Complete pre-project planning	Core team drafts vision statement and mission statement
5	Reflect on the results and the process	Reality check before committing to full development

Step 1 — Identify opportunities

An **opportunity** is an idea for a new product, arising from a **new need**, a **new technology** or a **new solution**. Opportunities are categorised by what the team already knows (solution or need) and by uncertainty:

Horizon	Risk	Nature	Typical content
Horizon 1	Low	Existing products	Improvement, variants, cost reduction
Horizon 2	Medium	Combination	Combining existing need with new solution, or new need with existing solution
Horizon 3	High	New to the world	New technology or new product

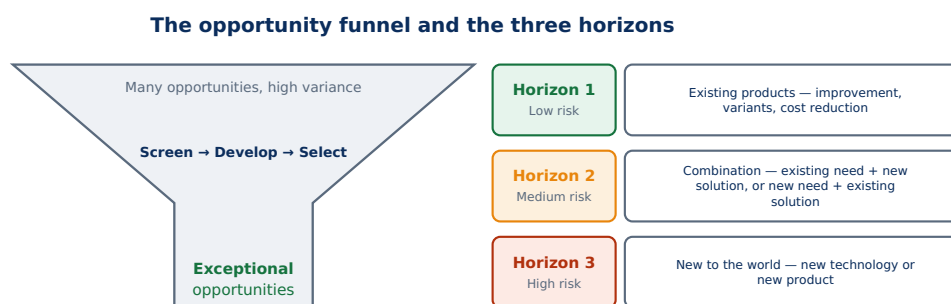


Fig 2.5 — The opportunity funnel and the three horizons

Opportunity identification process — six steps:

- Establish a charter (a **broader** charter generates more opportunities)
- Generate and sense many opportunities
- Screen opportunities
- Develop promising opportunities
- Select exceptional opportunities
- Reflect on the results and the process

Heuristics given in class: passion helps; **annoyance gives birth to opportunity** (the umbrella opening/closing example); exploit existing resources; study and listen to customers; follow trends; use external sources of ideas; prefer a **strong vote** over **most votes**; apply the **Real-Win-Worth it** screen.

Step 2 — Evaluate and prioritise projects: four perspectives

2.1 Competitive strategy — four generic strategies:

Technology leadership	Great emphasis on R&D of new technologies and their deployment through PD
Cost leadership	Focus on efficiency, superior manufacturing method, low-cost operation
Customer focus	Focus on customer needs and preferences
Imitative	Follow the market and competitors

2.2 Market segmentation — map competitors' products and the firm's own products onto segments; assess opportunities by identifying weaknesses in its own product line and in competitors' offerings.

2.3 Technology trajectories — answers "When to adopt a new basic technology in a product?" using the **S-curve**. The S-curve shows how a technology (e.g. additive manufacturing) matures from initial R&D on the lower flat part to high-volume production on the steep part.

2.4 Product platform planning — use of existing platforms to introduce a product rapidly, easily and economically. Platform-derived products typically cost 2×–10× less effort than new-platform products.

Technology roadmapping — a way to represent the expected availability and future use of various technologies relevant to the products being considered.

Evaluating fundamentally new product opportunities — criteria: market demand and growth rate; sustainability; knowledge and awareness of the market and technology; current product portfolio and capabilities; potential for patents, trade secrets or other barriers to competition.

Balancing the portfolio — map the portfolio on dimensions such as technical risk, financial return and market need; or on *extent of change in product line vs extent of change in production process*. Organisations often trade off investments of different types.

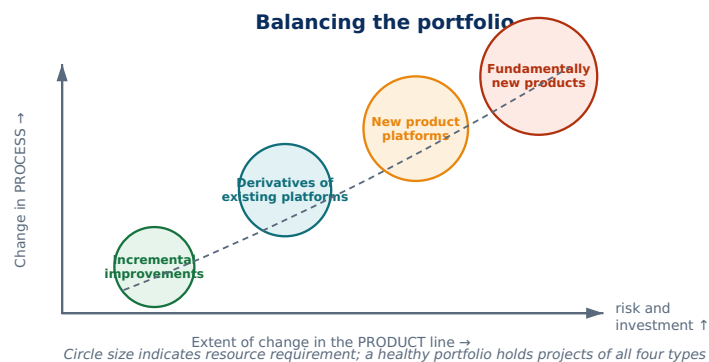


Fig 2.6 — Balancing the portfolio: product change vs process change

Step 3 — Allocate resources and plan timing

Too many projects without regard for resource availability results in skilled engineers and managers being allotted to too many projects. This causes:

- Drop in productivity → delayed deliveries → late to market → **loss in profit**
- Stressful job → disengaged staff → **loss of resources**

Rule of thumb from class: do not deploy a resource on more than three projects.

Project timing must consider: timing of product introduction (quality vs speed); technology readiness (robustness); market readiness (when to launch, when to improve); competition.

Step 4 — Complete pre-project planning

A **core team** — small and cross-functional — formulates:

- **Product vision statement** — very general
- **Mission statement** — detailed and more specific

Contents of a mission statement **MUST KNOW**

- Basic function of the product (brief product description)
- How it would benefit the customer (key business goals / value proposition)
- Business goals including time, cost and quality
- Primary and secondary markets
- Assumptions and constraints — e.g. manufacturing, service, sustainability
- List of stakeholders

Remember the one-line characterisation: a mission statement specifies the **direction**, but not a precise destination or a particular way to proceed.

Step 5 — Reflect on the results and the process

A reality check is required before proceeding, else the consequences are severe and expensive. Questions cover: the opportunity funnel; whether the product plan supports the firm's strategy and opportunities; resource allocation; creativity; acceptance of the mission statement; over-constrained objectives; improvement opportunities.

Key takeaways to memorise for Session 2

- Functional vs project vs matrix (lightweight/heavyweight) — at least 6 differentiators
- Four factors for choosing an organisation; heavyweight matrix generally best
- Four key elements of concurrent engineering + the sequential-vs-concurrent timeline diagram
- Five-step product planning process
- Four types of PD projects; three opportunity horizons
- Four perspectives for prioritising: competitive strategy, market segmentation, technology trajectories, platform planning
- Six contents of a mission statement

2.5 Expected Questions — Session 2

Q1. Explain the difference between traditional product development and the concurrent engineering approach. [15]
Why is a concurrent engineering team best suited for the modern product development process in today's competitive world? **PROF** **PYQ** **MUST DRAW**

This is the instructor's own answer-key question. She walked through it on screen and said: draw both diagrams (3–4 minutes), list **at least 5–6 differentiating aspects** using PQCDSM, then the justification. Total ~15–20 minutes.

Step 1 — draw the two timelines (Fig 2.2): sequential engineering as a descending staircase of non-overlapping boxes (Marketing → Concept Design → Detail Design → Prototyping → Process Design → Process Manufacture → Production), and concurrent engineering as the same boxes **overlapping**, with the gap between the two production start points marked "**Time Compression**".

Step 2 — the differentiating table (use PQCDSM as the spine):

Aspect	Traditional / Sequential	Concurrent Engineering
P — Productivity	Each function works alone; rework loops between departments; low throughput	Cross-functional team works in parallel; fewer rework loops; higher throughput
Q — Quality	Manufacturability problems discovered late; quality "inspected in"	Manufacturing and service inputs available from concept stage; quality "designed in"
C — Cost	Late design changes are expensive; high cost of poor quality	Change cost low because changes happen early; lower development and product cost
D — Delivery	Long cycle time — stages strictly in series	Significant time compression through overlapping stages → faster to market
S — Safety	Safety and regulatory issues surface during validation	Safety, regulatory and DFX considerations embedded from the start
M — Morale / Maintainability	"Over the wall" culture, blame between departments; serviceability an afterthought	Shared ownership, better communication and morale; maintainability designed in
Communication	Formal, document-driven, one-directional	Continuous, informal, multi-directional within the team
Supplier role	Supplier engaged after design freeze	Vendor partnering, RSP/RRSP arrangements from the outset

Step 3 — why concurrent engineering suits today's competitive world:

- **Shortened product life cycles** — being late to market costs more than being slightly over budget.
- **Cost is committed early** — because ~70–80% of life-cycle cost is locked in during design, having manufacturing, purchasing and service in the room at concept stage is where the savings are.
- **Product complexity** — modern electromechanical products span mechanical, electronics, software and regulatory domains; no single function can design them alone.
- **Global supply chains** — RSP/RRSP partners bring technology and share risk, which requires parallel rather than serial engagement.
- **Fewer engineering changes** — issues surface while the design is still fluid, so the change curve is front-loaded.
- **Regulatory and sustainability pressure** — compliance cannot be retro-fitted cheaply.

Closing line from the answer key: *traditional development suits simpler projects where predictability and control are prioritised; concurrent engineering is ideal for complex, competitive environments where speed, quality and innovation are critical.*

Q2. Explain the product planning process with a neat block diagram. Discuss the four types of product development projects an organisation must balance in its portfolio. [15]

- Draw the five-step flow (Fig 2.4), with the inputs shown feeding step 1: corporate strategy, technology availability, market objectives, customer/field data, competitive environment.
- Explain each step in two or three lines — use the table in section 2.4.
- Give the four project types in a table (section 2.3) with an automotive example each: new platform = a new vehicle architecture; derivative = a new body style on that architecture; incremental = a mid-life facelift or cost-reduced variant; fundamentally new = the shift to a battery-electric powertrain.
- Add the portfolio map (Fig 2.4) to show how the four types are balanced against technical risk and return.
- Close with the slide's own emphasis: **the ability to adjust the product plan is vital to success.**

Q3. Compare functional, project and matrix organisations for product development. Which would you recommend [15] for a company developing a complex electromechanical product, and why?

- Draw the organisation charts (Fig 2.1 and Fig 2.1) — functional grid, project grid, and the matrix with both.

- Give at least six differentiators from the table in section 2.1.
- Cover matrix advantages (5) and disadvantages (4) from section 2.1.
- Distinguish **lightweight** matrix (functional managers hold authority; project manager coordinates) from **heavyweight** matrix (project manager holds significant authority over resources and budget).
- **Recommendation: heavyweight matrix.** Justify against the four choice factors: a complex electromechanical product needs high **cross-functional integration** (mechanical, electronics, software, manufacturing), but also needs deep **functional expertise** retained in home departments; a matrix gives good **resource utilisation** across concurrent projects; and the heavyweight variant gives the project manager enough authority to protect **development speed**. Note the trade-off to watch: dual reporting and overhead cost.

Q4. What is an opportunity in product planning? Explain the opportunity identification process and how opportunities are screened and prioritised.

[15]

- Define: an opportunity is an idea for a new product arising from a new need, new technology or new solution.
- Draw the **opportunity funnel** (Fig 2.5) — many opportunities in, high variance, high quality out.
- List the six steps of the identification process.
- Present the three horizons as a table with risk level and content.
- Prioritisation: explain the four perspectives — competitive strategy (four generic strategies), market segmentation (segment map), technology trajectories (S-curve), product platform planning (2x-10x effort multiplier).
- Mention the **Real-Win-Worth it** screen and the criteria for evaluating fundamentally new opportunities.

Q5. Write a mission statement for a new product of your choice. Explain each element and state why the mission statement must specify direction rather than destination.

[15]

Use a two-column table — this is how the instructor's template is laid out. Example for a **compact electric kettle**:

Element	Content
Product description	A compact, lightweight electric kettle with reduced heat loss
Benefit proposition / key business goals	Faster boil with lower energy use; product introduced in Q3; 40% gross margin; 8% share of the premium small-appliance segment in year 2
Primary market	Urban single-person and two-person households
Secondary markets	Office pantries; student hostels; travel/portable segment
Assumptions and constraints	Uses existing 230 V single-phase supply; double-walled stainless body; manufactured on existing small-appliance assembly line; recyclable materials; complies with applicable electrical safety standards
Stakeholders	Buyers, end users, retailers, service network, manufacturing operations, component suppliers, regulatory bodies

Why direction and not destination: the mission statement is written **before** concept development. If it prescribed a specific solution it would foreclose the concept generation space and force the team into premature commitment to a form — exactly the failure mode that functional decomposition is designed to avoid. By bounding the problem (markets, goals, constraints) without specifying how, it gives the team maximum design freedom at the point where design freedom is most valuable.

Session 3 — Identifying Customer Requirements

3.1 Where it sits in the PD process

Customer need identification is the **first activity of Phase 1, Concept Development**. It follows the mission statement and feeds directly into target specifications.

Technical questioning — clarifying the task before gathering needs

Before interviewing customers, the team interrogates the task statement itself:

- What is the problem really about?
- What implicit expectations and desires are involved?
- Are the stated customer needs, functional requirements and constraints truly appropriate?
- What avenues are open for creative design and inventive problem solving? What avenues are limited or closed? Limitations on scope?
- What characteristics/properties **must** the product have? What **must it not** have?
- What aspects of the design task can and should be quantified now?
- Do any biases exist with the chosen task statement or terminology? Has the design been posed at the appropriate **level of abstraction**?
- What are the technical and technological conflicts inherent in the design task?

Worked example from class — fingernail clipper

What is the problem really about?	Clumsy operation of a typical clipper
Implicit expectations and desires?	Remain a manual clipper operable by oneself
Are stated needs appropriate?	Many groups, in trade magazines and customer studies, have noted clumsy operation; detailed assessment in post-customer interviews
Avenues open for creative design?	Can modify any and all parts; introduce different materials; combine functional parts
Limitations?	No electrical power; no service-based solution
Must have?	Easy to use, durable, safe, low cost
Must not have?	Should not be bulky

3.2 The Five-Step Customer Need Identification Process **MUST DRAW**

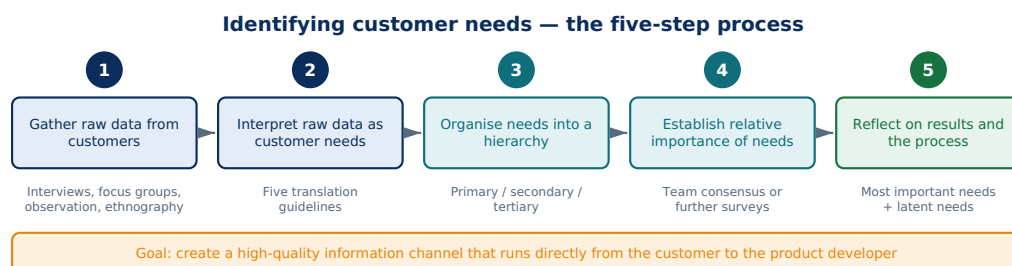


Fig 3.1 — The five-step customer need identification process

1. **Gather raw data from customers**
2. **Interpret the raw data** in terms of customer needs
3. **Organise the needs into a hierarchy** of primary, secondary and (if necessary) tertiary needs
4. **Establish the relative importance** of the needs
5. **Reflect on the results and the process**

Simple illustration used in class — T-shirt: size, colour, material, sleeves, cost.

Expressed vs latent needs **HIGH**

Expressed needs	What customers say they want
Latent needs	What the customer is unaware of — but which, if fulfilled, produces greater customer satisfaction

Example: the camera in mobile phones (Nokia, Motorola) — customers did not ask for it. *Paying special attention to what customers **do**, not what they **say**, is key to the successful development and marketing of new products.* Recognising latent needs is critical in product development.

3.3 Step 1 — Gather Raw Data from Customers

Ethnographic studies **PROF** **HIGH**

Ethnographic Studies



- Way people behave in their regular environments
- How a customer uses (or misuses) a product →
- Actual end users interacting with the product under typical use conditions
- Way of conducting the study:
 - Video
 - Photographs
 - Sketches
 - Interview

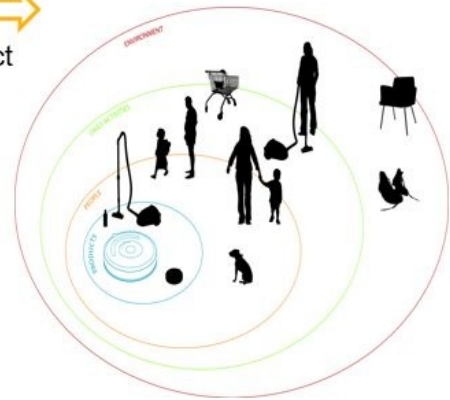


Fig 3.2 — *From the instructor's answer key.* Ethnographic studies

- The way people behave in their **regular environments**
- How a customer **uses (or misuses)** a product
- **Actual end users** interacting with the product under **typical use conditions**
- Ways of conducting the study: **video, photographs, sketches, interview**

Dimension	Contribution of Ethnography
Hidden Needs Discovery	Reveals unspoken behaviors, frustrations, and desires that inspire breakthrough ideas.
Contextual Understanding	Shows how products fit into daily routines, social interactions, and cultural practices.
Empathy Building	Helps designers and engineers see the world through the user's eyes, sparking human-centered solutions.
Idea Generation	Provides rich narratives and scenarios that fuel brainstorming beyond technical constraints.
Innovation Validation	Ensures creative concepts resonate with real users, not just theoretical models.
Cross-Disciplinary Inspiration	Anthropological insights encourage collaboration between design, engineering, and marketing teams.

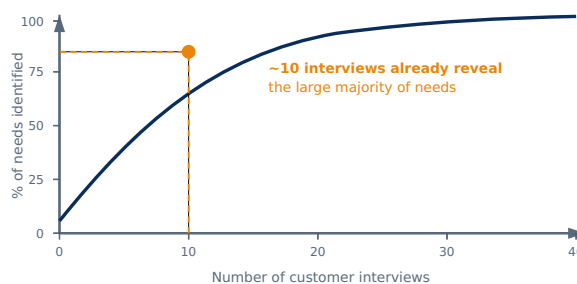
Fig 3.3 — *From the instructor's answer key.* How ethnography contributes to product development — reproduce these six dimensions

Methods to gather data

1. **Interviewing customers individually** — most information per unit effort
2. **Group discussion / focus group** with customers — efficient and less costly
3. **Observing customers** using the product or performing the intended task — UX sessions, beta events

Other methods: surveys, or surveys combined with the above. The aim is to create a **high-quality information channel directly from the customer** to the product developer.

Sample size vs proportion of needs identified



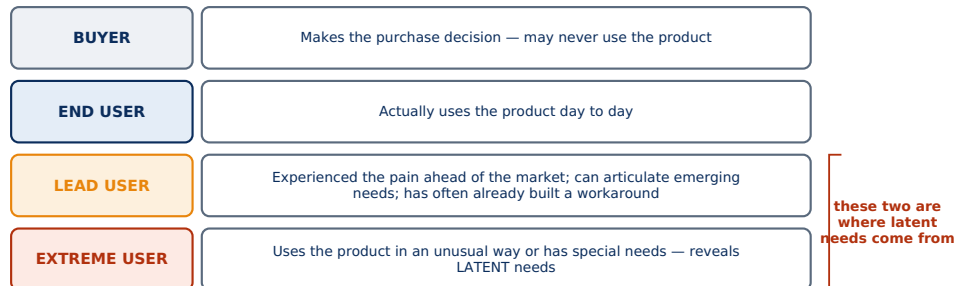
Strongly diminishing returns — a modest number of well-chosen interviews is sufficient

Fig 3.4 — *Sample size vs the proportion of needs identified*

Choosing the type of customer HIGH

Type	Characteristics
Buyer	Makes the purchase decision — may not be the user
End user	Actually uses the product
Lead user	Has experienced the pain ahead of the market; can articulate emerging needs ; has often already found a solution for themselves
Extreme user	Uses the product in an unusual way ; has special needs ; reveals latent needs

Which customers to talk to



Expressed needs = what customers SAY. Latent needs = what they are unaware of, but which delight when fulfilled.

Fig 3.5 — Which customers to talk to

Constructing a survey instrument

- Determine the survey purpose and the data-collection method
- Identify what specific information is needed
- Design questions **unbiased**; avoid jargon and sophisticated vocabulary; use simple sentences
- **Do not lead** the customer toward the answer you want
- In any list of options, include **"Other"** with space for a write-in answer
- Always include at least one **open-ended** question
- Length should be answerable in **15-30 minutes**
- Include instructions for completing and returning it

Interviewing techniques — questions to ask and to provoke

When and why do you use the product? • Walk us through the usage • Likes and dislikes • Issues • Improvement ideas. **Document the outcome as soon as possible.**

3.4 Step 2 — Interpret Raw Data in Terms of Customer Needs

Raw customer statements are converted into need statements written in a consistent form:

Raw customer statement	Need statement — RIGHT	Need statement — WRONG, and why
"The battery dies halfway through a job"	The drill operates continuously through a typical job	"Fit a lithium-ion battery" — that is a solution (HOW), not a need (WHAT)
"I hate it when the bit slips in the chuck"	The drill holds the bit securely under load	"The bit must not slip" — negative phrasing, and uses "must"
"It's awkward over my head for long"	The drill is comfortable to use overhead for extended periods	"The drill is light" — loses the detail that was in the raw data
"I can never tell how much charge is left"	The user can determine the remaining charge at a glance	"Add an LED charge indicator" — again a solution

Five guidelines for writing need statements MUST KNOW

1. Express the need in terms of **what the product has to do**, not **how** it might do it
2. Express the need **as specifically as the raw data** — do not lose detail
3. Use **positive**, not **negative** phrasing
4. Express the need as an **attribute of the product**
5. Avoid the words **"must"** and **"should"**

Also: **do not write the solution**, and **do not prioritise at this stage**.

3.5 Step 3 — Organise the Needs into a Hierarchy

Basis: customer survey analysis, designer team knowledge/experience/intuition, and competitor products. Procedure:

1. Print or write each need statement on a separate card or self-stick note
2. Eliminate redundant statements
3. Group the cards according to the similarity of the needs they express
4. For each group, choose a label

5. Consider creating supergroups consisting of two to five groups
5. Review and edit the organised need statements

Organising needs into a hierarchy (cordless drill example)

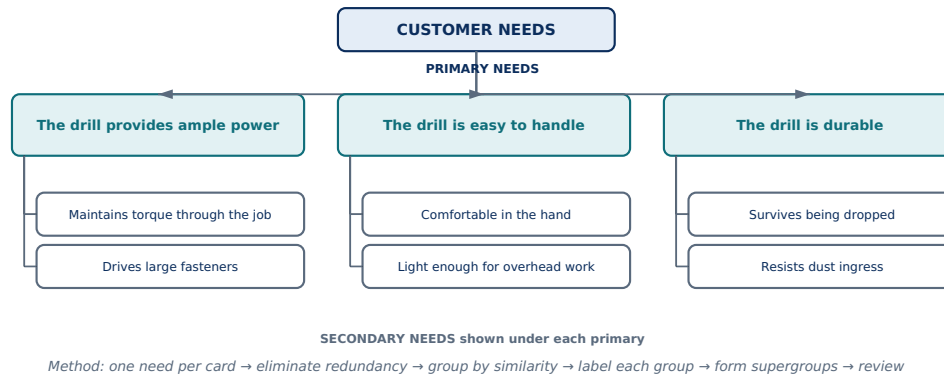


Fig 3.6 — Organising needs into a primary / secondary hierarchy

3.6 Step 4 — Establish the Relative Importance of the Needs

It is hard to satisfy all needs, so importance must be determined through trade-offs. Two methods: **(1) consensus of the development team** (quick) and **(2) further customer surveys** (more accurate, slower).

Challenges in knowing the customer's voice PROF

- The customer must understand the product being designed
- Difficult situation in rapid technology change
- The typical problem is that customers will always discuss the **failings** of the product; their experience of using it is never discussed easily
- Kano recognised **four levels** of customer requirements: **(1) Expecters (2) Spoken (3) Unspoken (4) Exciters**

Issue	Why It's a Problem
Surface-Level Insights	Surveys usually capture what customers say they want, not what they <i>actually</i> need or do in real life.
Response Bias	Customers may give socially desirable answers or what they think the company wants to hear.
Limited Context	Surveys rarely capture the environment, emotions, or cultural factors influencing customer behavior.
Low Engagement	Many respondents rush through surveys, leading to incomplete or inaccurate data.
Static Questions	Predefined questions restrict exploration of unexpected insights; they don't adapt to evolving responses.
Sampling Issues	Surveys often rely on small or non-representative samples, skewing results.
Delayed Feedback	By the time survey results are analyzed, market conditions or customer preferences may have shifted.
Quantitative Overemphasis	Surveys focus on numbers and ratings, missing qualitative richness like stories, motivations, and subtle frustrations.

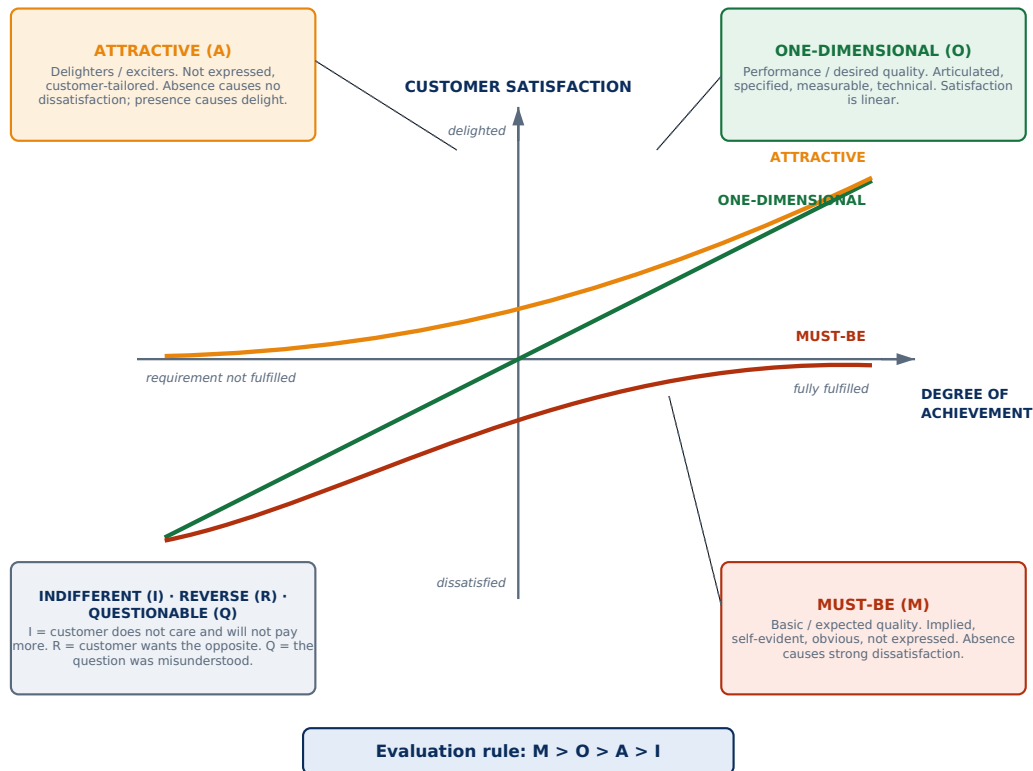
Fig 3.7 — **From the instructor's answer key.** Common issues with traditional customer surveys — eight issues, reproduce as a table

3.7 The Kano Model PROF MUST DRAW HIGH

Instructor's instruction

"If the question on Kano model comes, it is mandatory to draw this." — the satisfaction vs degree-of-achievement graph with all three curves labelled. Draw it before writing anything else.

The Kano model of customer satisfaction



Attributes migrate over time — today's delighter becomes tomorrow's performance attribute, and eventually a must-be

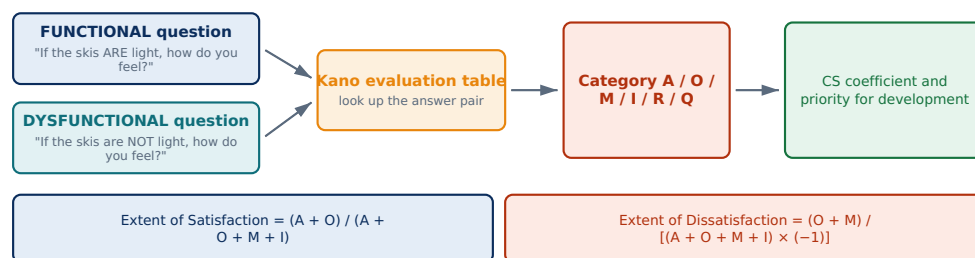
Fig 3.8 — The Kano model. THIS is the diagram to reproduce in the exam.

Category	Characteristics	Effect on satisfaction
Must-be (M) Basic / expected quality / Expecters	Implied, self-evident, not expressed, obvious	Presence causes no satisfaction; absence causes strong dissatisfaction . Curve stays below the axis and saturates.
One-dimensional (O) Performance / desired quality / Spoken	Articulated, specified, measurable, technical	Linear — satisfaction is proportional to degree of achievement. This is what customers openly ask for.
Attractive (A) Delighters / Exciters	Not expressed, customer-tailored, causes customer delight	Absence causes no dissatisfaction; presence causes disproportionate delight . Curve rises steeply above the axis.
Indifferent (I)	Customer does not care whether present or not	No effect; customer is not willing to pay more for it
Questionable (Q)	Question phrased incorrectly, misunderstood, or a wrong answer crossed by mistake	Indicates a survey defect, not a requirement
Reverse (R)	Feature is not wanted; the customer expects the reverse	Presence causes dissatisfaction

The Kano evaluation table

Each requirement is probed with a **functional** question ("If the skis are light, how do you feel?") and a **dysfunctional** question ("If the skis are **not** light, how do you feel?"). The answer pair is looked up in the evaluation table:

How a requirement is classified in the Kano method



Answer options for both questions: 1 Like, 2 Must be, 3 Neutral, 4 Live with, 5 Dislike

Fig 3.9 — How a requirement is classified in the Kano method

The Kano evaluation table — rows are the answer to the functional question, columns the answer to the dysfunctional question:

Functional answer ↓	Dysfunctional answer →				
	1. Like	2. Must be	3. Neutral	4. Live with	5. Dislike
1. Like	Questionable	Attractive	Attractive	Attractive	One-Dimensional
2. Must be	Reverse	Indifferent	Indifferent	Indifferent	Must-Be
3. Neutral	Reverse	Indifferent	Indifferent	Indifferent	Must-Be
4. Live with	Reverse	Indifferent	Indifferent	Indifferent	Must-Be
5. Dislike	Reverse	Reverse	Reverse	Reverse	One-Dimensional

Evaluation rule: $M > O > A > I$ – when categorising a requirement across a customer population, the category with the highest count wins, and where counts are close, this priority order applies.

Customer satisfaction (CS) coefficient

Extent of Satisfaction = $(A + O) / (A + O + M + I)$

Extent of Dissatisfaction = $(O + M) / [(A + O + M + I) \times (-1)]$

The satisfaction coefficient runs 0 to +1 (closer to 1 = stronger influence on satisfaction). The dissatisfaction coefficient runs 0 to –1 (closer to –1 = stronger influence on dissatisfaction if not fulfilled).

Advantages of Kano's method

- Prioritises customer requirements for product development
- Product requirements are better understood
- Provides valuable help in **trade-off situations** in the PD stage
- Discovering and fulfilling attractive requirements creates a wide range of possibilities for **differentiation**

Disadvantages of Kano's method

- Can only be used for **analysing effects**, not for **suggesting a new product feature**
- Based on competition over quality rather than on innovation
- Steve Jobs' objection quoted in class: *"It's really hard to design products by focus groups. A lot of times, people don't know what they want until you show it to them."*

Critical dynamic to state in any Kano answer: attributes migrate over time. Today's delighter becomes tomorrow's performance attribute and eventually a must-be. A rear-view camera in a car was attractive a decade ago; it is now expected.

3.8 Benchmarking

- Sun Tzu, 500 BC: *"Know your enemy before you know yourself."*
- Xerox's worry of 1979 about the Canon copier — the origin of modern competitive benchmarking.
- Benchmarking also means identifying **intra-company best practices**, not only external comparison.

Resistance to benchmarking — four common sources: fear of being perceived as copiers; fear of yielding competitive advantage if information is traded or shared; arrogance; impatience.

3.9 Reverse Engineering and Redesign HIGH

Reverse engineering and redesign — six steps

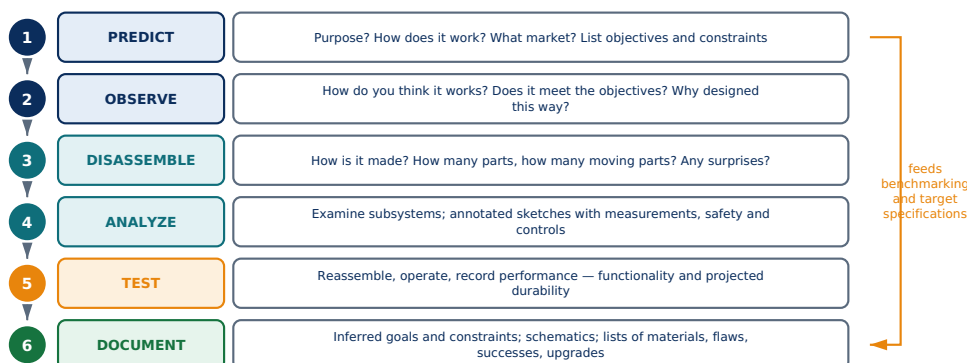


Fig 3.10 — Reverse engineering and redesign: six steps

Step	Questions and activities
1. Prediction	What is the purpose of this product? How does it work? What market was it designed to appeal to? List design objectives. List constraints that may have influenced the design.
2. Observation	How do you think it works? How does it meet design objectives overall? Why is it designed the way it is?
3. Disassemble	How does it work? How is it made? How many parts? How many moving parts? Any surprises?
4. Analyze	Carefully examine and analyse subsystems — structural, mechanical, electrical. Develop annotated sketches with measurements and notes on components, system design, safety and controls.
5. Test	Carefully reassemble the product. Operate the device and record observations on performance — functionality (operational and ergonomic) and projected durability.
6. Documentation	Inferred design goals; inferred constraints; design (functionality, form/geometry, materials); schematic diagrams; lists (materials, components, critical components, flaws, successes); refinements that might enhance usefulness; upgrades and changes.

3.10 Step 5 — Reflect on the Results and the Process

Seven aspects to review: (1) important customers (2) latent needs identification (3) pending inquiries or follow-ups (4) customers to involve during PD efforts (5) any surprising needs (6) involvement of relevant stakeholders (7) any improvement in the process. **Summarise the results by listing the most important needs and the latent needs.**

Key takeaways to memorise for Session 3

- Five-step need identification process
- Expressed vs latent needs, with the mobile-phone-camera example
- Ethnographic studies — definition, four methods, six contributions
- Buyer / end user / lead user / extreme user
- Five guidelines for writing need statements
- Kano: the graph, six categories, the evaluation table, $M > O > A > I$, CS coefficient formulas, advantages and disadvantages
- Benchmarking types, resistance sources, pros and cons
- Reverse engineering — six steps

3.11 Expected Questions — Session 3

Q1. What are the common issues with traditional customer surveys? List down the various quality attributes of the [15] Kano model. Discuss possible methods for prioritising customer requirements. PROF PYQ MUST DRAW

Part A — issues with traditional customer surveys (present as a two-column table, per the answer key):

Issue	Why it is a problem
Surface-level insights	Surveys capture what customers say they want, not what they actually need or do in real life
Response bias	Customers give socially desirable answers, or what they think the company wants to hear
Limited context	Surveys rarely capture the environment, emotions or cultural factors influencing behaviour
Low engagement	Respondents rush through, leading to incomplete or inaccurate data
Static questions	Predefined questions restrict exploration of unexpected insights; they do not adapt to responses
Sampling issues	Small or non-representative samples skew results
Delayed feedback	By the time results are analysed, market conditions or preferences may have shifted
Quantitative overemphasis	Focus on numbers and ratings misses qualitative richness — stories, motivations, subtle frustrations

Add the four "challenges in knowing the customer's voice" points from section 3.6 — especially that customers discuss failings readily but never describe their experience of usage.

Part B — Kano quality attributes. Draw the Kano graph first (Fig 3.8), then the six-category table from section 3.7. Include the four levels Kano originally recognised: Expecters, Spoken, Unspoken, Exciters. Give a worked product example — for a smartphone: must-be = call connects and battery lasts a day; one-dimensional = camera resolution, storage; attractive = on-device AI features. Mention attribute migration over time.

Part C — methods for prioritising customer requirements:

1. **Team consensus** — quick, uses development team judgement
2. **Further customer surveys** — direct importance ratings on a 1-5 or 1-9 scale
3. **Kano categorisation** — classify M/O/A/I and apply the $M > O > A > I$ rule; satisfy all must-bes first, be competitive on one-dimensionals, differentiate with a few attractives
4. **CS coefficient** — compute extent of satisfaction and dissatisfaction to rank quantitatively
5. **AHP** — pairwise comparison of needs to derive weights (least arbitrary method)
6. **House of Quality importance ratings** — carry customer importance into Room 1 and let the relationship matrix propagate it into engineering characteristics

Q2. In recent days, designers have been paying more attention to ethnographic studies. Why is this trend significant? In which ways can ethnographic studies enhance creativity in product development? [15] PROF PYQ

Part A — what ethnographic study is (bullets from section 3.3, plus the four methods: video, photographs, sketches, interview).

Part B — why the trend is significant:

- Surveys and focus groups capture only **expressed** needs; competitive advantage now comes from **latent** needs, which only observation reveals
- Customers cannot articulate what they have adapted to — workarounds and misuse are invisible in a questionnaire but obvious on video
- Rapid technology change means customers cannot evaluate what does not yet exist
- Products increasingly compete on experience rather than specification, and experience is contextual

Part C — how it enhances creativity (the six dimensions from the answer key, Fig 3.3):

Hidden needs discovery	Reveals unspoken behaviours, frustrations and desires that inspire breakthrough ideas
Contextual understanding	Shows how products fit into daily routines, social interactions and cultural practices
Empathy building	Helps designers and engineers see the world through the user's eyes, sparking human-centred solutions
Idea generation	Provides rich narratives and scenarios that fuel brainstorming beyond technical constraints
Innovation validation	Ensures creative concepts resonate with real users, not just theoretical models
Cross-disciplinary inspiration	Anthropological insights encourage collaboration between design, engineering and marketing teams

Examples (the answer key's own):

- **Consumer electronics:** ethnographic study revealed that people often use smartphones one-handed while multitasking — this led to gesture-based interfaces and thumb-friendly designs.
- **Automotive:** observing families during long trips inspired modular seating and integrated entertainment systems.

Q3. Explain the customer need identification process with a neat block diagram. Apply it to a product of your choice and show the resulting need hierarchy. [15] MUST DRAW

- Draw the five-step flow.
- Under step 1, cover the three gathering methods, sample size behaviour, and the four customer types.
- Under step 2, give the five translation guidelines with a right/wrong pair. Example for a **cordless drill**: raw statement "the battery dies halfway through a job" → need "the drill operates continuously through a typical job" (positive, what-not-how, no "must").
- Under step 3, build a small hierarchy. For the cordless drill:

Primary	Secondary
The drill provides ample power	Maintains torque through the job; drives large fasteners; works through hardwood
The drill is easy to handle	Comfortable in hand; light enough for overhead work; balanced; usable one-handed
The drill is durable	Survives being dropped; resists dust ingress; chuck does not wear
The drill is convenient to power	Battery charges quickly; charge state visible; batteries interchangeable across tools
The drill is safe	No accidental start; no kickback injury; insulated against shock

- Under step 4, describe team consensus vs further surveys; mention Kano.
- Under step 5, list the seven reflection aspects and note the output: most important needs plus latent needs.

Q4. What is benchmarking? Explain its types and the benchmarking process. What are the common sources of resistance to benchmarking within organisations, and how would you overcome them? [15]

- Definition plus the two historical anchors — Sun Tzu and the Xerox/Canon episode of 1979.
- Types: **internal** (between units of the same firm), **competitive/product** (direct competitors), **functional** (same function in a different industry), **generic** (best-in-class processes anywhere). Draw Fig 3.10.
- Process steps: plan → collect → analyse gap → set targets → implement → monitor and recalibrate.
- Resistance: fear of being perceived as copiers; fear of yielding competitive advantage if information is shared; arrogance; impatience.
- **Overcoming it:** frame benchmarking as learning rather than copying; use non-competing firms for functional benchmarking so no advantage is traded; secure senior sponsorship to counter arrogance ("not invented here"); set realistic timelines and stage the study so impatience does not kill it; start with **internal** benchmarking to build the habit at low risk.
- Close with a short pros-and-cons table for benchmarking.

Q5. Explain the reverse engineering and redesign process. How does it serve as a source of customer need and specification information? [15]

- Give the six-step table from section 3.9 — Prediction, Observation, Disassemble, Analyze, Test, Documentation — drawn as a flow.
- **How it informs needs and specifications:** disassembly reveals the **inferred design goals and constraints** the competitor worked to; part count and moving-part count indicate the architecture and assembly cost; materials and processes indicate the cost target; measured performance provides **competitive benchmarking data** for the needs-metrics matrix; observed flaws become opportunities and latent needs.
- Link forward: the output feeds directly into Session 4's benchmarking charts (both the metric-based chart and the perception-based

chart) and into target specification setting.

- Quote the Session 1 summary line: *reverse engineering and redesign is a forum for learning, experimenting and living the product design process — it helps you understand how products are executed within their "black boxes".*

Q6. Distinguish between lead users and extreme users. Why are both valuable in identifying latent needs? Illustrate with an example.

[15]

Aspect	Lead user	Extreme user
Position in market	Ahead of the mainstream — experiences the pain before others do	At the edge of the usage distribution — uses the product in an unusual way
Articulation	Can articulate emerging needs clearly	May not articulate; the need is inferred from observed behaviour
Solutions	Has often already built a workaround or solution	Adapts or misuses the product to cope
What they reveal	The direction the mainstream market will move in	Latent needs and the boundaries of the design space
How to use them	Co-development; their workaround often becomes the product feature	Ethnographic observation; design for the extreme often improves the mainstream product

Example — a kitchen mixer: the **lead user** is a small commercial baker who has already fitted a larger bowl and an external cooling fan because the motor overheats on long doses — that workaround points directly at a duty-cycle requirement the mass market will need next. The **extreme user** is an elderly customer with limited grip strength who wedges the bowl against the wall to lock it — revealing a latent need for one-handed bowl locking, which when solved makes the product easier for **everyone**. That last point — designing for the extreme improves the mainstream — is the key insight to state.

Session 4 — Developing Product Specifications and QFD

4.1 Customer Need vs Specification **MUST DRAW**

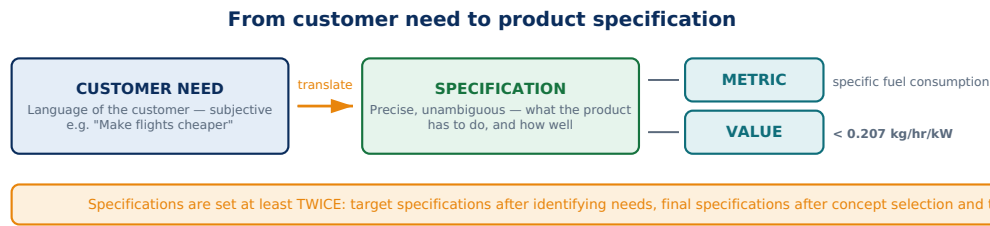


Fig 4.1 — From customer need to product specification

Customer Need	Specification
Expressed in the language of the customer	States what the product has to do, and how well
Subjective	Unambiguous — consists of a metric and a value

A **product specification** is a precise description of what the product has to do. Also called product requirement, technical specification, or **engineering characteristic**.

The canonical example

Customer Need	Specification
"Make flights cheaper"	Specific fuel consumption is less than 0.207 kg/hr/kW ← a metric and a value

Why specifications are needed: translate needs to precise targets; give metrics for success or failure; build confidence; resolve inevitable trade-offs.

Specifications are set at least twice. Target specifications immediately after identifying customer needs, and final specifications after concept selection and testing.

4.2 Establishing Target Specifications — Four Steps

Target specifications are "preliminary" specifications set as **goals** by the PD team. They cannot be arbitrary, as they may not be technically feasible. Example: 1400 kg mass + manufacturing cost of Rs 10 lakh + power of 100 kW.

1. Prepare the list of metrics
2. Collect competitive benchmarking information
3. Set ideal and marginally acceptable target values
4. Reflect on the result and the process

Step 1 — Prepare the list of metrics

Metrics **reflect** customer needs and turn them into precise and measurable specifications. Those establishing specifications should have been directly involved in identifying customer needs.

Guidelines for constructing metrics

- **Complete** — every need reflected
- **Dependent**, not independent — metrics are dependent variables; design variables are independent
- **Practical** — easily measurable
- Should **correlate perfectly** with the need
- **What-not-how** — leave freedom to designers (e.g. specify "fuel consumption", not "engine displacement")

Metrics **may be subjective** (e.g. "feels premium", "instils pride") and should include criteria from independently published evaluations, e.g. the **ICOTY** (Indian Car of the Year) award criteria.

Units of a metric can be: engineering units • a list • subjective • binary (pass/fail). Importance ratings of metrics are usually derived from the importance ratings of the needs.

Step 2 — Collect competitive benchmarking information

Establishes the relationship of the new product with competitive products. Two chart types:

Step 3 — Set ideal and marginally acceptable target values

Ideal value	The best result the team could hope for
Marginal value	The value that would just barely make the product commercially viable

Values can be expressed as: **exact**, **discrete**, **at least**, **at most**, or **between**.

Step 4 — Reflect

(1) Is the team influenced by marketing and sales — i.e. are specifications driven by assumptions rather than objective user research? (2) Single vs multiple products to match needs of various markets (3) Missing specifications (4) Do the specifications dictate commercial

success?

4.3 Setting Final Specifications — Five Steps

Once the concept is finalised, specifications that were only targets expressed as broad ranges are refined and made more precise. The most challenging part is handling **trade-offs**, which arise from inverse relationships between two specifications:

- **Cost vs Mass** — lighter is costlier
- **Power vs Mileage** — higher power consumes more fuel

Step	Activity	Purpose
1	Develop technical models of the product	Predict the value of the metrics for a particular set of design decisions
2	Develop a cost model of the product	Make sure the target cost can be met
3	Refine the specifications, making trade-offs where necessary	Converge on precise values
4	Flow down the specifications	Cascade system specs to subsystems
5	Reflect on the result and the process	Reality check

Step 1 — Technical model

An analytical or physical model that predicts the values of the metrics for a particular set of design decisions. Ideally implemented as spreadsheet equations or computer simulation, allowing rapid prediction of performance from a choice of design variables **without costly physical experimentation**.

Design Variables (Concept) → [Analytical or Physical Model] → Metrics

Step 2 — Cost model

The goal is to ensure the product can be produced at the target cost. First estimates of manufacturing cost are made by drafting a **bill of materials** (a list of all the parts) and estimating a purchase price or fabrication cost for each part.

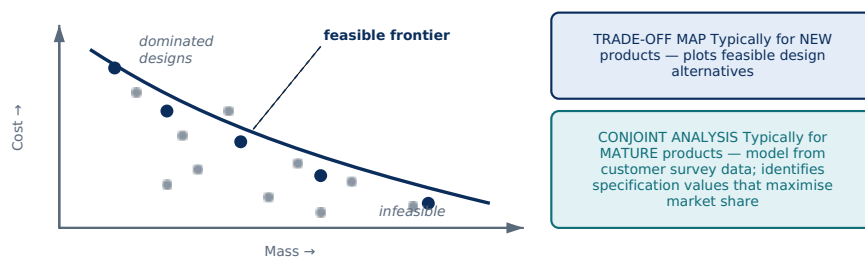
Manufacturing Cost (Company) + Adequate Profit (Company, partners, distributors) → Competitive Price (Customer)

Step 3 — Refine the specifications

Technical model + cost model + performance models feed a group discussion which converges, balancing competitive position, customer need and profit, to give **refined specifications**.

Tool	What it does	When used
Trade-off map	Plots one metric against another for all feasible design alternatives, showing the frontier	Typically for new products
Conjoint analysis	A model constructed from customer survey data in which hypothetical products are evaluated; identifies specification values that maximise market share	Typically for mature products

Refining specifications: the trade-off map



Common trade-offs: cost vs mass, power vs mileage — an inverse relationship between two specifications

Fig 4.2 — Refining specifications: the trade-off map and conjoint analysis

Setting target specifications, then final specifications

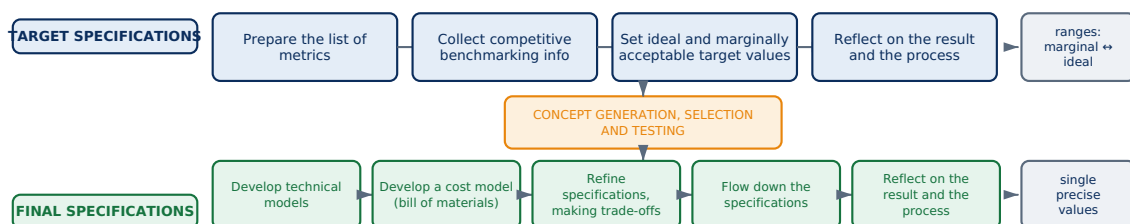


Fig 4.3 — The two-stage process: target specifications, then final specifications

Step 4 — Flow down the specifications

System-level specifications are cascaded into subsystem metrics. The subsystem specifications must **reflect the overall specification** and be **equally challenging**.

Car specification (system)	Engine metrics (subsystem)
Fuel economy	Power
Acceleration	Torque
Turning radius	Fuel consumption

Flowing specifications down to subsystems

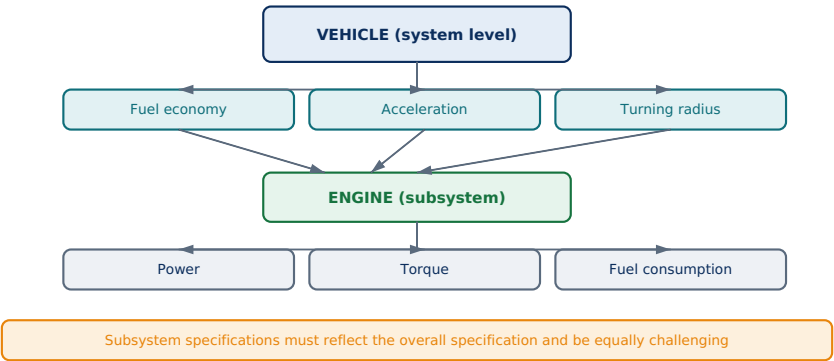


Fig 4.4 — Flowing specifications down to subsystems

Step 5 — Reflect

(1) Is the product a winner — will it meet customer needs and excel competitively? (2) Uncertainty in the models (3) Is the concept suited to the target market — is it a dramatically superior product? (4) Scope of improvement in the future.

4.4 Quality Function Deployment and the House of Quality PYQ HIGH

Quality Function Deployment (QFD) is a popular technique used to generate engineering specifications. It was developed in **Japan in the mid-1970s**. Goals:

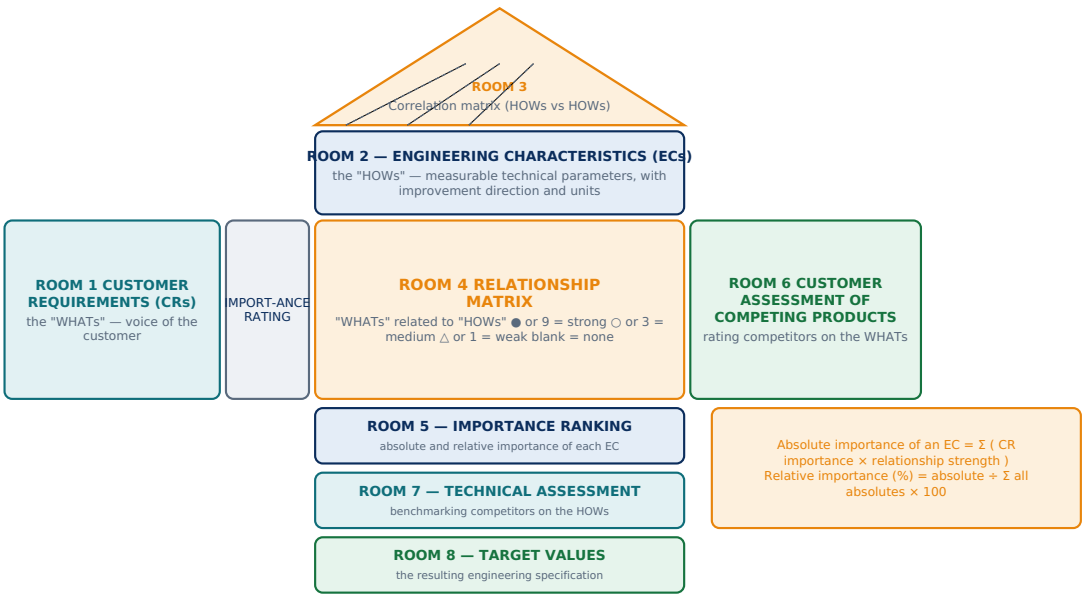
- Increase customer satisfaction
- Reduce the cycle time of product development
- Increase competitiveness

Toyota use case: Toyota reduced the cost of bringing a new car model to market by **over 60%** and decreased development time by **one-third**.

QFD is a **planning and team problem-solving tool**, adopted by a wide variety of companies as the tool of choice for focusing a design team's attention on satisfying customer needs throughout the product development process.

The House of Quality — eight rooms MUST DRAW

The House of Quality — the eight rooms



comes from the Session 3 need hierarchy; Room 2 from the Session 4 metrics list; Rooms 6 and 7 are the two benchmarking charts; Room 8 becomes the target spec.

Fig 4.5 — The House of Quality: the eight rooms. THIS is the diagram to reproduce.

Room	Name	Content
Room 1	Customer Requirements (CRs) — the WHATs	Voice of the customer, with an importance rating column
Room 2	Engineering Characteristics (ECs) — the HOWs	Measurable technical parameters, with improvement direction (↑/↓) and units
Room 3	Correlation matrix (the roof)	HOWs vs HOWs — positive and negative correlations between engineering characteristics; this is where trade-offs surface
Room 4	Relationship matrix	WHATs related to HOWs. Strength codes: ● or 9 = Strong , ○ or 3 = Medium , △ or 1 = Weak , blank = none
Room 5	Importance ranking	Absolute and relative importance of each engineering characteristic
Room 6	Customer assessment of competing products	Rating competitors on the WHATs — the perception-based benchmarking chart
Room 7	Technical assessment	Benchmarking competitors on the HOWs — the metric-based benchmarking chart
Room 8	Target values	The measures of the HOWs — the resulting engineering specification

Absolute importance of $EC_j = \sum_i (\text{Importance of } CR_i \times \text{Relationship strength}_{ij})$

Relative importance (%) = Absolute importance of $EC_j \div \sum \text{all absolute importances} \times 100$

Advantages of QFD

1. Reduces the time required for the planning process
2. Focuses completely on customer needs and requirements
3. Reduces design changes and alterations
4. Improved quality of the products
5. High customer satisfaction
6. Decreased design and manufacturing costs and overheads
7. Reduces time to market
8. Helps in **prioritising** various design parameters
9. Aids in **benchmarking** the processes

Disadvantages of QFD

1. Ambiguity in the voice of the customer — qualitative inputs are hard to convert reliably
2. Customer needs change over time; the matrix is a snapshot
3. Matrices become very large and unwieldy for complex products
4. Time consuming and resource intensive to build and maintain
5. Requires genuine cross-functional participation — fails if done by one function
6. Subjectivity in assigning relationship strengths
7. Not well suited to radically new products with no comparable customer experience

Key takeaways to memorise for Session 4

- Need vs specification — metric + value; the fuel-consumption example
- Four steps to target specifications; five steps to final specifications
- Needs-metrics matrix; five guidelines for metrics; four unit types
- Two benchmarking chart types (metrics vs customer perception)
- Ideal vs marginal value; five value expressions
- Trade-off map (new products) vs conjoint analysis (mature products)
- Flow-down of specifications with the car/engine example
- **HOQ: eight rooms, 9/3/1 relationship codes, the absolute-importance formula**
- QFD advantages (9) and disadvantages; Toyota 60% / one-third figures

4.5 Expected Questions — Session 4

Q1. Prepare a detailed House of Quality (HOQ) to prioritise the engineering characteristics for the development of [15] an automotive front bumper for a passenger vehicle. The customer requirements have already been identified and listed in Table 1 along with their importance ratings. Also discuss any two possible outcomes from this HOQ study.

PYQ **MUST DRAW**

Given (from the past paper): CR1 High impact resistance (5), CR2 Lightweight (4), CR3 Low manufacturing cost (4), CR4 Corrosion resistance (3), CR5 Aesthetic surface finish (3), CR6 Easy assembly (4).

Step 1 — Define engineering characteristics (the HOWs) with units and improvement direction:

EC	Engineering characteristic	Unit	Direction
EC1	Impact energy absorption	kJ	Maximise ↑
EC2	Bumper assembly mass	kg	Minimise ↓
EC3	Section wall thickness / crush-can geometry	mm	Target (nominal)
EC4	Material cost per part	Rs/part	Minimise ↓
EC5	Moulding / forming cycle time	s	Minimise ↓
EC6	Salt-spray corrosion life	hours	Maximise ↑
EC7	Surface gloss / roughness Ra	GU / μm	Target
EC8	Number of parts and fasteners	count	Minimise ↓

Step 2 — Build the relationship matrix (Room 4) using 9 = strong, 3 = medium, 1 = weak, blank = none:

Customer Requirement	Imp.	EC1	EC2	EC3	EC4	EC5	EC6	EC7	EC8
CR1 High impact resistance	5	9	3	9	3	-	-	-	1
CR2 Lightweight	4	3	9	9	3	1	-	-	3
CR3 Low manufacturing cost	4	1	3	3	9	9	1	3	3
CR4 Corrosion resistance	3	-	1	1	3	-	9	1	-
CR5 Aesthetic surface finish	3	-	-	3	1	3	3	9	1
CR6 Easy assembly	4	-	3	1	-	1	-	-	9

Step 3 — Compute the importance ranking (Room 5). Absolute importance = Σ (CR importance $\times$ relationship):

EC	Calculation	Absolute	Relative %	Rank
EC1	$(5 \times 9) + (4 \times 3) + (4 \times 1) = 45 + 12 + 4$	61	11.6	5
EC2	$(5 \times 3) + (4 \times 9) + (4 \times 3) + (3 \times 1) + (4 \times 3) = 15 + 36 + 12 + 3 + 12$	78	14.8	2
EC3	$(5 \times 9) + (4 \times 9) + (4 \times 3) + (3 \times 1) + (3 \times 3) + (4 \times 1) = 45 + 36 + 12 + 3 + 9 + 4$	109	20.7	1
EC4	$(5 \times 3) + (4 \times 3) + (4 \times 9) + (3 \times 3) + (3 \times 1) = 15 + 12 + 36 + 9 + 3$	75	14.3	3
EC5	$(4 \times 1) + (4 \times 9) + (3 \times 3) + (4 \times 1) = 4 + 36 + 9 + 4$	53	10.1	6
EC6	$(4 \times 1) + (3 \times 9) + (3 \times 3) = 4 + 27 + 9$	40	7.6	8
EC7	$(4 \times 3) + (3 \times 1) + (3 \times 9) = 12 + 3 + 27$	42	8.0	7
EC8	$(5 \times 1) + (4 \times 3) + (4 \times 3) + (3 \times 1) + (4 \times 9) = 5 + 12 + 12 + 3 + 36$	68	12.9	4
Total		526	100.0	

Step 4 — Roof: correlation matrix (Room 3). Mark these on the triangular roof:

EC pair	Correlation	Reason
EC1 Energy absorption $\leftrightarrow$ EC2 Mass	Strong negative	More absorption normally needs more material $\rightarrow$ heavier
EC3 Wall thickness $\leftrightarrow$ EC2 Mass	Strong negative	Thicker section directly adds mass
EC1 Energy absorption $\leftrightarrow$ EC4 Material cost	Negative	High-absorption materials (composites, high-strength steel) cost more
EC5 Cycle time $\leftrightarrow$ EC4 Material cost	Positive	Both reduce total part cost together
EC6 Corrosion life $\leftrightarrow$ EC7 Surface finish	Positive	A good coating system delivers both
EC8 Part count $\leftrightarrow$ EC5 Cycle time	Positive	Fewer parts means less assembly time

Step 5 — Target values (Room 8). Example set: EC1 ≥ 12 kJ at 15 km/h; EC2 ≤ 6 kg; EC3 2.8–3.2 mm; EC4 $\leq$ Rs 1,450/part; EC5 ≤ 70 s; EC6 ≥ 720 h salt spray; EC7 ≥ 85 GU gloss; EC8 ≤ 8 parts.

Two outcomes from this HOQ study (the question asks for two)

Outcome 1 — Design effort should concentrate on section geometry and wall thickness (EC3, 20.7%). It is the single highest-ranked characteristic because it simultaneously drives three high-importance requirements: impact resistance (9), lightweight (9) and cost (3). The practical consequence is that CAE-based crush-can and rib optimisation delivers more value per engineering hour than any other activity. Mass (EC2, 14.8%) and material cost (EC4, 14.3%) follow, confirming that material selection is the second lever.

Outcome 2 — The roof exposes a hard trade-off between impact energy absorption and mass that cannot be resolved by compromise. Improving EC1 worsens EC2, and both are highly weighted customer requirements. This tells the team that a conventional single-material steel bumper will not satisfy both, and that the solution must come from either a **material/architecture change** (thermoplastic energy absorber with a high-strength steel or composite beam) or from a **TRIZ-style contradiction resolution** rather than from splitting the difference. This is exactly the kind of finding the roof exists to surface, and it should trigger the concept generation activity of Session 5.

A third, if space permits: corrosion resistance (EC6, 7.6%) ranks lowest, so heavy investment in exotic corrosion protection is not justified by the customer importance data — a standard coating system meeting the 720 h target is sufficient, and resources should be redirected to EC3 and EC2.

Note on marks: the instructor said ratings are subjective and "whatever rating you feel and you mention is acceptable" — but you must **state your assumption** and the arithmetic must be internally consistent. Show the calculation column; that is where the method marks are.

Q2. Differentiate between target specifications and final specifications. Explain the five-step process for setting final specifications with a suitable example. [15]

Aspect	Target specifications	Final specifications
When set	Immediately after identifying customer needs	After concept selection and testing
Nature	Preliminary; hopes and aspirations of the team	Refined; reflect actual technological constraints and expected production cost
Form of values	Ranges — marginal value and ideal value	Single precise values
Basis	Customer needs + competitive benchmarking	Technical models + cost models + trade-off analysis
Certainty	May not be technically feasible	Demonstrated feasible and affordable

Then give the five steps (table in section 4.3), each with the car example: technical model predicts fuel economy from engine displacement and mass; cost model built from the BOM; refinement handles the cost-vs-mass trade-off using a trade-off map; flow-down cascades fuel economy, acceleration and turning radius into engine power, torque and fuel consumption; reflection asks whether the product is a winner.

Q3. What is QFD? Explain the House of Quality with a neat labelled diagram. State its advantages and limitations. [15]

MUST DRAW

- Definition, origin (Japan, mid-1970s), three goals, Toyota figures (60% cost reduction, one-third time reduction).
- **Draw the eight-room House of Quality** and label every room — this is the bulk of the marks.
- Explain each room in one line (table in section 4.4).
- State the 9/3/1 relationship convention and the absolute-importance formula.
- Give the nine advantages and at least five limitations.
- Close by connecting QFD to the wider process: Room 1 comes from Session 3's need hierarchy, Room 2 from Session 4's metrics list, Rooms 6 and 7 are the two benchmarking charts, and Room 8 becomes the target specification table.

Q4. What is a needs-metrics matrix? Construct one for a product of your choice and explain the guidelines used to establish good metrics. [15]

Definition: a matrix with customer needs as rows and metrics as columns, marked where a metric reflects a need. Its purpose is to guarantee **completeness** — every need must be captured by at least one metric — and to expose needs that are hard to measure.

Example for a **portable electric kettle**:

Customer need	Boil time (s)	Mass (g)	Energy per litre (Wh)	Outer surface temp (°C)	Noise (dB)	Cleaning time (s)
Boils water quickly	●		○			
Is easy to carry		●				
Wastes little energy	○		●	○		
Is safe to touch when hot			○	●		
Operates quietly					●	
Is easy to clean		△				●

Then give the metric guidelines from section 4.2 — complete, dependent not independent, practical, correlate perfectly, what-not-how — explaining each with a one-line example. Note that metrics may be subjective, and list the four unit types (engineering units, list, subjective, binary pass/fail).

Session 5 — Establishing Product Function and Concept Generation

5.1 What is a Concept?

A **concept** is an approximate description of the **technology**, **working principles** and **form** of a product. It is often expressed as a **sketch** or a **3D model**. A concept is not a product or system attribute — it is a **mapping** from function to form.

Concept generation is **inexpensive and quick**: typically about **5% of budget** and **15% of PD time** — yet it determines most of the product's eventual cost and quality.

5.2 Creative Thinking and Mental Blocks

Aids to creative thinking: develop a creative attitude • unlock your imagination • be persistent • develop an open mind • suspend your judgment • set problem boundaries.

Block	Nature and examples
Perceptual	Stereotyping — thinking conventionally, or in a formulaic way about an event, person or way of doing something. Information overload — so much unreliable data that you are unable to sort out the critical aspects of the problem. Limiting the problem unnecessarily — broad observation of the problem may keep the mind open to a wider range of ideas. Fixation — thinking that we have influenced as greatly by their previous experience or other data that they cannot recognise alternative ideas.
Emotional	Fear of risk taking — this is the hardest of proposing an idea that will ultimately be found to be faulty. Unease with chaos — people in general, and many engineers are uncomfortable with highly unstructured situations. Inability or unwillingness to incubate new ideas — in our busy lives we often don't take the time to let ideas be dormant so they can incubate properly. Motivation — lack of motivation to seek creative solutions to challenging problems.
Cultural	Thought patterns — people acquire a set of thought patterns from living in a culture. Tradition bound — culture provides tradition and are reluctant to change, even though profitability is decreasing. Attitudes — countries even differ in their attitudes toward creative problem solving (political, educational, business cultures).
Intellectual	Poor choice of problem-solving language, or problem representation; memory blocks — hindrance due to variations in remembering based on prior information; insufficient knowledge base; incorrect information — it is obvious that using incorrect information can lead to poor design.
Environmental	Physical environment — ambience, weather, space. Criticism — non-supportive remarks about your ideas can be personally hurtful and harmful to your creativity; it is important for the team to maintain an atmosphere of support and trust.

5.3 The Five-Step Concept Generation Method **MUST DRAW**

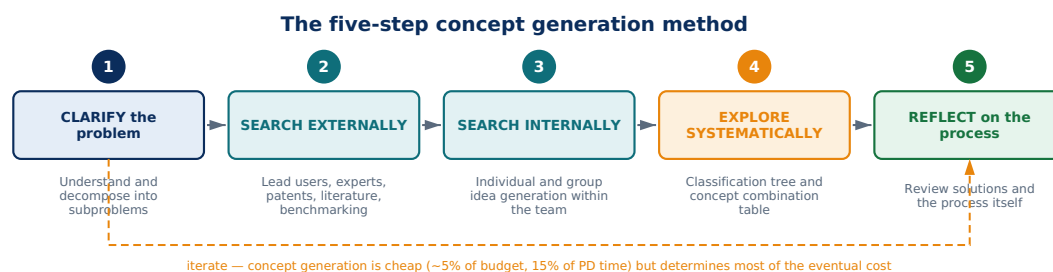


Fig 5.1 — The five-step concept generation method

1. **Clarify the problem** — understand and decompose into subproblems
2. **Search externally** — lead users, experts, patents, literature, benchmarking
3. **Search internally** — individual and group idea generation within the team
4. **Explore systematically** — classification tree and concept combination table
5. **Reflect** on the solutions and the process

5.4 Classification of Concept Generation Methods **PYQ** **HIGH**

Classification of concept generation methods

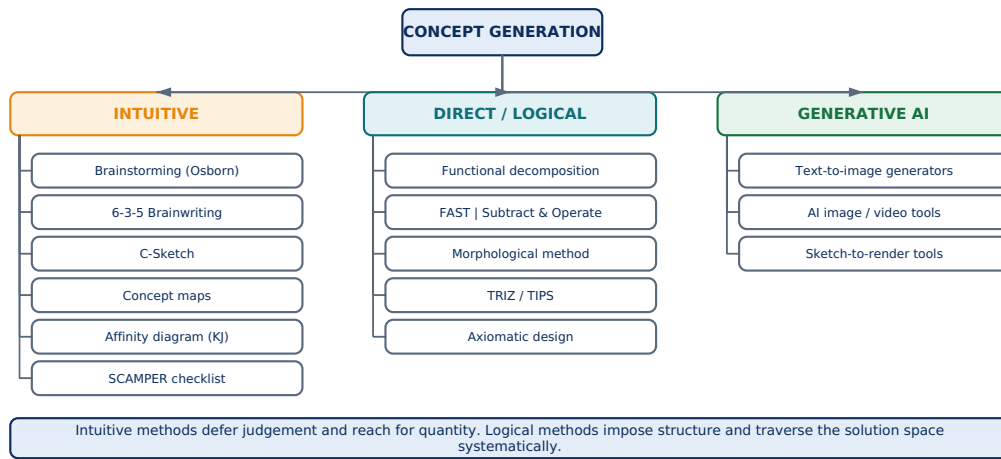


Fig 5.2 — Classification of concept generation methods

The classification to reproduce

Class	Methods
Intuitive methods	Brainstorming • 6-3-5 Method / Brainwriting • C-Sketch (Collaborative Sketching) • Concept maps / mind mapping • Affinity diagram (KJ Method) • SCAMPER checklist
Direct or logical methods	Functional decomposition (FAST, Subtract and Operate) • Morphological method • Theory of Inventive Problem Solving (TRIZ / TIPS) • Axiomatic design
Generative AI tools	Text-to-image generators, AI image/video generators, sketch-to-render tools (Vizcom and similar)

5.5 Intuitive Methods

Brainstorming — Osborn's method HIGH

Two principles contribute to **ideative efficacy**: **defer judgment** and **reach for quantity**. Four rules:

1. **Go for quantity**
2. **Withhold criticism**
3. **Welcome wild ideas**
4. **Combine and improve ideas**

Conduct: typically **5-15 people**, sessions of **30-45 minutes**. Avoid hierarchically structured groups. All ideas are recorded and displayed; practicality is ignored. The wilder the idea the better — it is easier to tone down a wild idea than to improve a mediocre one. Killer phrases are forbidden.

Improving brainstorming sessions

- Stick to the rules
- Pay attention to everyone's ideas
- Include both individual and group approaches
- Take breaks
- Do not rush
- Stay persistent
- Facilitate the session

Drawbacks of traditional brainstorming

- The right idea may not come at the right time
- Group conventions may sidetrack original ideas
- Distractions by misdirected focus
- Certain team members may **dominate the discussion**
- The team may not be open to new ideas
- **Production blocking** — only one person can speak at a time
- **Evaluation apprehension** — introverts self-censor

SCAMPER checklist

Used as a stimulant for new ideas during brainstorming:

Substitute	Replace a material, process or component — petrol to E20 / alternative fuels; mechanical actuation replaced by electronic
Combine	Merge functions or products — the laptop (screen + keyboard + CPU + battery); combination alloys; two-flavour softy
Adapt	Borrow a solution from elsewhere — biomimicry
Modify / Magnify / Minify	Change meaning, colour, motion, odour, form, shape — same container in different colours; kids' vs adults' toothbrushes
Put to other uses	New markets or applications for the existing product
Eliminate	Remove excess weight or parts; simplify
Rearrange / Reverse	Transpose cause and effect; interchange or rearrange components; change the sequence or schedule

6-3-5 Method (Brainwriting) HIGH

The 6-3-5 brainwriting method

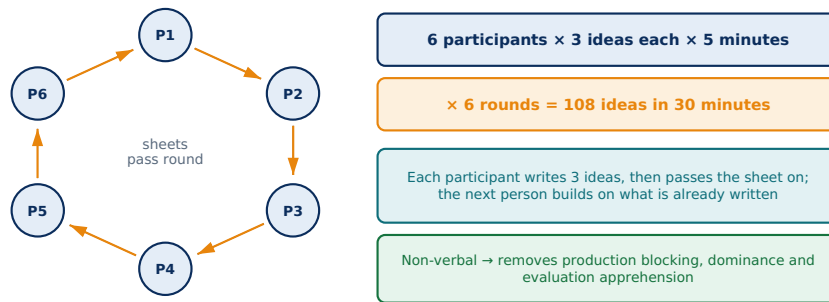


Fig 5.3 — The 6-3-5 brainwriting method

6 participants × 3 ideas each × 5 minutes per round × 6 rounds = **108 ideas** in 30 minutes

Each participant writes three ideas on a sheet in five minutes, then passes the sheet to the next participant, who builds on or adds to what is written. Repeated until each sheet has visited everyone.

Advantages

- Procedure and rules are easy to communicate
- Non-verbal — promotes creative and open exchange
- **Introverted employees are also encouraged** to participate
- No minute-taker needed; no idea lost due to lack of documentation
- Very straightforward, easy and quick to learn

Disadvantages

- Lack of spontaneity
- The need for concrete formulation
- Possibility of **multiple entries** due to simultaneous, anonymous writing — risk of clash of similar ideas
- Issues in clarity of ideas
- Stress from time constraints may reduce idea quality
- Requires all participants to share deep background knowledge of the topic

C-Sketch (Collaborative Sketching)

Team members draw sketches of possible design solutions. Each member starts by drawing a sketch; every couple of minutes the sketch is passed to a different team member, who adds to it. This continues until each drawing has visited every team member, creating a progressive sketch. The ideas fostered drive the final design.

Concept maps

An effective way to record the results of a brainstorming session as it happens (memory mind mapping). The facilitator writes the problem statement in the middle of a clean sheet **as two words** and draws a box around it. Ideas are then recorded quickly with circles around them, each new idea connected to the original problem statement.

Pros

- More detailed information
- Illustrates multiple problems
- Activates critical thinking
- Good interface; optimal for analysing work
- Explicit relationships among factors
- Solves various problems at the same time

Cons

- Unable to memorise information
- Takes much effort
- Complicated layout
- Difficult to change one block

Affinity diagram (KJ Method)

A business tool used to organise ideas and data. The term was devised by **Jiro Kawakita in the 1960s**, so it is also called the **KJ Method**. Used widely in project management, it allows large numbers of ideas from brainstorming to be sorted into groups based on their natural relationships. Procedure: record each idea on a card or note → look for ideas that seem related → sort cards into groups until all cards have been used → label each group.

5.6 Systematic Exploration — Concept Combination Table

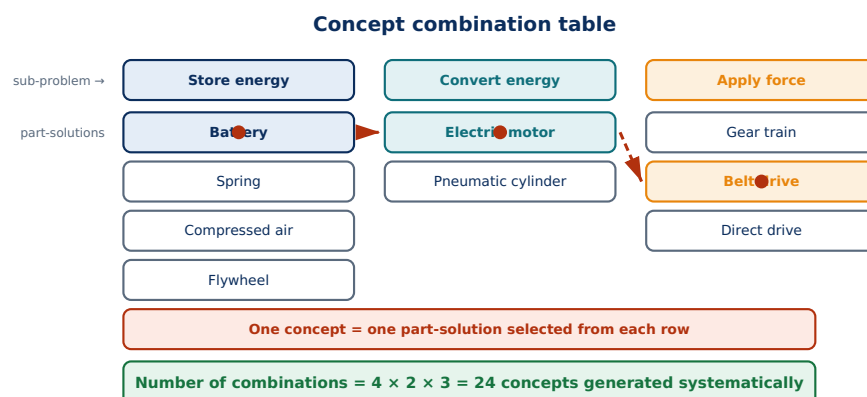


Fig 5.4 — Concept combination table

The overall problem is decomposed into subproblems (columns). Solution fragments for each subproblem are listed beneath. A concept is formed by selecting one fragment from each column. The number of combinations is the product of the fragment counts.

5.7 Functional Decomposition PROF MUST DRAW HIGH

Why functional decomposition

- Concentrates on **"What"** has to be achieved by a new concept, not **"How"** it is to be achieved
- Basis for organising the design team, tasks and process
- **Creativity enhancement**
- More acceptable solutions can be generated by mapping **Needs → Function → Form**
- Removes excess reliance on the designer's previous experience
- Avoids jumping to conclusions (form) too quickly

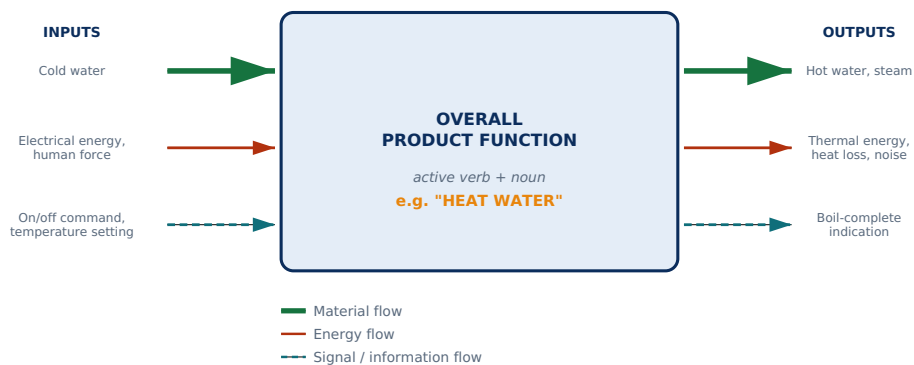
Functional modelling basics — definitions to quote

Function	A statement of a clear, reproducible relationship between the available input and the desired output of a product, independent of any particular form . Example: input electricity and hand force → output ironing of clothes.
Product function	The intended overall function of the product, expressed as just an active verb + noun — "wash clothes", "cut beans", "cools air", "heats water".
Sub-function	A component of the product function corresponding to subtasks.
Abstraction	The process of ignoring what is particular or incidental and emphasising what is essential and general.
Constraint	A statement of a clear criterion that must be satisfied by a product, requiring consideration of the entire product to determine the criterion value. Cost is a constraint.

Key distinction: functions are satisfied through the operation of the product; constraints are satisfied by the properties of the entire product (cost, compactness, mass, reliability).

The black box model MUST DRAW

Black-box function structure — the M-E-S convention



If a question says "black box function structure", all three flows **MUST** be drawn, with distinct line styles, crossing the boundary

Fig 5.5 — Black-box function structure: the mandatory M-E-S convention

Instructor's instruction on black-box questions

"If it is specific that black box function structure, that means MES — material, energy and signal. Input to output, that you have to list down." The three flows must be drawn with distinct line styles:

→ thin solid arrow = **Energy** • → thick solid arrow = **Material** • - - → dashed arrow = **Signal / Information**

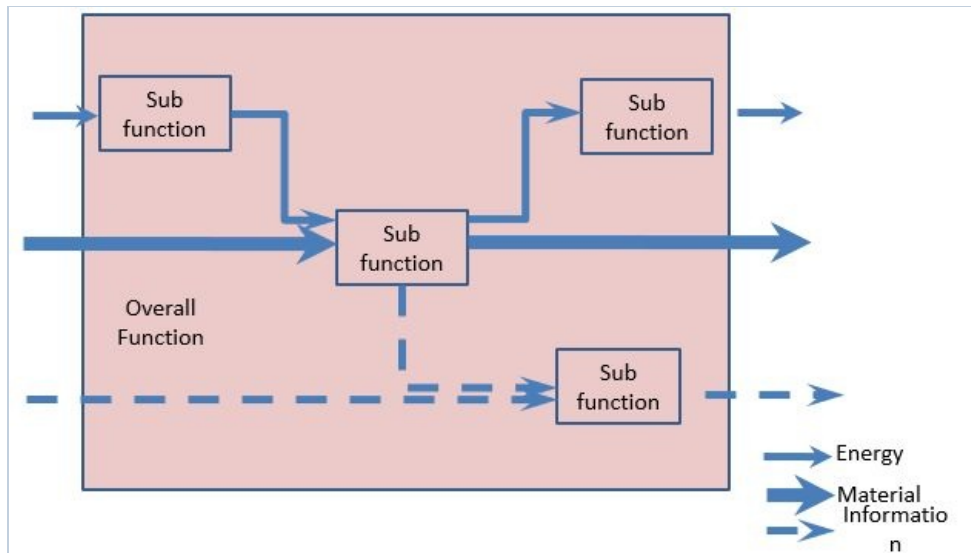


Fig 5.6 — **From the instructor's answer key.** The overall function decomposed into sub-functions, with energy, material and information flows distinguished by line style. Reproduce this convention.

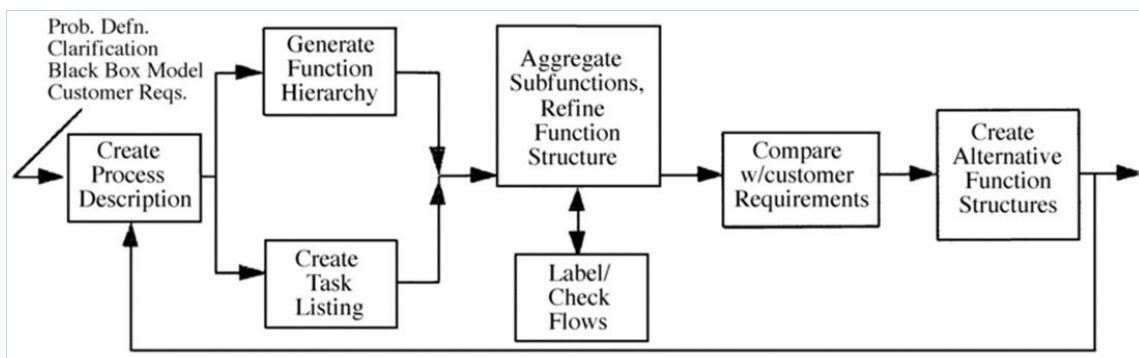


Fig 5.7 — **From the instructor's answer key.** The functional decomposition process: problem definition/clarification/black-box model/customer requirements → create process description → generate function hierarchy and create task listing → aggregate sub-functions and refine function structure (with label/check flows) → compare with customer requirements → create alternative function structures.

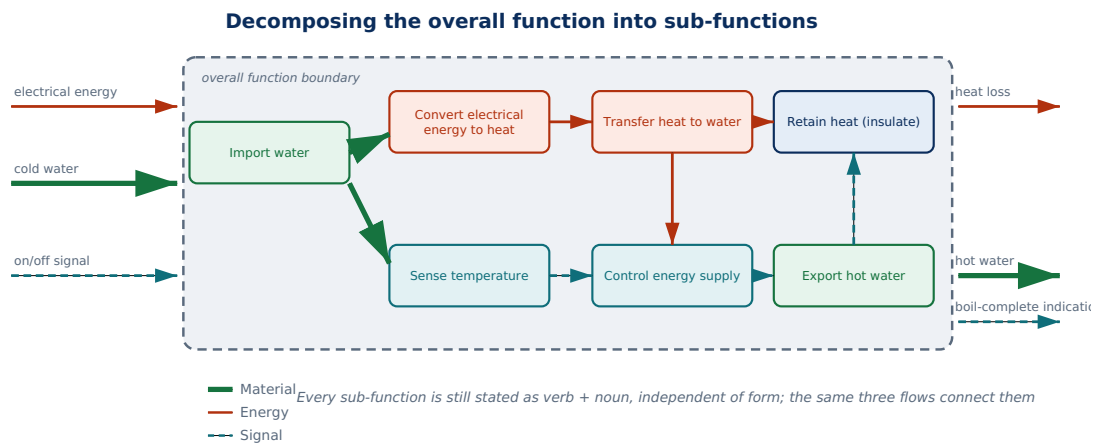


Fig 5.8 — Decomposing the overall function into sub-functions

5.8 Function Trees — FAST and Subtract-and-Operate

Function trees decompose the prime function hierarchically into sub-functions. They are fast and simple to construct. Two approaches:

FAST method	Top-down approach
Subtract and Operate (SOP)	Bottom-up approach

FAST — Function Analysis System Technique

- Displays functions in a **logical sequence**, prioritises them and tests their dependency
- Uses **HOW-WHY analysis**
- **Higher order function** — e.g. "Iron the clothes"
- **Basic function** — e.g. "Heats the base of the iron box"
- **Secondary functions**, of three types:
 - **Required** — "heater assembly which converts electricity to heat"

- **Aesthetic** — "styling of outer grip or handle"
- **Unwanted** — "heating elements lose heat to handles; handles need to be insulated"
- **Critical path** = Higher order function → Basic function → Required secondary function

The FAST reading rule — state this in any FAST answer

Each node to the **right** on the critical path must answer **HOW** the node to its left is achieved. Each node to the **left** must answer **WHY** the node to its right is being achieved.

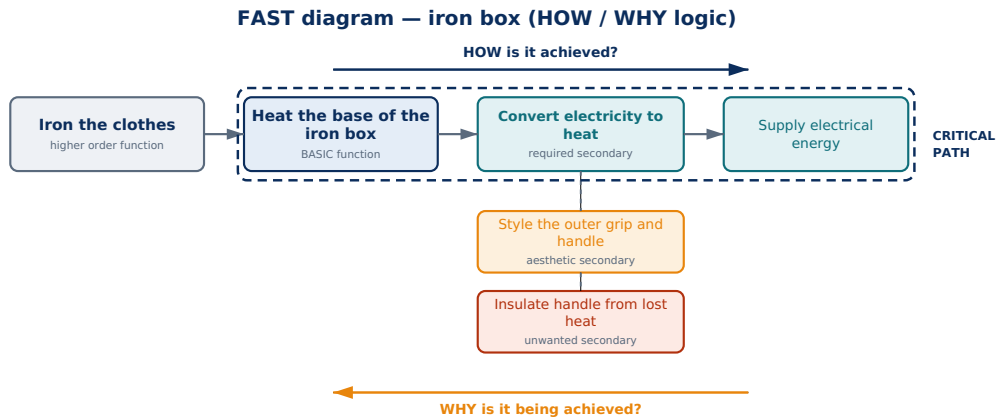


Fig 5.9 — FAST diagram for an iron box, with the HOW / WHY directions

Subtract and Operate (SOP) Method PYQ HIGH

- **Bottom-up** approach to developing a function tree
- **Assumption:** a form concept or an actual product already exists; this product is then **reverse engineered**
- A better way to figure out the function of a component is by **removing it** and operating the system without it, conceptually
- Starts by considering the **smallest isolatable function or feature** — ones which are not further decomposable

SOP procedure — six steps

1. **Disassemble** one component of the assembly
2. **Operate** the system through its full range
3. **Analyze** the effect
4. **Deduce** the sub-function of the missing component
5. **Replace** the component and repeat the procedure "**n**" times, n being the number of components in the assembly
6. **Translate** the collection of sub-functions into a function tree

Subtract and Operate (SOP) — bottom-up function analysis

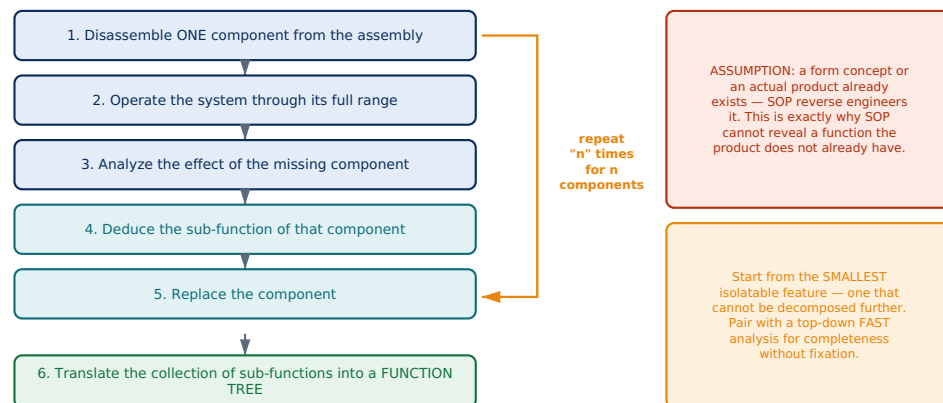


Fig 5.10 — Subtract and Operate: bottom-up function analysis

Key takeaways to memorise for Session 5

- Definition of a concept; 5% budget / 15% time
- Five types of mental block with examples
- Five-step concept generation method
- Classification: intuitive / logical / generative AI, with the methods under each
- Osborn's four brainstorming rules, 5-15 people, 30-45 min, drawbacks
- 6-3-5: the arithmetic (108 ideas), advantages, disadvantages
- SCAMPER expansion
- Affinity diagram = KJ Method, Jiro Kawakita, 1960s
- Function / product function / sub-function / abstraction / constraint definitions
- **Black box with M-E-S flows and line conventions**
- FAST: HOW-WHY rule, higher order / basic / secondary (required, aesthetic, unwanted), critical path
- **SOP: six-step procedure, assumption, bottom-up**

5.9 Expected Questions — Session 5

Q1. Explain the Subtract and Operate (SOP) analysis for identification of weaknesses in product design. What [5+5+5] are its shortcomings in terms of innovation and the emergence of new functions? Illustrate with a practical example.

PYQ MUST DRAW

Part A [5] — What SOP is and how it works. Give the definition and the six-step procedure (box in section 5.8). Emphasise: bottom-up, assumes an existing product, reverse-engineering based, repeated n times for n components, output is a function tree.

Draw the procedure as a small loop flow chart: Disassemble → Operate → Analyze effect → Deduce sub-function → Replace component → (repeat n times) → Translate into function tree.

Part B [5] — Shortcomings in terms of innovation and emergence of new functions. This is the discriminating part of the question; write it as a list:

Shortcoming	Why it limits innovation
Bounded by the existing embodiment	SOP can only discover functions that the present product already performs. It cannot reveal a function the product never had. The solution space is capped by the artefact on the bench.
Cannot reveal latent or emergent needs	A removed component tells you what stops working — not what the customer wishes the product could also do. New functions arise from customer need identification and ethnography, not from disassembly.
Incremental by construction	The output is a function tree of the current design, so the concepts generated from it tend to be re-designs and variants. It supports Horizon-1 opportunities, not Horizon-3.
Misses emergent / system-level functions	Some functions arise only from the interaction of several components. Removing one at a time attributes the whole effect to that single part and can mislead. Incidental interactions are invisible to the method.
Multi-function components confound the result	In an integral architecture , one component performs several functions, so its removal produces a tangle of effects that is hard to decompose cleanly. SOP works best on modular products with one-to-one function-component mapping.
Practical limits	Some components cannot be removed without destroying the product (welded, bonded, moulded-in features); electronic and software functions cannot be "subtracted" physically. Effort grows linearly with part count.
Risk of fixation	Working from the existing product reinforces design fixation — a perceptual block — making it harder to conceive alternative architectures.

The remedy to state: SOP should be paired with a **top-down FAST** analysis and with external search. FAST works from the higher-order function downward and is therefore free of the existing form; SOP works from the existing form upward and is therefore grounded in reality. Using both gives completeness without fixation.

Part C [5] — Practical example. Use the **coffee mill** from class, laid out as the SOP table:

Component removed	Observed effect when operated	Deduced sub-function
Chamber	No defined way of holding contents; no measurable volume; no body to contain or hold the apparatus; aesthetic appearance reduced; difficult to clean; cannot pour out contents	Contain beans; hold beans; measure amount; support structure; enable cleaning; provide aesthetic form
Seal	No protection against contents splattering; no protection against the spinning blade; safety issues; chamber cannot be closed; power cannot be actuated because the electric circuit is not closed; impact noise not enclosed	Contain chopping; separate blade from hands; close the circuit (interlock); contain chop noise
Slicing blade	Contents will not be chopped; no resistance to torque	Cut beans
Shaft	Slicing blade not attachable; contents will not be chopped; no resistance to torque	Rotate shaft; transmit torque; support motor
Motor	Shaft does not spin; electricity is not transformed into mechanical energy	Convert electricity to rotation

Then show the resulting bottom-up function tree: prime function "**Grind Coffee**", decomposed into Input Beans, Input On/Off Signal, Input Power, Hold, Rotate Shaft, Provide Coffee, Chop Beans, Permit Cleaning — each further decomposed (Accept Beans, Indicate

Amount, Transmit On/Off Signal, Support Motor, Contain Noise, Separate Blade from Hands, Cut Beans, Contain Chopping, Dispense Coffee, and so on).

Innovation limitation made concrete: SOP on this coffee mill will never suggest **grind-size adjustment**, **burr grinding instead of blade chopping**, or a **built-in scale** — because none of those functions exists in the product being dismantled. Those come from customer needs and from TRIZ/morphological exploration.

Q2. Classify the various methods of concept generation. Discuss the limitations of traditional brainstorming techniques. How can structured approaches overcome these limitations? **[15]**
PYQ **HIGH**

Part A — classification. Draw a three-branch tree (Fig 5.2) and populate it using the table in section 5.4: intuitive methods, direct/logical methods, generative AI tools. Give one line on each method. Note the underlying distinction: **intuitive methods defer judgment and reach for quantity; logical methods impose structure and traverse the solution space systematically.**

Part B — limitations of traditional brainstorming:

Limitation	Mechanism
Dominance by individuals	Senior or extroverted members occupy the airtime; hierarchical groups amplify this
Production blocking	Only one person can speak at a time, so others' ideas decay while they wait
Evaluation apprehension	Fear of appearing foolish suppresses wild ideas — an emotional block
Group conventions and anchoring	Early ideas set a frame; later ideas cluster around it, narrowing the search
Timing	The right idea may not arrive within the 30–45 minute window; incubation is not allowed for
Misdirected focus / distraction	The session drifts from the problem statement
Poor documentation	Ideas are lost if a facilitator or recorder is not disciplined
No coverage guarantee	Nothing ensures the solution space has been systematically explored — whole regions can be missed

Part C — how structured approaches overcome them (the part that earns the marks — map each remedy to a specific limitation):

Structured method	Limitation it removes
6-3-5 brainwriting	Removes dominance, production blocking and evaluation apprehension — ideas are written silently and simultaneously, so introverts contribute equally and no one waits to speak. Also guarantees documentation.
C-Sketch	Same anonymity benefit, plus it forces embodiment thinking rather than verbal abstraction, and each pass builds progressively on others' work.
Affinity diagram (KJ)	Converts an unstructured pile of ideas into labelled groups, so the team can see which regions are crowded and which are empty — a coverage check.
Concept combination table	Guarantees systematic coverage : decomposing into subproblems and taking the cross-product enumerates combinations no one would have voiced aloud ($4 \times 2 \times 3 = 24$).
Functional decomposition / FAST	Attacks fixation by forcing the team to think in "what" terms before "how", removing reliance on previous experience.
Morphological chart	Explicit part-solutions per function; combining them mechanically produces concepts free of group anchoring.
TRIZ	Replaces reliance on the group's collective memory with a database of prior invention — "somebody, sometime, somewhere has already solved your problem". Resolves contradictions instead of compromising.
SCAMPER checklist	Provides external stimulus when the group stalls, systematically prompting substitution, combination, reversal, and so on.

Conclusion to write: the two classes are complementary, not competing. Best practice is to use intuitive methods to generate raw variety, then logical methods to ensure the solution space has been systematically covered, then affinity grouping to organise before moving to concept screening.

Q3. What is functional decomposition? Why is it essential for concept generation? A designer wishes to modify an electric kettle to be more compact, lightweight and aesthetically pleasing while minimising heat loss. Draw a generic black-box function structure for the electric kettle. **[15]**
PROF **PYQ** **MUST DRAW**

Part A — what it is. Functional decomposition is the process of breaking the overall product function into a hierarchy of sub-functions, expressed as **verb + noun** pairs and stated **independently of any particular form**. It maps **Needs → Function → Form**.

Part B — why essential (six reasons from the box in section 5.7). Emphasise the two that matter most: it concentrates on **what** rather than **how**, and it **avoids jumping to a form conclusion too quickly**, which is what preserves the size of the concept space.

Part C — the black box for the electric kettle. Draw the outer box first:

Flow type	Inputs	Outputs
Material (thick arrow)	Cold water, limescale/impurities, air	Hot water, steam, scale residue
Energy (thin arrow)	Electrical energy, human (hand) force for lifting/pouring/switching	Thermal energy in water, heat loss to surroundings , sound/noise, mechanical work of pouring
Signal (dashed arrow)	On/off command, desired temperature setting, water-level observation	Power-on indication, boil-complete indication, water-level display, audible click

Overall function in the centre: **"HEAT WATER"** (verb + noun).

Then decompose into sub-functions inside the box, connected by the three flow types:

Sub-function	Notes tied to the stated design objectives
Import water	Accept and contain the water; measure level
Import electrical energy	Connect to supply; ensure isolation when lifted from base
Import human signal	Accept on/off and temperature setting
Convert electrical energy to thermal energy	Heating element; efficiency drives energy use
Transfer heat to water	Element-to-water interface area
Retain / insulate heat	Directly serves "minimise heat loss" — double-wall or vacuum construction, insulated lid and spout
Contain water and steam	Vessel, lid seal, steam vent
Sense boiling / temperature	Thermostat or sensor
Control energy supply	Switch off at set point; prevent dry boiling
Indicate status	Signal output to user
Support structure and loads	Serves "compact and lightweight" — minimise material and wall thickness consistent with strength
Enable handling and pouring	Handle, spout, balance
Provide aesthetic form	Serves "aesthetically pleasing" — secondary aesthetic function in FAST terms
Export hot water	Pour without drip
Permit cleaning / descaling	Access to interior

Closing point that earns the mark: note that "retain heat", "compact/lightweight" and "aesthetic form" are in tension — insulation adds wall thickness and mass, which fights compactness. Identifying that contradiction at the function stage is exactly the value of decomposition: it hands a well-posed problem to TRIZ or morphological analysis rather than to guesswork.

Q4. Explain the FAST method with the HOW-WHY logic. Construct a FAST diagram for an electric iron box and identify the basic function, secondary functions and the critical path. [15] **MUST DRAW**

- Definition of FAST: displays functions in a logical sequence, prioritises them, tests their dependency; top-down.
- State the HOW-WHY rule (box in section 5.8) and mark the two direction arrows on the diagram.
- For the iron box: **higher order function** = "Iron the clothes"; **basic function** = "Heats the base of the iron box"; **secondary functions**: required = "heater assembly converts electricity to heat"; aesthetic = "styling of outer grip/handle"; unwanted = "heating elements lose heat to handles → handles need insulation".
- **Critical path** = Higher order function → Basic function → Required secondary function. Mark it on the diagram.
- Contrast with SOP in two lines: FAST is top-down and form-independent; SOP is bottom-up and requires an existing artefact. Use both for completeness.

Q5. What are mental blocks in design? Classify them and explain how a design team can overcome each type. [15]

- Give the five-way classification table from section 5.2 with the named sub-blocks (stereotyping, information overload, limiting the problem unnecessarily, fixation; fear of risk taking, unease with chaos, inability to incubate, motivation; thought patterns, tradition bound, attitudes; poor problem-solving language, memory blocks, insufficient knowledge base, incorrect information; physical environment, criticism).
- **Countermeasures, mapped:** perceptual → restate the problem at a higher level of abstraction, use functional decomposition to break fixation; emotional → defer judgment (Osborn rule 2), welcome wild ideas, allow incubation time and breaks; cultural → use cross-functional and cross-geography teams, avoid hierarchically structured groups; intellectual → external search, patents and literature, benchmarking, bring in experts; environmental → maintain an atmosphere of support and trust, ban killer phrases, provide suitable physical space.
- Link to the aids to creative thinking (six points, section 5.2) and to the 6-3-5 method as the structural fix for emotional and environmental blocks.

Session 6 — Concept Development: Morphological, TRIZ, Axiomatic — and the Basis of Concept Selection

6.1 Morphological Analysis HIGH

The morphological method is essentially **an orderly way of looking at things**. Morphological charts originate from the concept of the **n-dimensional morphological box (Zwicky box, 1948)**.

Morphological chart (Zwicky box) — water purifier example

FUNCTION	Part-solution 1	Part-solution 2	Part-solution 3	Part-solution 4
Remove suspended solids	Sediment cartridge	Ceramic candle	Cyclone separator	Cloth pre-filter
Remove dissolved salts	Reverse osmosis	Ion exchange resin	Distillation	
Disinfect	UV lamp	Chlorination	Ozonation	Boiling
Move water	Booster pump	Gravity head	Mains pressure	Hand pump
Indicate status	LED panel	Mechanical flag	Mobile app	Audible alert
Concept A — mains-powered premium unit Concept B — gravity unit for intermittent supply				

A concept is one part-solution taken from each row. Combinations = $4 \times 3 \times 4 \times 4 \times 4 = 768$ — which is why a screening step must follow

Fig 6.1 — Morphological chart, with two distinct concepts traced through it

Steps involved in morphological analysis

- Each individual designer makes a list of the **most important functions** that need to be fulfilled, based on the design brief. These are translated from the programme of demands into functions and aspects, and inserted into the **first column**.
- The designer adds the possible **part-solutions** to the related rows of each function/aspect.
- Individual morphological charts are combined by the design team; functions and aspects are discussed and the team decides which will be placed in the **morphological overview**.
- All participants contribute their solutions for these functions and aspects by filling in the rows within the morphological overview.
- A concept is formed by **selecting one part-solution from each row** and combining them.

Putting the individual charts together enables individual perspectives from each discipline to be put forward, which in turn highlights the implications of design choices for each discipline.

Simple illustration from class: for the function "provide rotary drive", the part-solutions listed are: electrical motor • mechanism • hydraulic • electro-mechanical • chemical.

Number of possible concepts = product of the number of part-solutions in each row.

Advantages

- Systematic and exhaustive coverage of the solution space
- Forces separation of function from form
- Surfaces combinations nobody would have voiced
- Multi-disciplinary contributions made visible

Limitations

- Combinatorial explosion — many combinations are infeasible or absurd
- Does not evaluate compatibility between part-solutions
- Quality depends entirely on the completeness of the function list
- Needs a screening step afterwards (Pugh)

6.2 TRIZ / TIPS — Theory of Inventive Problem Solving HIGH

TRIZ is a Russian acronym: *Teoriya Resheniya Izobretatelskikh Zadatch* — Theory of Inventive Problem Solving (also written TIPS). Developed by **Genrich Altshuller** from the study of patents.

The two basic principles of TRIZ — quote these

- "**Somebody, sometime, somewhere has already solved your problem or one similar to it. Creativity means finding that solution and adapting it to the current problem.**"
- "**Don't accept contradictions. Resolve them.**"

Altshuller's observations:

- Three categories of inventive problem: identifying conflicts and solving them with known physical principles; identifying new principles; identifying new product functions and solving them with known or new principles
- A number of **trends in historical invention** were observed
- Evolution of engineering systems follows the **same pattern, independent of engineering discipline**
- Conflicts (contradictions) are the key drivers for product invention**
- Compromise is unacceptable**

The TRIZ toolkit

39 Engineering Parameters	The standard vocabulary for describing what improves and what worsens — weight of moving object, length, area, volume, speed, force, stress/pressure, shape, strength, durability, temperature, power, energy loss, loss of substance, loss of information, loss of time, reliability, measurement accuracy, ease of manufacture, ease of operation, ease of repair, adaptability, complexity of device, productivity, and so on
40 Inventive Principles	The standard solution patterns (segmentation, extraction, local quality, asymmetry, combining, universality, nesting, counterweight, prior action, inversion, spheroidality, dynamicity, mechanical vibration, periodic action, rushing through, convert harm into benefit, feedback, mediator, self-service, copying, cheap disposable objects, pneumatics/hydraulics, flexible membranes, porous materials, changing colour, homogeneity, rejection and regeneration, phase transformation, thermal expansion, inert environment, composite materials ...)
Contradiction Matrix	A 39 × 39 lookup: row = parameter to be improved, column = parameter that worsens; the cell lists the inventive principle numbers most often used historically to resolve that contradiction

TRIZ – resolving a contradiction instead of compromising it

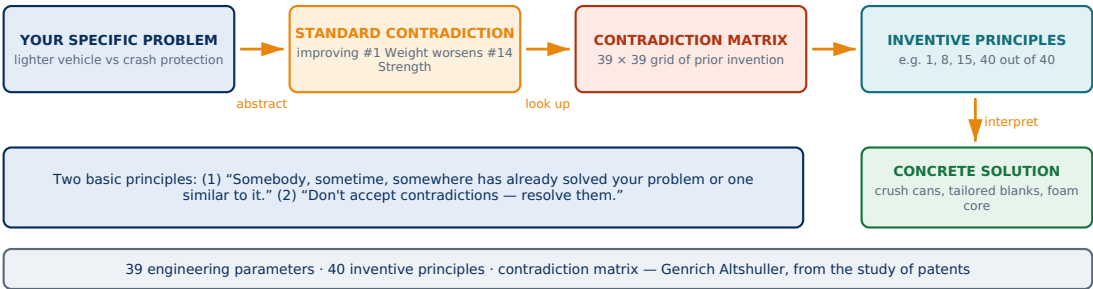
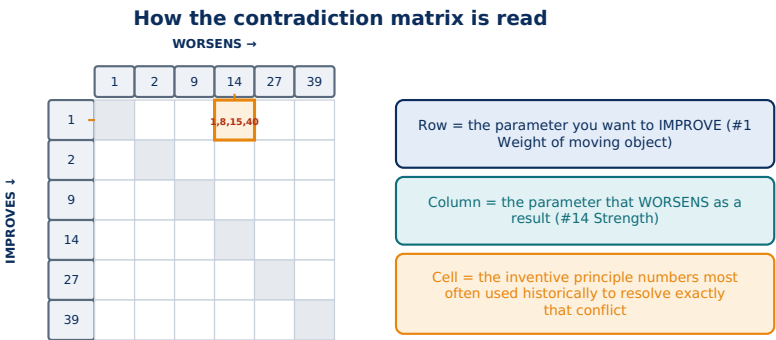


Fig 6.2 — TRIZ: resolving a contradiction instead of compromising it



Diagonal cells are blank — a parameter cannot contradict itself

Fig 6.3 — How the contradiction matrix is read

TRIZ procedure — the four steps to write in any TRIZ answer

1. **Identify the contradiction** — state which parameter **improves** and which **worsens**
2. **Map onto the 39 engineering parameters** — find the parameter numbers
3. **Look up the contradiction matrix** — obtain the recommended inventive principle numbers
4. **Interpret each principle** into a concrete concept for your specific problem

Worked TRIZ examples from class

Problem	Contradiction	Principles obtained and interpretation
Bulletproof vest — should be strong but not heavy	Strength (#14) improves vs Weight (#2) worsens	1 Segmentation — divide the object into independent parts, make it easy to disassemble. 26 Copying — instead of an unavailable, expensive or fragile object, use simpler and inexpensive copies. 27 Cheap short-living objects — replace an expensive object with multiple inexpensive objects having certain qualities such as service life. 40 Composite materials — polymers (Kevlar) reinforced with carbon nanofibres as a strong lightweight alternative to steel.
Electric iron — heavy iron removes wrinkles better, but heavy iron is bad ergonomically	Removal effectiveness improves vs ergonomics/weight worsens	8 Counterweight — attach an object with lifting power or use interactions with the environment (aerodynamic lift). 1 Segmentation — divide into independent parts, increase the degree of segmentation. 37 Thermal expansion — use expansion or contraction of material by heat; use materials with different coefficients of thermal expansion. 18 Mechanical vibration — make the object vibrate; increase the frequency of vibration (this is the origin of the steam/vibration iron).
Airbag — must inflate before contacting occupants to prevent forward motion, but faster inflation causes injury	Speed of inflation improves vs harmful side effects worsen	16 Partial or excessive action — use a lower-powered airbag (less acceleration, fewer injuries) or smaller bags with higher power to reach full inflation sooner. 21 Rushing through — inflate faster than current practice. 40 Composite materials — airbag material that cannot grab skin as it deploys.
Refrigerator — increasing internal volume while keeping energy consumption low	Volume of stationary object improves vs energy consumption worsens	10 Prior action , 15 Dynamicity , 14 Spheroidality , 7 Nesting — leading to nested compartments, variable-speed compressors, and curved internal geometry with better insulation-to-volume ratio.

Benefits of TRIZ

- Resolves contradictions rather than compromising them — no trade-off penalty
- Draws on a distilled database of prior invention, so the team is not limited to its own experience
- Systematic and repeatable — reduces dependence on individual genius
- Discipline-independent — the same patterns apply across mechanical, electrical, chemical domains
- Fast — narrows a vast solution space to a handful of candidate principles
- Breaks psychological inertia and fixation

Limitations of TRIZ

- Requires skill to correctly abstract a real problem into the 39 standard parameters
- The principles are abstract; interpreting them into a concrete concept still needs creativity
- The matrix is derived from historical (largely mechanical) patents
- Not all problems reduce to a clean two-parameter contradiction
- Steep learning curve; limited support for cost and aesthetics

6.3 Axiomatic Design

Design models claiming legitimacy from **first principles** include **Nam P. Suh's** texts on Axiomatic Design, which articulate the **Design Independence** and **Information** axioms.

The method translates a design task into **functional requirements** (the engineering equivalent of what the customer wants) and uses those to identify **design parameters** — the physical components of the design. Suh's principles lead to theorems and corollaries that help designers diagnose a candidate solution, represented as a **matrix equation** relating functional requirements and design parameters.

The four domains **MUST DRAW**

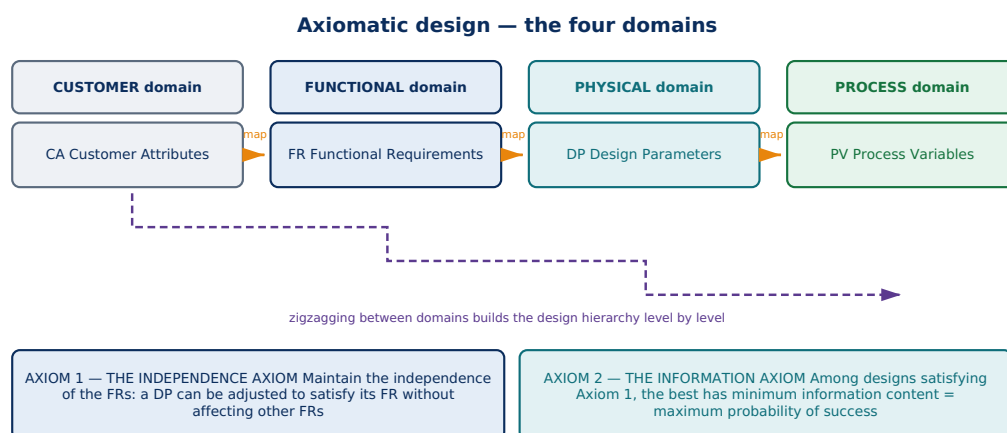


Fig 6.4 — Axiomatic design: the four domains and the two axioms

Domain	Contains	Symbol
Customer domain	The benefits customers seek	CA — Customer Attributes
Functional domain	Functional requirements of the design solution	FR — Functional Requirements
Physical domain	Design parameters of the design solution	DP — Design Parameters
Process domain	Process variables	PV — Process Variables

CA → FR → DP → PV (each arrow is a mapping, and **zigzagging** between domains produces the hierarchy)

The two axioms

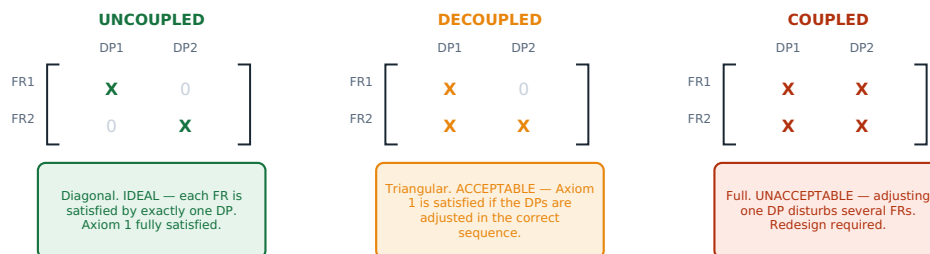
Axiom 1 — The Independence Axiom: *Maintain the independence of FRs.* In an acceptable design, the DPs and the FRs are related in such a way that a specific DP can be adjusted to satisfy its corresponding FR **without affecting other FRs**.

Axiom 2 — The Information Axiom: *Minimise the information content.* Among alternative designs which satisfy Axiom 1, the best has the **minimum information content**, which means the **maximum probability of success**.

$\{FR\} = [A] \{DP\}$ where $[A]$ is the design matrix

Design matrix form	Classification	Meaning
Diagonal	Uncoupled design	Ideal — each FR is satisfied by exactly one DP; Axiom 1 fully satisfied
Triangular	Decoupled design	Acceptable — Axiom 1 satisfied if the DPs are adjusted in the correct sequence
Full	Coupled design	Unacceptable — adjusting one DP disturbs several FRs; redesign required

Design matrices: uncoupled, decoupled and coupled



Worked example — FR1 control flow rate, FR2 control temperature. Two-tap basin: both tap angles affect both FRs → COUPLED. Single-lever mixer: tilt = flow, rotation = temperature → UNCOUPLED.

$$\{FR\} = [A] \{DP\}$$

Fig 6.5 — Uncoupled, decoupled and coupled design matrices

The classic example to quote: a traditional two-tap basin has FR1 = control flow rate and FR2 = control temperature, but both DPs (hot tap angle, cold tap angle) affect both FRs — a **coupled** design. A single-lever mixer tap assigns lever tilt to flow and lever rotation to temperature — an **uncoupled** design satisfying Axiom 1.

6.4 Concept Selection — Basis and Filters

Selecting a concept requires three things:

1. A set of **design selection criteria** (from the specifications)
2. A set of **design alternatives** (from concept generation)
3. A **means to evaluate** the alternatives against each criterion (concept selection)

Concept selection is an **iterative** process closely related to concept generation and testing. Concept screening and scoring help the team refine and improve concepts, leading to one or more promising concepts for further testing and development.

Patterns observed in people when deciding

Five decision-making patterns, only one of which is sound:

- Decide to continue with the current action and ignore information about risk of losses
- Uncritically adopt whichever course of action is most strongly recommended
- **Search painstakingly for relevant information that is assimilated in an unbiased manner, and appraised carefully before a decision is made** — the desirable pattern
- Evade conflict by procrastinating, shifting responsibility to someone else, and remaining inattentive to corrective information
- Search in a hurried or "excited" way for an immediate problem solution

Benefits of a structured concept selection method

A customer-focused product • a competitive design • better product-process coordination • reduced time to product introduction • effective group decision making • **documentation of the decision process**

Absolute vs relative comparison HIGH

Type	Definition	Example
Absolute comparison	The concept is compared directly with a fixed and known set of requirements , such as a product specification	Checking whether a design alternative would be under the weight limit
Relative comparison	The concepts are compared with each other on the basis of a metric	Checking which of the possible designs would be the lightest design

The absolute evaluation filters — applied in sequence

1. **Functional feasibility of the design** — concepts are placed into one of three categories: **(1) not feasible (2) feasible under certain conditions (3) feasible**
2. **Technology readiness**
3. **Go / No-go screening** — check whether the concept satisfies the constraints of the problem, eliminating concepts that fail to meet: **(1) constraints** and **(2) the minimum acceptable level of important engineering characteristics**

Important nuance from the slides: the weak areas (No-go) in a concept may be fixable by borrowing ideas from another concept — which triggers a new idea. Screening is therefore not only elimination; it is also a generator.

Geometric modelling and FEA in concept evaluation

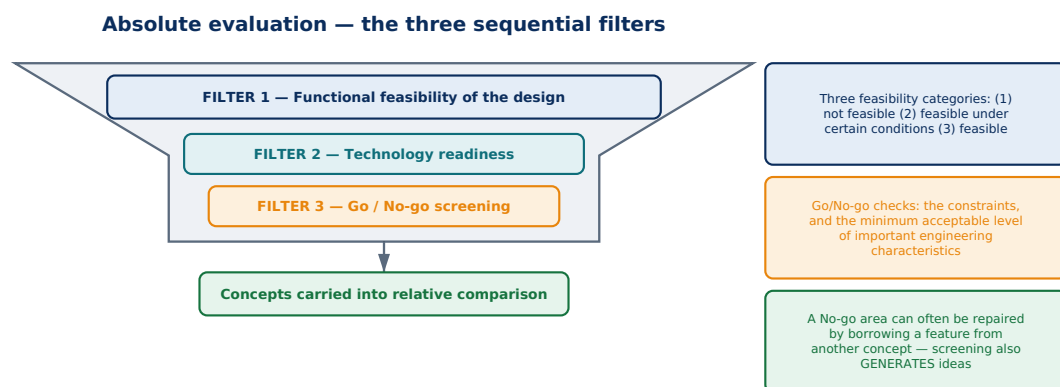


Fig 6.6 — Absolute evaluation: the three sequential filters

Associativity, compatibility, assembly structure with constraints → geometry de-featuring and simplification → meshing and set-up for solving → apply loading conditions/scenarios → solve and extract results → margin of safety, life analysis

Key takeaways to memorise for Session 6

- Morphological analysis — Zwicky box 1948, five steps, concept = one part-solution per row
- TRIZ — full name, Altshuller, two basic principles, 39 parameters / 40 principles / contradiction matrix, four-step procedure, three worked examples, benefits
- Axiomatic design — Suh, four domains, Axiom 1 and Axiom 2 statements, uncoupled/decoupled/coupled
- Three requirements for concept selection; absolute vs relative comparison
- Three feasibility categories; the three sequential filters
- Six benefits of a structured method

6.5 Expected Questions — Session 6

Q1. OEMs aim to reduce vehicle mass to improve fuel economy and EV range, but lighter structures often reduce crash energy absorption and occupant protection. Suggest possible solutions using the TRIZ approach. Discuss the benefits of the TRIZ method. PYQ HIGH [15]

Step 1 — State the contradiction explicitly.

Improving parameter	Weight of a moving object (#1) — we wish to reduce vehicle mass
Worsening parameter	Strength (#14) / Durability of a moving object (#15) / Reliability (#27) — crash energy absorption and occupant protection degrade

Write the TRIZ position clearly: **compromise is unacceptable** — we do not accept "a bit heavier and a bit safer"; we resolve the contradiction.

Step 2 — Look up the contradiction matrix for Weight of moving object vs Strength. The principles recommended are typically **1, 8, 15, 40** (and for the durability variant, **27, 28, 35**).

Step 3 — Interpret each principle into a concrete automotive solution. This is where the marks are:

No.	Inventive principle	Applied solution for the vehicle structure
1	Segmentation	Break the monolithic structure into discrete, tuned segments: dedicated crush cans , progressive crumple zones with graded section thickness, multi-cell extruded aluminium crash boxes. Each segment absorbs energy at a designed rate rather than the whole body being made thicker.
40	Composite materials	Multi-material body-in-white : ultra-high-strength boron steel in the safety cage, aluminium in the closures, CFRP or SMC in non-load-path panels. Also carbon-fibre passenger cells as used in high-end EVs. Strength is placed only where the load path demands it.
3	Local quality	Tailor-welded and tailor-rolled blanks — a single part with locally varying thickness or grade, thick at the joints and thin elsewhere. Also hot-stamped patch reinforcements at B-pillar high-load zones.
15	Dynamicity	Adaptive structures : active bonnet lift, deployable bumper beams, adaptive load-limiting seat belts and multi-stage airbags that vary restraint force with occupant mass and crash severity. Protection becomes a function of the event rather than of steel thickness.
9 / 10	Prior counteraction / Prior action	Pre-crash systems : radar and camera sensing triggers pre-tensioning of belts, closing of windows and pre-positioning of seats before impact. Energy management starts before the structure is loaded.
2	Extraction / separation	Separate the two functions the structure is being asked to perform: use a stiff, light survival cell for occupant protection and separate, sacrificial, easily-replaceable energy absorbers ahead of it.
17	Moving to a new dimension	Move load paths out of plane — multi-load-path front ends , and honeycomb or lattice cores that absorb energy through the thickness direction rather than requiring more material in-plane.
31	Porous materials	Aluminium foam and honeycomb fillers inside hollow sections — very high specific energy absorption at very low added mass.
22	Convert harm into benefit	Use the battery pack structure of an EV, otherwise dead mass, as a stressed structural member and side-impact barrier. The heaviest component becomes part of the protection.
28	Replace mechanical system with fields	Replace part of the mechanical protection with electronic active safety — AEB, lane keeping, stability control — which avoids or reduces the collision severity, so less structural mass is needed for the same occupant outcome.

Step 4 — Benefits of the TRIZ method (the second half of the question — use the table in section 6.2). Emphasise the ones this example demonstrates:

- **It removes the compromise.** Conventional optimisation would trade mass against safety along a single curve; TRIZ generated ten concepts that move the curve instead.
- **It draws on solved problems from other fields** — aluminium foam and honeycomb came from aerospace, not automotive.
- **It is systematic and repeatable**, not dependent on a single creative individual.
- **It is fast** — four steps narrow a vast space to a handful of directions.
- **It breaks psychological inertia** — the instinct "make it thicker" is exactly the fixation TRIZ defeats.

Mention the limitations briefly for balance: abstraction skill is required, the matrix is patent-derived and mechanically biased, and the principles still need creative interpretation.

Q2. Explain the morphological method of concept generation. Construct a morphological chart for a product of your[15] choice and derive two distinct concepts from it. MUST DRAW

Definition, Zwicky box (1948), the five steps from section 6.1. Then build the chart. Example for a **domestic water purifier**:

Function	Solution 1	Solution 2	Solution 3	Solution 4
Remove suspended solids	Sediment cartridge	Ceramic candle	Cyclone separator	Cloth pre-filter
Remove dissolved salts	Reverse osmosis	Ion exchange resin	Distillation	—
Disinfect	UV lamp	Chlorination	Ozonation	Boiling
Move water	Booster pump	Gravity head	Mains pressure	Hand pump
Store treated water	Pressure vessel	Open tank	Direct flow (no storage)	—
Indicate status	LED panel	Mechanical flag	Mobile app	Audible alert

Total combinations = $4 \times 3 \times 4 \times 4 \times 3 \times 4 = 2,304$ theoretical concepts

Concept A — "Mains-powered premium unit": sediment cartridge + reverse osmosis + UV lamp + booster pump + pressure vessel + LED panel. High performance, needs electricity and drainage, higher cost.

Concept B — "Gravity unit for intermittent supply": ceramic candle + ion exchange resin + chlorination + gravity head + open tank + mechanical flag. No electricity required, low cost, lower dissolved-salt performance — suited to areas with unreliable power.

Close with the limitation: the chart does not check **compatibility** between part-solutions (a booster pump with gravity head is nonsense), so the enumerated set must pass through Go/No-go screening and then Pugh screening.

Q3. State and explain the two axioms of Axiomatic Design. Explain the four domains of the design world. [15]
Distinguish between uncoupled, decoupled and coupled designs with an example. MUST DRAW

- Attribute to **Nam P. Suh**. Draw the four-domain diagram (Fig 6.4) with the mapping arrows $CA \rightarrow FR \rightarrow DP \rightarrow PV$ and mention **zigzagging** to build the hierarchy.
- State both axioms verbatim (box in section 6.3).

- Write the design equation $\{FR\} = [A]\{DP\}$ and give the three matrix forms with their classification table.
- **Example — the water tap.** FR1 = control flow rate, FR2 = control water temperature.
 - *Two-tap basin:* DP1 = hot valve angle, DP2 = cold valve angle. Both DPs affect both FRs → **full matrix** → **coupled** → **unacceptable**. This is why adjusting temperature at a two-tap basin also changes the flow, and you iterate.
 - *Single-lever mixer:* DP1 = lever tilt (flow), DP2 = lever rotation (temperature) → **diagonal matrix** → **uncoupled** → **ideal**.
- Add a decoupled illustration: a design where temperature must be set first and flow second works acceptably provided the sequence is respected — triangular matrix.
- Close by connecting Axiom 2 to reliability: minimum information content = maximum probability of success, so among Axiom-1-compliant alternatives, choose the one with the tightest tolerance-to-design-range ratio.

Q4. Differentiate between absolute and relative comparison in concept evaluation. Explain the sequential filters used in the absolute evaluation process. [15]

- Give the comparison table from section 6.4 with both worked examples (under the weight limit vs lightest design).
- Draw the three filters as a funnel: Functional feasibility → Technology readiness → Go/No-go screening.
- Under functional feasibility, give the three categories: not feasible / feasible under certain conditions / feasible. Explain that "not feasible" concepts are discarded only after proper examination — the judgement is based on the team's experience.
- Under technology readiness, note that a concept may be functionally sound but rely on technology not yet mature enough for the launch timeline — connect to the S-curve from Session 2.
- Under Go/No-go, give the two checks: constraints, and minimum acceptable level of important engineering characteristics.
- **The point that earns the mark:** screening is not purely eliminative. The weak (No-go) areas of a concept can often be repaired by borrowing a feature from a different concept, which triggers a new hybrid idea. This is the same logic that Pugh's "attack the negatives" step formalises in Session 7.
- Close with the six benefits of a structured method.

Session 7 — Concept Selection and Testing

The three selection methods, in increasing rigour

Method	Also called	Nature
Concept screening	Pugh concept selection / Pugh matrix	Coarse, relative, uses +/0/- against a datum. Quick, no weights.
Concept scoring	Weighted decision matrix	Finer, uses weighting factors and numerical ratings.
Analytic Hierarchy Process	AHP	Most rigorous. Derives weights from pairwise comparison. Least arbitrary method.

7.1 Concept Screening — the Pugh Matrix PROF MUST DRAW HIGH

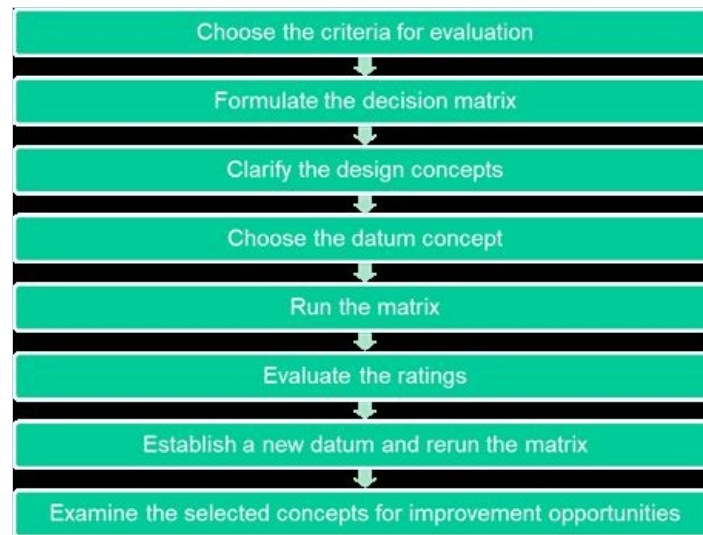


Fig 7.1 — From the instructor's answer key. The eight steps of Pugh concept selection — reproduce this flow

1. Choose the criteria for evaluation
2. Formulate the decision matrix
3. Clarify the design concepts
4. Choose the datum concept
5. Run the matrix
6. Evaluate the ratings
7. Establish a new datum and rerun the matrix
8. Examine the selected concepts for improvement opportunities

Scoring convention: + = better than datum • 0 or S = same as datum • - = worse than datum

Then compute: $\Sigma+$ • $\Sigma 0$ (or ΣS) • $\Sigma-$ • Net score = $\Sigma+ - \Sigma-$ • Rank

Guidelines for running a Pugh matrix — quote these

- Do not become too quantitative — the method is deliberately coarse
- Do not reject before proper examination
- Examine highly rated concepts for what makes them strong
- Look for "positive" features that can be transplanted into other concepts
- If most concepts get the **same rating on a certain criterion**, review the **clarity** of that criterion and of the evaluation — the criterion may be non-discriminating or poorly defined

Why establish a new datum and re-run

- To establish a new datum, usually the **highest-rated concept**, and run the matrix again
- Not to verify the selection of round 1, but to **gain more insight**
- A new datum gives different perspectives and more clarity on relative strengths and weaknesses

Examining the selected concept then helps to: **generate a new approach; change negatives to positives; identify design modifications.** This is the "attack the negatives" step.

Pugh concept selection — eight steps

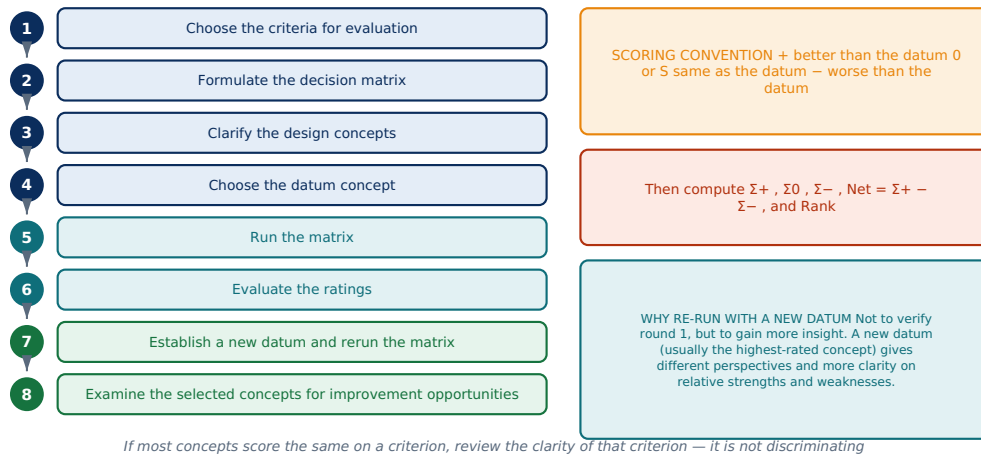


Fig 7.2 — Pugh concept selection: the eight steps and the scoring convention

7.2 Concept Scoring — the Weighted Decision Matrix PYQ HIGH

A **weighted decision matrix** evaluates concepts by ranking the design criteria with **weighting factors** and scoring the degree to which each design concept meets the criteria.

Three concept selection methods, in increasing rigour

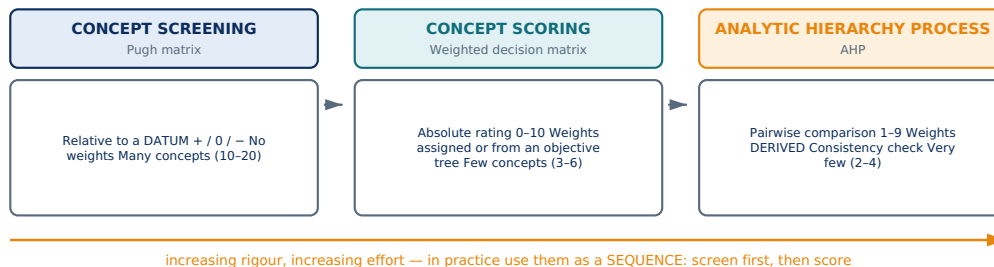


Fig 7.3 — The three concept selection methods, in increasing rigour

Three systematic methods for determining weighting factors

1. Direct assignment	Based on experience, expertise and engineering judgement
2. Objective tree	Hierarchical objective tree; weights multiply down the branches. Based on experience and expertise
3. Analytic Hierarchy Process	Least arbitrary method — weights derived from pairwise comparison

The objective tree

Objective tree — deriving weighting factors (SUV example)

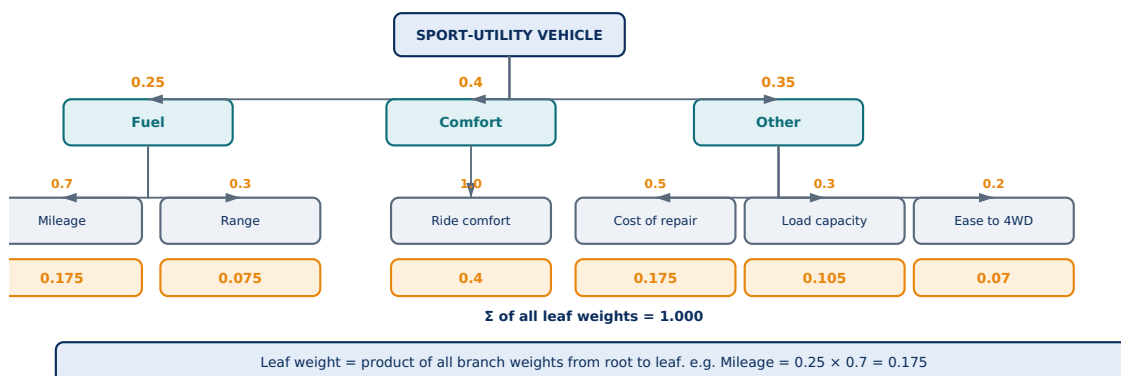


Fig 7.4 — Objective tree deriving the SUV weighting factors

Leaf weight = product of all branch weights from root to leaf. All leaf weights must sum to 1.0.

Worked example — Sport-Utility Vehicle PROF

Four preliminary designs for a sport-utility vehicle. Weighting factors are derived from the objective tree, then each design is rated on a 0–10 scale.

Deriving the weights from the objective tree:

Characteristic	Branch calculation	Weight factor
Gas mileage	Fuel 0.25 × Mileage 0.7	0.175
Range	Fuel 0.25 × Range 0.3	0.075
Ride comfort	Comfort 0.40	0.400
Cost of repair	Other 0.35 × 0.5	0.175
Load capacity	Other 0.35 × 0.3	0.105
Ease to convert to 4-wheel drive	Other 0.35 × 0.2	0.070
Total		1.000

The completed matrix (0-10 rating scale):

Characteristic	Weight	Design A		Design B		Design C		Design D	
		Rate	Score	Rate	Score	Rate	Score	Rate	Score
Gas mileage	0.175	9	1.575	7	1.225	6	1.050	9	1.575
Range	0.075	8	0.600	5	0.375	6	0.450	9	0.675
Ride comfort	0.400	3	1.200	8	3.200	7	2.800	5	2.000
Ease to convert to 4WD	0.070	8	0.560	7	0.490	7	0.490	3	0.210
Load capacity	0.105	9	0.945	7	0.735	9	0.945	6	0.630
Cost of repair	0.175	3	0.525	5	0.875	5	0.875	9	1.575
Sum of weighted scores		5.405		6.900		6.610		6.665	
Rank		4		1		3		2	

Conclusion: Design B is the most promising (6.900), driven by its strong ride-comfort rating against the heaviest weight factor (0.400). Design D is a close second on economy and repair cost but is penalised by poor 4WD conversion and ride comfort.

7.3 The Analytic Hierarchy Process (AHP) PROF MUST DRAW HIGH

AHP is a problem-solving methodology for making a choice from among a set of alternatives when the selection criteria represent multiple objectives, have a natural hierarchical structure, or consist of qualitative and quantitative measurements.

- AHP builds on the **mathematical properties of matrices** for making consistent pairwise comparisons
- It defines a pairwise, comparison-based method for determining relative ratings for the degree to which each option fulfils each criterion
- It includes the calculation of an **inconsistency measurement** and threshold values that determine whether the comparison process has remained consistent

AHP is relevant for choice problems in these categories: comparing untested concepts • structuring a decision-making process for a new situation • evaluating non-commensurate trade-offs • performing and tracking group decision making • integrating results from different sources (analytical calculations, HOQ relative values, group consensus, expert opinion) • performing strategic decision making.

Instructor's exam constraints on AHP — important

- Never more than **3 alternatives** and a maximum of **4 criteria** — "otherwise the matrix becomes bigger, I know, I am aware about it."
- **The consistency check may be skipped.** "I have given instructions to my TA that consistency check is eliminated and evaluated based on only the procedure followed. Till the normalising of the matrix C you have to perform and choose the best combination."

The consistency method is included below anyway — it appears in the lecture slides and in the instructor's own handwritten solution, and it is worth knowing for the Comprehensive exam.

The AHP hierarchy MUST DRAW

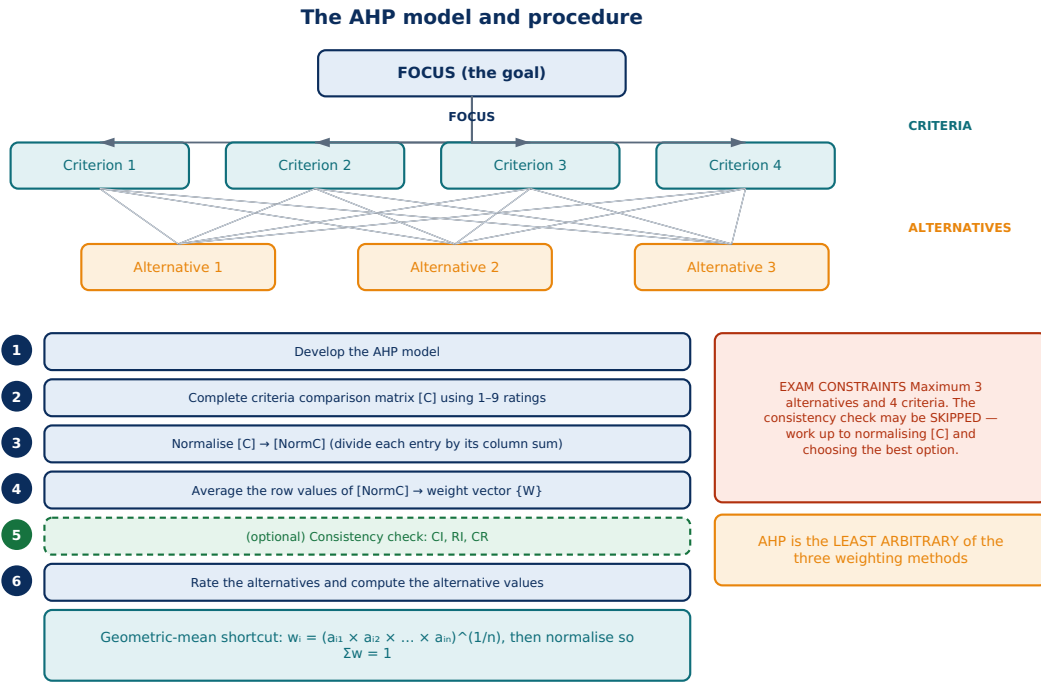


Fig 7.5 — The AHP model and the six-step procedure

The 1-9 rating scale **MUST KNOW**

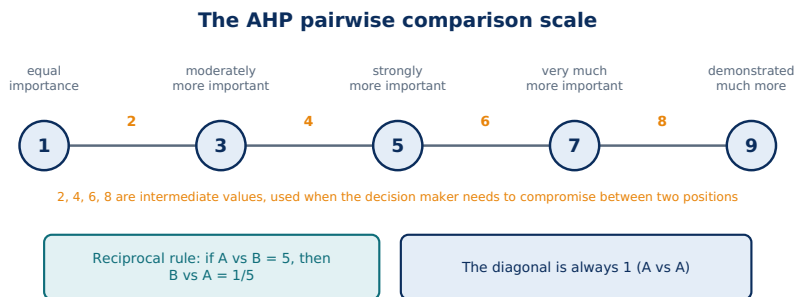


Fig 7.6 — The AHP pairwise comparison scale

Rating	Meaning
1	A and B have equal importance — both contribute equally to the product's overall success
3	A is moderately more important than B — A is slightly more important to product success
5	A is strongly more important than B
7	A is very much more important than B, or is demonstrated to be more important — A's dominance over B has been demonstrated
9	A is demonstrated to have much more importance than B — the highest possible degree of evidence
2, 4, 6, 8	Intermediate values, used when the decision maker needs to compromise between two positions

Reciprocal rule: the rating of B to A is the reciprocal of the rating of A to B. If A is strongly more important than B (5), then B to A is $1/5 = 0.20$. The diagonal is always 1.

The AHP procedure

1. **Develop the AHP model** — Focus (goal), Criteria, Alternatives
2. **Complete the criteria comparison matrix [C]** using the 1-9 ratings
3. **Normalise the matrix [C]** to give [NormC] — divide each entry by its column sum
4. **Average the row values** of [NormC]. This is the **weight vector {W}**
5. (Optional in this exam) **Perform a consistency check** on [C]
6. **Rate the alternatives** against each criterion and compute the alternative values

Alternative to step 3-4 — the geometric-mean method used in the instructor's handwritten solution: for each row, compute the n-th root of the product of the row entries, then normalise those n values so they sum to 1. Both methods give the same answer to 2-3 decimal places; the geometric-mean method is faster by hand.

$$w_i^{\text{raw}} = (a_{i1} \times a_{i2} \times \dots \times a_{in})^{1/n} \quad \text{then} \quad w_i = w_i^{\text{raw}} / \sum w^{\text{raw}}$$

Consistency check — the formulas

$$A3 = [C] \times \{W\} \quad \bullet \quad A4 = A3 \div \{W\} \text{ (element-wise)} \quad \bullet \quad \lambda_{\max} = \text{mean of } A4$$

Consistency Index $CI = (\lambda_{\max} - n) / (n - 1)$

Consistency Ratio $CR = CI / RI$ – acceptable if $CR < 0.1$

n (attributes)	3	4	5	6	7	8	9	10
Random Index RI	0.52	0.89	1.11	1.25	1.35	1.40	1.45	1.49

Worked AHP — mobile phone selection PROF

This is the instructor's own fully worked example. Learn the shape of it; the exam question will be structurally identical.

Step 1 — Model. Focus: Selection of Mobile Phone. Criteria: Cost, Battery backup, Screen size, Ease of use. Alternatives: Phone 1, Phone 2, Phone 3.

Step 2 — Criteria comparison matrix [A1]:

	Cost	Battery backup	Screen size	Ease of use
Cost	1	1/7	1/5	1/3
Battery backup	7	1	2	3
Screen size	5	1/2	1	3
Ease of use	3	1/3	1/3	1

Step 3 — Geometric means (4th root of the row product):

Criterion	Calculation	Geometric mean	Weight {W}
Cost	$(1 \times 1/7 \times 1/5 \times 1/3)^{1/4}$	0.31239	0.059246
Battery backup	$(7 \times 1 \times 2 \times 3)^{1/4}$	2.5457	0.4828
Screen size	$(5 \times 1/2 \times 1 \times 3)^{1/4}$	1.6549	0.31385
Ease of use	$(3 \times 1/3 \times 1/3 \times 1)^{1/4}$	0.75984	0.1441
Sum		5.2728	1.0000

Priority order: Battery backup > Screen size > Ease of use > Cost

Step 4 — Consistency check (shown for completeness; may be skipped in the exam):

$A3 = [A1] \times \{W\} = \{0.23902, 1.9575, 1.2638, 0.58731\} \rightarrow A4 = A3 / \{W\} = \{4.0344, 4.0545, 4.0265, 4.0762\}$

$\lambda_{\max} = 4.0639$ • $CI = (4.0639 - 4) / (4 - 1) = 0.0213$ • $RI (n = 4) = 0.89$ • $CR = 0.0213 / 0.89 = 0.0239 < 0.1 \rightarrow$ the proposed weights are consistent

Step 5 — Rate the alternatives. Raw data, with the direction of preference marked:

	Cost (Rs) ↓	Battery backup (mAh) ↑	Screen size (inches) ↓	Ease of use ↑
Phone 1	15,000	4,000	5.6	Better (5)
Phone 2	11,000	3,000	6.0	Poor (3)
Phone 3	16,000	2,500	5.2	Excellent (8)

Normalisation rule used by the instructor: for a criterion where lower is better, divide the smallest value by each value. For a criterion where higher is better, divide each value by the largest value. Every normalised value then lies between 0 and 1, with 1.0 being best.

	Cost ↓	Battery ↑	Screen ↓	Ease of use ↑
Phone 1	$11000/15000 = 0.7333$	$4000/4000 = 1.0$	$5.2/5.6 = 0.9285$	$5/8 = 0.625$
Phone 2	$11000/11000 = 1.0$	$3000/4000 = 0.75$	$5.2/6.0 = 0.86667$	$3/8 = 0.375$
Phone 3	$11000/16000 = 0.6875$	$2500/4000 = 0.625$	$5.2/5.2 = 1.0$	$8/8 = 1.0$

Step 6 — Synthesise the model. Alternative value = Σ (normalised rating $\times$ criterion weight):

	Cost (0.059246)	Battery (0.4828)	Screen (0.31385)	Ease of use (0.1441)	Alternative value	Rank
Phone 1	0.7333	1.0	0.9285	0.625	0.90774	1
Phone 2	1.0	0.75	0.86667	0.375	0.7474	3
Phone 3	0.6875	0.625	1.0	1.0	0.80043	2

Phone 1 = $(0.7333 \times 0.059246) + (1.0 \times 0.4828) + (0.9285 \times 0.31385) + (0.625 \times 0.1441) = 0.90774$

Result: Phone 1 is selected (0.90774), because it scores maximum on battery backup, which carries by far the heaviest weight (0.4828).

A second worked AHP — the crane hook

The slides carry a second example with **six criteria** and **three concepts** (welded plates, riveted plates, monolithic cast steel). The structure is identical; only the size of the matrix changes. The results are worth remembering as a sanity check on your own arithmetic:

Criterion	Weight from [NormC]	Comment
Material cost	0.047	Durability and Reliability together carry 71.7% of the decision weight, so they determine the outcome almost entirely.
Manufacturing cost	0.079	
Reparability	0.124	
Durability	0.445	
Reliability	0.272	
Time to produce	0.033	

Final synthesis: $[FRating]^T \{W\} = \{Alternative\ Value\}$

Alternative	Alternative value	Result
Welded plate design	0.336	second
Riveted plate design	0.520	selected
Monolithic casting	0.144	rejected

Note the arithmetic check available to you in the exam: the criteria weights must sum to **1.000**, and every column of [NormC] must also sum to **1.000**. If either fails, the normalisation is wrong.

Pros of AHP

- Least arbitrary of the weighting methods
- Handles qualitative and quantitative criteria together
- Pairwise comparison is cognitively easier than assigning absolute weights
- Built-in consistency measurement catches contradictory judgements
- Natural hierarchical structure; supports group decision making
- Transparent and auditable — documents the decision

Cons of AHP

- Number of comparisons grows as $n(n-1)/2$ — tedious for many criteria
- Still depends on subjective judgement at the pairwise stage
- Rank reversal can occur when an alternative is added or removed
- Time consuming relative to Pugh screening
- Assumes criteria are independent

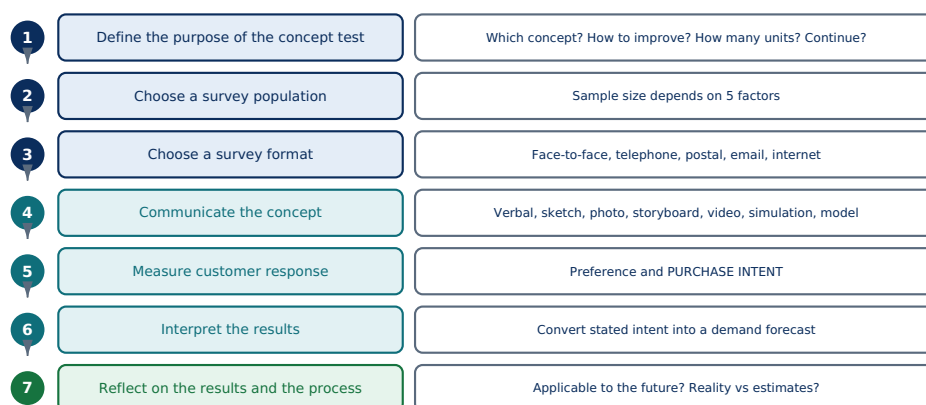
7.4 Concept Testing HIGH

Concept testing is based on **listening to the audience's views** and exploring how viable the concept is for them — without predefined parameters or expectations. It is an open learning process in which you continue to engage with the same target audience.

The seven-step concept testing process MUST DRAW

1. **Define the purpose of the concept test**
2. **Choose a survey population**
3. **Choose a survey format**
4. **Communicate the concept**
5. **Measure customer response**
6. **Interpret the results**
7. **Reflect on the results and the process**

The concept testing process — seven steps



Concept testing relies on data from potential CUSTOMERS; concept selection relies on the development TEAM's judgement

Fig 7.7 — The concept testing process: seven steps

Step	Content
1. Purpose	Which of several alternative concepts should be pursued? How can the concept be improved to better meet customer needs? Approximately how many units are likely to be sold? Should development be continued?
2. Survey population	Sample size varies from a few to thousands. Factors affecting sample size: (1) the stage of product development (2) cost to conduct the survey (3) nature and intent of the survey (4) budget of the development project (5) how possible it is to collect the intended information. Multiple surveys with different objectives can be structured at different stages.
3. Survey format	Face-to-face interaction • telephone • postal mail • electronic mail • internet (a test site). Each has its pros and cons, and each has its bias.
4. Communicate the concept	Verbal description • sketch • photos and renderings • storyboard • video • simulation • interactive multimedia • physical appearance models
5. Measure response	Measure preferences among alternative concepts; understand why and how customers respond; attempt to measure purchase intent (likelihood of buying)
6. Interpret the results	Convert stated purchase intent into a forecast of demand, discounting for the well-known optimism bias in stated intent
7. Reflect	Applicable to the future? Considering alternate markets? Was the concept communicated properly or not? Reality vs estimates (sales)?

The verbal concept description — the class example

"The product is a lightweight electric scooter that can be easily folded and taken with you inside a building or on public transportation. The scooter weighs about 25 pounds. It travels at speeds of up to 15 miles per hour and can go about 12 miles on a single charge. The scooter can be recharged in about two hours from a standard electric outlet. The scooter is easy to ride and has simple controls — just an accelerator button and a brake."

Note the structure: what it is → key physical attributes → performance figures → ease of use. If asked to write a concept description, follow this template.

Concept testing vs concept selection — state this distinction: concept testing is based on data gathered **directly from potential customers** and relies to a lesser degree on judgements made by the development team; concept selection is a team judgement exercise against criteria.

Key takeaways to memorise for Session 7

- The three methods and their increasing rigour; AHP is the least arbitrary
- Pugh: eight steps, $+/-$, $\Sigma +$ $\Sigma 0$ $\Sigma -$, five guidelines, why re-run with a new datum
- Three methods of determining weighting factors; objective tree multiplication rule
- Weighted decision matrix layout and the SUV worked numbers
- **AHP: hierarchy, 1-9 scale with meanings, reciprocal rule, normalise-and-average, geometric-mean shortcut, CI/RI/CR formulas, RI table**
- AHP normalisation rule for alternatives (min/value for ↓, value/max for ↑)
- Concept testing: seven steps, five sample-size factors, five survey formats, eight ways to communicate a concept

7.5 Expected Questions — Session 7

Q1. An organisation is selecting a marketing strategy for a new product. Option A: Social Media Ads, Option B: Influencer Marketing, Option C: Trade Shows. Use a weighted decision matrix to evaluate and choose the right option. **[15]**

PYQ

Note: in the original paper the criteria and weights were supplied in Table 2. If the table is given, use the given weights. If not, state your assumed criteria and weights clearly at the start — the instructor confirmed that assumed ratings are acceptable provided they are stated.

Step 1 — State the criteria and weighting factors (assumed here; sum must be 1.0):

Criterion	Weight	Rationale
Cost effectiveness (cost per acquisition)	0.30	Budget is the binding constraint for a new product launch
Reach / audience size	0.25	A new product needs awareness before conversion
Conversion / engagement rate	0.20	Awareness must translate into purchase
Brand credibility built	0.15	Longer-term equity, less urgent at launch
Speed of implementation	0.10	Time to market matters but is least critical
Total	1.00	

Step 2 — Rate each option on a 0-10 scale and compute weighted scores:

Criterion	Weight	A: Social Media Ads		B: Influencer Marketing		C: Trade Shows	
		Rate	Score	Rate	Score	Rate	Score
Cost effectiveness	0.30	8	2.40	6	1.80	3	0.90
Reach / audience size	0.25	9	2.25	7	1.75	4	1.00
Conversion / engagement	0.20	6	1.20	8	1.60	9	1.80
Brand credibility	0.15	5	0.75	8	1.20	9	1.35
Speed of implementation	0.10	9	0.90	6	0.60	3	0.30
Total weighted score		7.50		6.95		5.35	
Rank		1		2		3	

Step 3 — Decision and justification. Option A, Social Media Ads, is selected with a weighted score of 7.50. It wins because it scores highest on the two heaviest criteria — cost effectiveness (0.30) and reach (0.25) — which together account for 55% of the decision weight. Trade shows score best on conversion and credibility but are penalised heavily on cost and reach, the two criteria that matter most at launch.

Step 4 — Sensitivity comment (add this; it shows judgement): the result is not robust. Option B is only 0.55 behind. If brand credibility were weighted above 0.30 — which would be reasonable for a premium or a trust-sensitive product such as a medical device — influencer marketing or trade shows would overtake social media ads. State that the recommendation is conditional on the launch objective being **awareness at low cost**. A practical recommendation is a **phased combination**: social media ads for reach at launch, influencer marketing to build credibility in phase two.

If you have time, add the **objective tree** that produced the weights — root "Successful launch" branching into Cost (0.30), Market impact (0.45, splitting into reach 0.556 and conversion 0.444) and Brand/execution (0.25, splitting into credibility 0.6 and speed 0.4). Showing the derivation of weights rather than asserting them is worth marks.

Q2. Select the best material for a car body panel from Steel, Aluminium and Composite, using the criteria Cost, Strength and Weight. Apply the Analytic Hierarchy Process. (Consistency check may be skipped.) **[15]**

MUST DRAW

This is question 6 from the instructor's own answer key. Her stated reasoning: "Strength is most critical (safety). Weight is next (fuel efficiency, performance). Cost is important but less than the other two." Build your pairwise matrix to reflect exactly that ordering.

Step 1 — Define the goal and draw the hierarchy.

FOCUS: Select the most suitable material for a car body panel

CRITERIA: C1 Cost • C2 Strength • C3 Weight

ALTERNATIVES: A1 Steel • A2 Aluminium • A3 Composite

Draw it as the three-level tree (Fig 7.5 structure).

Step 2 — Criteria comparison matrix [C]. Judgements: Strength is **strongly** more important than Cost (5); Strength is **moderately** more important than Weight (3); Weight is **moderately** more important than Cost (3). Reciprocals fill the lower triangle.

[C]	Cost	Strength	Weight
Cost	1	1/5 = 0.2000	1/3 = 0.3333
Strength	5	1	3
Weight	3	1/3 = 0.3333	1
Column sum	9.0000	1.5333	4.3333

Step 3 — Normalise [C] to give [NormC] (divide each entry by its column sum) and **average the row values** to obtain the weight vector {W}:

[NormC]	Cost	Strength	Weight	Row average = {W}
Cost	1/9 = 0.1111	0.2/1.5333 = 0.1304	0.3333/4.3333 = 0.0769	0.1061
Strength	5/9 = 0.5556	1/1.5333 = 0.6522	3/4.3333 = 0.6923	0.6334
Weight	3/9 = 0.3333	0.3333/1.5333 = 0.2174	1/4.3333 = 0.2308	0.2605
Sum	1.0000	1.0000	1.0000	1.0000

Criteria priority: Strength (0.6334) > Weight (0.2605) > Cost (0.1061) — which matches the stated reasoning. This is where the instructor said you may stop the [C] work and move on.

Step 4 (optional) — Consistency check. $A3 = [C]\{W\} = \{0.3196, 1.9454, 0.7899\}$; $A4 = A3/\{W\} = \{3.0124, 3.0714, 3.0324\}$; $\lambda_{\max} = 3.0387$.

$CI = (3.0387 - 3)/(3 - 1) = 0.01935$; RI for $n = 3$ is **0.52**; $CR = 0.01935/0.52 = 0.0372 < 0.1 \rightarrow$ judgements are consistent.

Step 5 — Rate the alternatives. State your assumed material data:

	Cost (Rs/kg) ↓	Strength (MPa) ↑	Panel mass (kg) ↓
A1 Steel	60	350	12
A2 Aluminium	220	250	7
A3 Composite	450	400	5

Normalise using the instructor's rule — min/value where lower is better, value/max where higher is better:

	Cost ↓	Strength ↑	Weight ↓
Steel	60/60 = 1.0000	350/400 = 0.8750	5/12 = 0.4167
Aluminium	60/220 = 0.2727	250/400 = 0.6250	5/7 = 0.7143
Composite	60/450 = 0.1333	400/400 = 1.0000	5/5 = 1.0000

Step 6 — Synthesise. Alternative value = Σ (normalised rating $\times$ criterion weight):

Alternative	Cost (0.1061)	Strength (0.6334)	Weight (0.2605)	Alternative value	Rank
Steel	0.1061	0.5542	0.1085	0.7688	2
Aluminium	0.0289	0.3959	0.1861	0.6109	3
Composite	0.0141	0.6334	0.2605	0.9080	1

Composite = $(0.1333 \times 0.1061) + (1.0000 \times 0.6334) + (1.0000 \times 0.2605) = 0.0141 + 0.6334 + 0.2605 = \mathbf{0.9080}$

Decision: Composite is selected (0.9080). It scores maximum on both Strength (weight 0.6334) and Weight (0.2605), which together carry 89% of the decision weight. Its very poor cost score (0.1333) barely matters because Cost carries only 0.1061.

Closing observation worth a mark: the answer is highly sensitive to the criteria weights, not to the material data. If Cost were weighted first — as it would be for a mass-market small car rather than a performance or EV application — Steel would win. Always state that the AHP result is a function of the stated priorities, and that the priorities came from the design brief.

Q3. Perform concept screening for four concepts of a portable laptop stand. Assume the stand will be used by professionals who frequently travel. Attack the negatives to formulate improvisation. Use the evaluation criteria: (a) ease of use (b) stability (c) durability (d) aesthetic appeal (e) manufacturability. PROF PYQ MUST DRAW **[15]**

Another of the instructor's own answer-key questions. Her note: "Student answer may be different and acceptable." The method matters, not which concept wins.



Fig 7.8 — The three concepts illustrated in the answer key. Concept A: simple folding multi-position stand. Concept B: solid angled base stand. Concept C: multi-joint fully adjustable stand.

Step 1 — Clarify the concepts.

Concept A	Simple folding laminar stand — light, flat-packing, few parts (chosen as DATUM)
Concept B	Solid angled base with integrated support — sleek, single-piece, fixed angle
Concept C	Multi-joint fully adjustable stand — height and tilt adjustable, many joints
Concept D	Inflatable / collapsible fabric-and-frame stand — packs to near zero volume (added as a fourth concept)

Step 2 — Run the Pugh matrix with Concept A as datum:

Criterion	A (Datum)	B	C	D
Ease of use	0 (simple foldable)	+ (compact, quick setup)	– (multi-joint, more complex to adjust)	– (inflation/assembly step needed)
Stability	0 (basic stability)	+ (solid base, angled support)	+ (adjustable joints allow stable positioning)	– (flexible structure, less rigid)
Durability	0 (light frame, moderate)	+ (solid construction, fewer moving parts)	– (multiple joints prone to wear)	– (fabric/bladder vulnerable to puncture)
Aesthetic appeal	0 (functional, minimal)	+ (sleek integrated look)	0 (functional but bulky)	– (utilitarian appearance)
Manufacturability	0 (simple design, easy to produce)	– (complex base, higher cost)	– (multi-joint, precision manufacturing needed)	+ (few rigid parts, low tooling cost)
$\Sigma +$	—	4	1	1
$\Sigma 0$	5	0	1	0
$\Sigma -$	—	1	3	4
Net ($\Sigma+ - \Sigma-$)	0	+3	-2	-3
Rank	2	1	3	4

Step 3 — Evaluate the ratings. Following the answer key's own reading: Concept A is the neutral reference; **Concept B has 4 positives and 1 negative → strong candidate**; Concept C has mixed performance; Concept D is weakest overall but has one distinctive strength.

Step 4 — Attack the negatives (this is what the question specifically asks for):

Concept	Negative to attack	Improvisation
B (leading concept)	Manufacturability — — complex base, higher cost	Replace the machined/moulded solid base with a single folded sheet-metal or extruded aluminium profile with a bent foot. Use Principle 1, Segmentation — split the base into two identical mirrored extrusions so one die serves both, halving tooling cost. Borrow Concept D's low part count.
C	Ease of use — — multi-joint, complex to adjust	Replace continuous joints with discrete detented positions (3 or 4 clicks) so the user selects rather than adjusts. Borrow Concept A's simple foldable logic.
C	Durability — — joints prone to wear	Reduce the joint count from four to two; use self-lubricating polymer bushes at the pivots and a friction hinge rated for the expected cycle count.
C	Manufacturability —	Standardise on one hinge part number used at every joint — enabling standardisation in architecture terms.
D	Stability — and Durability —	Add a rigid spine to the collapsible structure — a hybrid of A and D. This keeps the packing advantage while restoring rigidity.

Step 5 — New datum and re-run. Make **Concept B the new datum** and re-run. The purpose, per the slides, is **not to verify round 1 but to gain more insight** — with B as reference, C and D are judged against a genuinely good design and their remaining relative strengths become visible (C's adjustability, D's packed volume).

Step 6 — Recommendation. Carry forward **Concept B improved by the Step 4 attack** — a solid angled base re-engineered as a two-part folded extrusion for manufacturability, with Concept A's flat-fold feature added for travel. For frequent travellers the decisive attributes are packed size and setup time, so add **packed thickness** and **setup time** as criteria in the next round — noting that if most concepts score the same on a criterion, the criterion set needs revision.

Q4. Explain the concept testing process with a neat flow chart. How does concept testing differ from concept selection? What are the factors that determine the survey sample size? [15] **MUST DRAW**

- Draw the seven-step flow chart (box in section 7.4).
- Expand each step using the table in section 7.4 — particularly the four purpose questions, the five survey formats, and the eight ways to communicate a concept.
- **Difference from concept selection:** concept testing solicits a direct response **from potential customers in the target market** and relies to a lesser degree on team judgement; concept selection is an internal team evaluation against criteria. Concept testing can **verify** that customer needs have been adequately met, **assess sales potential**, and **gather information for refining** the concept.
- **Five factors determining sample size:** (1) the stage of product development (2) cost to conduct the survey (3) nature and intent of the survey (4) budget of the development project (5) how possible it is to collect the intended information.
- Note that concept testing is appropriate at **several points** in the process: when identifying the original product opportunity; when selecting between two or more concepts; when assessing sales potential; and when deciding whether to continue development and commercialisation.
- Add the electric scooter verbal description as an example of Step 4.

Q5. Compare the Pugh concept screening method with the weighted decision matrix and AHP. When would you use each? [15] **HIGH**

Aspect	Pugh screening	Weighted decision matrix	AHP
Comparison basis	Relative to a datum	Absolute rating against a scale	Pairwise comparison
Scale	+ / 0 / –	0-5 or 0-10	1-9 with reciprocals
Weights	None — criteria treated equally	Assigned directly or via objective tree	Derived from pairwise comparison
Arbitrariness	Highest	Moderate	Lowest
Consistency check	None	None	CI / RI / CR available
Effort	Minutes	Moderate	$n(n-1)/2$ comparisons — highest
Number of concepts	Many (10-20) — a filter	Few (3-6)	Very few (2-4)
When to use	Early , to narrow a large concept set quickly and to generate hybrids by attacking negatives	Middle , once concepts are clarified and criteria can be weighted by judgement	Late or high-stakes , for a small set where criteria are non-commensurate, qualitative and quantitative mixed, or where the decision must be defensible and documented

The point to conclude with: these are not competing alternatives but a **sequence**. The recommended practice is a **two-stage process — screening then scoring**: use Pugh to reduce twenty concepts to three or four and to generate improved hybrids, then use the weighted decision matrix or AHP to make the final choice among the survivors. Applying AHP to twenty concepts would be prohibitively laborious; applying Pugh alone to a final high-stakes decision would be insufficiently rigorous and would leave no documented weighting rationale.

Session 8 — Product Architecture

8.1 Where it sits — the Preliminary / System-Level / Embodiment Design phase

The preliminary design phase is sub-divided into three activities:

(1) Product architecture	Setting the arrangement of the physical elements of the design into modules
(2) Configuration design	Designing special-purpose parts and the selection of standard components
(3) Parametric design	Determining exact values, dimensions or tolerances of the components/features that are critical-to-quality

8.2 What is Product Architecture?

Definition to quote

Product architecture is the arrangement of the physical elements of a product to carry out its required functions — equivalently, the scheme by which the functional elements of the product are arranged into physical chunks.

- A **mapping from product function to product form**
- Layout and architecture of the product and **their interfaces** (best possible architecture selection)
- Rough sketches of concepts; proof-of-concept model
- Key decisions on how the product will **physically operate**
- A key milestone for any class of products

The architecture is established during the concept development and system-level design phases of development.

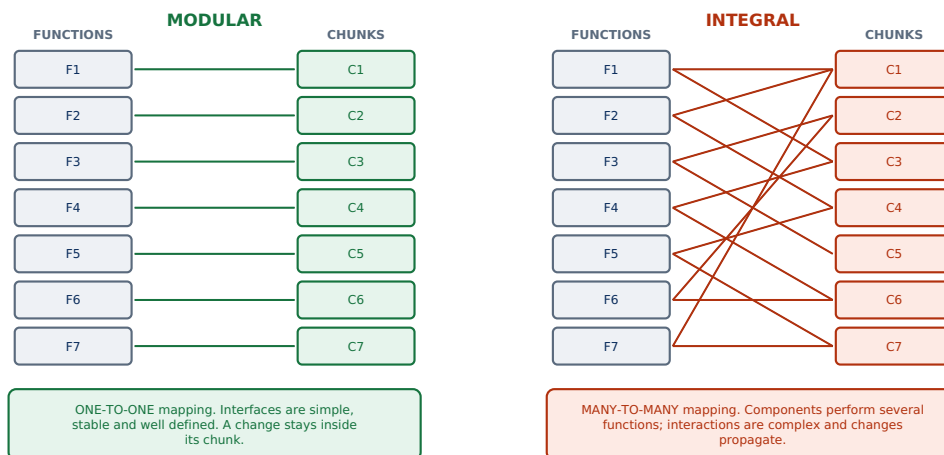
Implications of architecture — six factors **HIGH**

How much modularity to impose on the architecture depends upon:

Product change	e.g. product improvement programme — how easily can the product be upgraded in future?
Product variety	e.g. car model variants — how many versions must be offered?
Component standardisation	e.g. common components across multiple products
Product performance	Integral designs generally perform better for a given mass/cost
Manufacturability and reparability	e.g. fail-safe engine mount; reliability, repairability, maintainability, poka-yoke / foolproofing
PD management	e.g. team coordination, supplier engagement — managed through PLM

8.3 Integral vs Modular Architecture **MUST DRAW** **HIGH**

Modular vs integral architecture — the mapping test



The crossing lines ARE the definition — draw them; this is what the examiner is looking for

Fig 8.1 — Modular vs integral architecture: the mapping test

Aspect	Modular architecture	Integral architecture
Function-component mapping	One-to-one — each chunk implements one or a few functional elements	Many-to-many — components perform multiple functions
Number of modules	Several distinct building blocks	Only one or a few modules; no individual isolated components
Interfaces	Simple, stable and well defined (e.g. wear interfaces)	Complex interactions , not clearly bounded
Effect of a change	Contained within the chunk — easy design change	Changes to any component tend to propagate to many others
Development	Chunks designed independently and concurrently → short PD cycle	Tightly coupled; must be developed together
Component count	Higher	Fewer components → cost benefit
Performance	Lower for a given mass/cost (interfaces cost weight and space)	Higher performance ; suited to high-volume products
Typical example	Desktop computer (separate CPU, monitor, keyboard, mouse, speakers); Mecano/Meccano sets; Bosch handheld power tool family	Bicycle brake and shift controls integrated in one lever; aircraft engine main block; a multi-function spanner

Pros and cons

	Pros	Cons
Modular	Improves device re-configurability • increases device variety and speed of introduction for new devices • improves maintainability and serviceability • decouples development and manufacturing tasks	May make devices look similar • reduces device performance • more expensive
Integral	Harder for a competitor to copy the design • tighter coupling of teams with fewer interface problems • possible reduction in system cost	Hinders changes of design in production • reduces the variety of devices that can be produced

Further benefits of modular architecture

- Helps to develop a **product family**
- Modular products make **economic sense**
- Easier **assembly and disassembly**, modification and reassembly
- **Standardisation** of components
- **Re-configurability** of devices
- The product can be adapted to the needs of different customers by **adding or deleting modules**

Caution noted in the slides: difficulties may be faced in the **initial stages**, because the compatibility of the modules in the first product must be established with those in subsequent products. Once established, evolution becomes easy.

8.4 Types of Modularity — the Taxonomy MUST DRAW HIGH

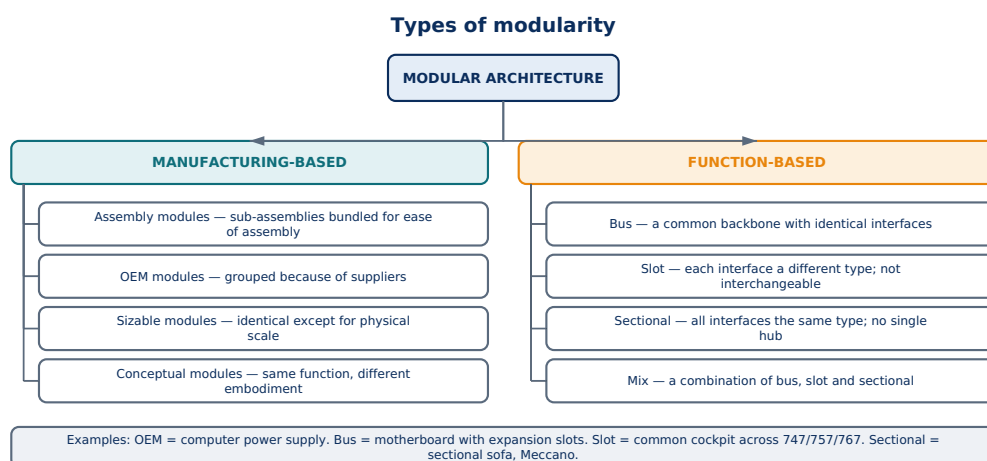


Fig 8.2 — Types of modularity

(A) Manufacturing-based modularity

Type	Definition and example
OEM modules	A group of components or functions grouped simply because of suppliers . Suppliers can provide them at less expense than could be developed and manufactured in-house. <i>Example: power supplies in computers, bought as commodity items; hard disks.</i>
Assembly modules	Modules or components that solve unrelated functions but are bundled to increase assembly ease . Typically called " sub-assemblies " in a manufacturing environment.
Sizable modules	Components that are exactly the same except for their physical scale . <i>Example: cutters of different sizes.</i>
Conceptual modules	Modules that solve the same function (plugging a hole, for example) but have different physical embodiments (shapes) .

(B) Function-based modularity

Type	Definition	Example from class
Bus	A common bus or backbone to which modules of different types attach through identical interfaces	The bus structure at the bottom of a modular table onto which sections are built; a computer motherboard with expansion slots
Slot	Each interface between chunks is of a different type , so chunks cannot be interchanged; a defined slot exists for each	The Boeing common nose/cockpit section fitted onto the 747, 757 and 767 — the same portion acts as a slot module across several aircraft
Sectional	All interfaces are of the same type and there is no single element to which all others attach; the assembly is built by joining sections together	A sectional table — storage section, support section and top section joined to form the final product; sectional sofas; Meccano
Mix	A combination of bus, slot and sectional types within one product	Most real complex products, e.g. Bosch handheld drilling/grinding machine family

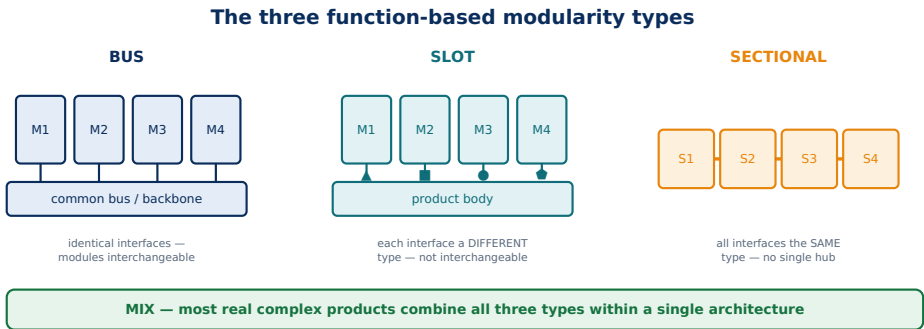


Fig 8.3 — The three function-based modularity types

Worked classification example from class — the table: a single moulded plastic one-piece table is integral; a table built onto a bottom bus structure is bus modular; a table made of separate storage, support and top sections joined together is sectional modular; a table whose top/legs are bought as sub-assemblies is assembly/OEM modular.

Function-based modularity — the trailer example (Ulrich)

Reference: Karl Ulrich, *The role of product architecture in the manufacturing firm*, Research Policy 24 (1995) 419–441.

8.5 Considerations at Product Architecting

Four questions the team must answer:

1. How will it affect the ability to offer **product variety**?
2. How will it affect the **product cost**?
3. How will it affect the **design lead time**?
4. How will it affect the **development process management**?

8.6 The Architecture Design Process — Four Steps MUST DRAW HIGH

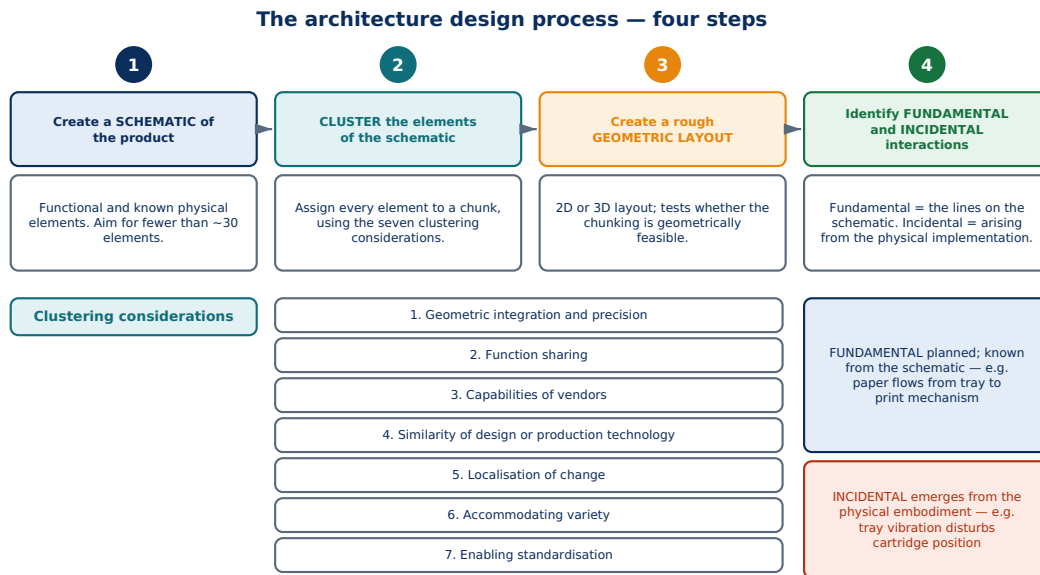


Fig 8.4 — The architecture design process and the clustering considerations

1. **Create a schematic of the product**
2. **Cluster the elements of the schematic**
3. **Create a rough geometric layout**
4. **Identify the fundamental and incidental interactions**

Step 1 — Create a schematic

The schematic is a diagram representing the team's understanding of the state of the product, showing both **functional elements** and **physical elements** (where key physical components are already known). It is essentially the **black box / function structure** from Session 5, extended with the physical elements already committed.

Rule of thumb given: aim for roughly **fewer than 30 elements** in the schematic for a product of modest complexity. A good rule is to use no more than about 30 elements in the schematic.

Step 2 — Cluster the elements into chunks

The goal of clustering is to assign each element of the schematic to a chunk. **Clustering is generally done based on the following considerations** — learn all five:

Consideration	Explanation
Geometric integration and precision	Assigning elements to the same chunk allows a single individual or group to control the physical relationships among those elements. Elements requiring tight geometric integration or precise location should be clustered.
Function sharing	When a single physical component can implement several functional elements, those elements belong in the same chunk. This is where a degree of integral architecture enters a mostly modular design.
Capabilities of vendors	A trusted vendor may have specific capabilities related to a project; to best take advantage of those, the team may cluster the elements about which the vendor has expertise into one chunk. <i>(In the Deskjet case an internal team did most of the engineering, so this was not a major consideration.)</i>
Similarity of design or production technology	When two or more functional elements are likely to be implemented using the same design and/or production technology, incorporating them into the same chunk allows more economical design and production. A common strategy is to combine all functions likely to involve electronics into the same chunk — allowing them all to be implemented on a single circuit board .
Localisation of change	When a team anticipates a great deal of change in some element, it makes sense to isolate that element into its own modular chunk , so required changes can be carried out without disrupting other chunks. Design-for-environment principles may support localisation-of-change clustering.
Accommodating variety	Elements should be clustered to enable the firm to vary the product in ways valuable to customers. <i>Example: the Deskjet printer was sold worldwide in regions with different electrical power standards, so the team created a separate chunk for the element supplying DC power.</i>
Enabling standardisation	If a set of elements will be useful in other products, they should be clustered into a single chunk, allowing the physical elements of that chunk to be produced in higher quantities .

Step 3 — Create a rough geometric layout

A two- or three-dimensional layout showing where each chunk physically sits. It tests whether the chunking is geometrically feasible and exposes packaging conflicts early.

Step 4 — Identify fundamental and incidental interactions HIGH

Interaction type	Definition and example
Fundamental interactions	Those corresponding to the lines on the schematic that connect the chunks to one another. <i>Example: a sheet of paper flows from the paper tray to the print mechanism.</i> This interaction is planned , and should be well understood even from the very earliest schematic, as it is fundamental to the system's operation.
Incidental interactions	Those that arise because of the particular physical implementation of functional elements, or because of the geometric arrangement of the chunks. <i>Example: vibrations induced by the actuators in the paper tray could interfere with the precise location of the print cartridge in the x-axis.</i>

Incidental interactions are the reason system engineering methods and special attention are needed — they are not visible from the function structure and only emerge from the physical embodiment. Typical incidental interactions to name in an answer: thermal, vibration, EMI/electrical noise, acoustic, and geometric interference / tolerance stack-up.

8.7 Fundamental Decisions and Practical Concerns

Five fundamental decisions:

1. Integral vs modular architecture?
2. What type of modularity?
3. How to assign functions to chunks?
4. How to assign chunks to teams?
5. Which chunks to outsource?

Function → Decomposed detail functions → Geometric layout → Interactions → Team and interaction → System level design

Practical concerns:

- **Planning is essential** to achieve the desired variety and product change capability
- **Coordination is difficult**, particularly across teams, companies or great distances
- **Special attention must be paid to handle complex interactions between chunks** (system engineering methods)

The team structure should mirror the chunk structure — a mechanical chunk team and an electrical chunk team, with a defined interface owner where the two chunks interact. This is the architecture-organisation link back to Session 2.

8.8 Delayed Differentiation (Postponement) HIGH

Definitions to quote exactly

- The **point of product differentiation** is the point in the manufacturing process where an item is limited to use for a **single specific product** — where a product can only be made into a specific SKU.
- Delaying that point is called **Late Point Product Differentiation**.
- **Delayed differentiation or postponement** is a concept in supply chain management where the manufacturing process starts by making a **generic or family product** that is later differentiated into a specific end-product. It is widely used in industries with **high demand uncertainty**, and can be effectively used to address final demand **even if forecasts cannot be improved**.

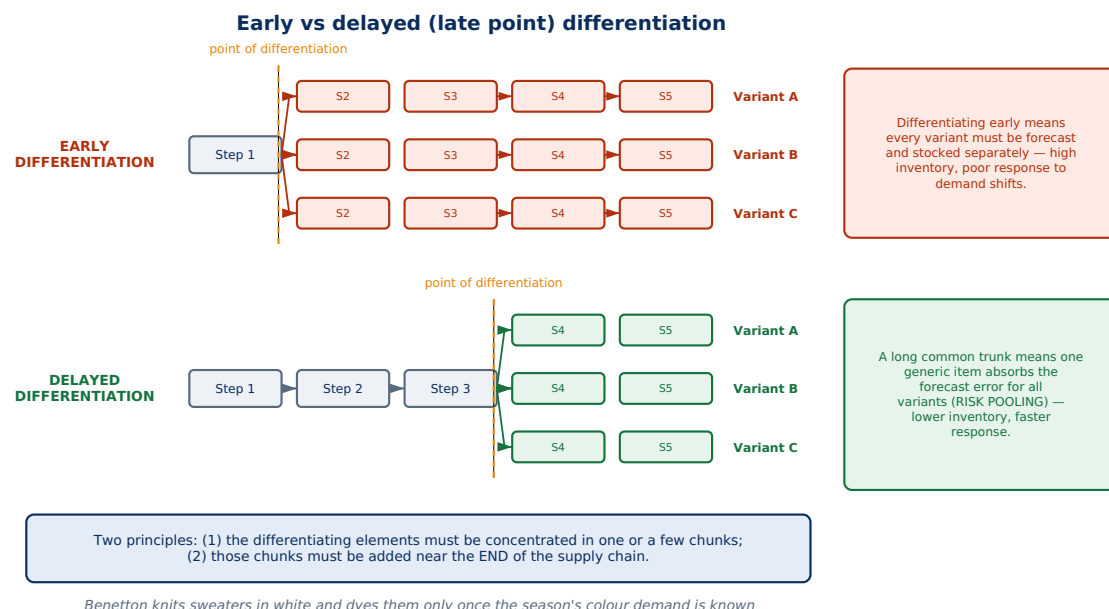


Fig 8.5 — Early vs delayed (late point) differentiation

The classic example

Benetton and their knitted sweaters: initially all knitted in **white**, then dyed into different colours only when the season's customer colour preference/demand is known. It is usually necessary to **redesign the products specifically for delayed differentiation**, and to **re-sequence** to modify the order of product manufacturing steps.

Further examples from the slides

Paint	Paint may be offered in hundreds of colours. Producing these at the factory tends to lead to overproduction of some colours and underproduction of others. Paints are therefore formulated from three or more base mixes , and the final colour is produced at the distributor or retailer .
Automotive	Manufacturers mass-produce base models and add minor customisation when the car is actually ordered. Customisation such as audio systems may be installed at the dealership . (The paint shop is also a classic delayed-differentiation point.)
Fresh packaged foods	Grocery stores run their own kitchens producing fresh foods such as salads and precooked entrees in the store . Local production reduces waste as the store creates things in small batches that typically sell.
Fashion	Clothing companies produce items in white and do not perform the final colouring until they know what colours are selling. Base items may be manufactured globally and colouring steps may occur at regional locations .

Advantages and principles

Advantages of late point differentiation

- **Reduced inventories** — one generic item replaces many SKUs
- **More easily respond to demand variation**
- Easier to control
- Faster reaction to customer requirement
- Lower inventory costs
- Fewer interfaces
- Risk pooling — forecast error on the aggregate is smaller than on individual variants

Late point differentiation principles

1. The **differentiating elements** of the product must be **concentrated in one or a few chunks**
2. The product and production process must be designed so that the **differentiating chunk(s) can be added to the product near the end of the supply chain**

Note the direct dependency: **modular design is what allows late point differentiation**. If the differentiating features were smeared across an integral architecture, they could not be deferred.

8.9 Platform Planning

Platform planning is the process of deciding **what should be shared across products and what should be unique across products**. *Example posed in class: how many driveshafts should you have for a Ford F150?*

It attempts to resolve the tension between:

1. **Differentiating** the product for various customers
2. Taking advantage of the **economic benefits of using common components**

Product architecture will determine what trade-offs are available. If no good options are available, see if the options can be improved by changing the product architecture.

Platform planning — commonality vs differentiation

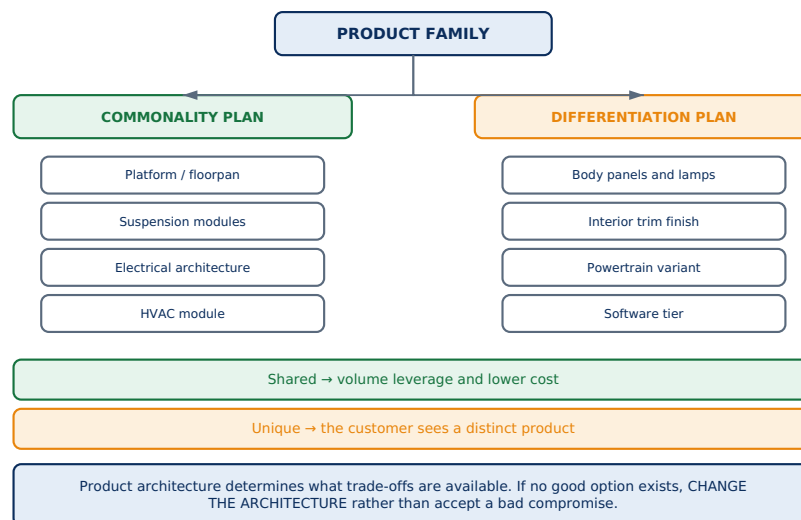


Fig 8.6 — Platform planning: commonality plan vs differentiation plan

Key takeaways to memorise for Session 8

- Preliminary design = architecture + configuration + parametric
- Definition of product architecture; six implications
- **Modular vs integral — the F1-F7 / C1-C7 mapping diagram, and the pros/cons table**
- **The modularity taxonomy tree:** manufacturing-based (assembly, OEM, sizable, conceptual) and function-based (bus, slot, sectional, mix), with an example each
- Four considerations at product architecting
- **Four-step architecture design process**, with the five to seven clustering considerations
- **Fundamental vs incidental interactions**, with the paper-tray/print-cartridge example
- Five fundamental decisions; three practical concerns
- **Delayed differentiation:** definition, Benetton, four industry examples, advantages, two principles
- Platform planning — differentiation plan vs commonality plan

8.10 Expected Questions — Session 8

Q1. Can a company offer a wide range of automobile variants without adopting modular product design? Why or why not? Discuss how modularity impacts cost, customisation and manufacturing efficiency. PYQ MUST DRAW [15]

Direct answer first: Technically **yes, but not economically or competitively**. A company *can* produce many variants from integral architectures — by designing and tooling each variant essentially as a separate product — but the cost, lead time and inventory penalties make it unsustainable in a competitive automobile market. In practice, wide variety without modularity is not viable.

Step 1 — Draw the modular vs integral mapping diagram (Fig 8.1) and define both.

Step 2 — Why integral architecture blocks variety. Use the six implications:

Mechanism	Consequence for an integral-architecture automobile
Change propagation	Changes to any component propagate to many others. Offering a new variant means re-engineering a large portion of the vehicle, not a chunk.
Tooling and investment	Each variant needs its own dies, fixtures and assembly line balance. Capital cost scales roughly with the number of variants instead of being amortised.
Component standardisation	No common components across the range → no volume leverage with suppliers → higher piece cost.
Design lead time	Chunks cannot be developed concurrently and independently ; PD cycle time rises per variant.
Manufacturability and serviceability	Variants cannot share a line or a service parts catalogue; aftermarket inventory multiplies.
Delayed differentiation	Impossible — differentiating features are smeared across the whole product, so the SKU is fixed early and inventory must be held as finished variants.

Step 3 — How modularity resolves each of these. Structure it under the three headings the question names:

Dimension	Impact of modularity
Cost	Component standardisation — one platform, one powertrain family, one HVAC module across many models; volume per part number rises and unit cost falls
	Development cost amortised — a platform costs 2x-10x a derivative, so each additional variant is cheap (the platform planning multiplier from Session 2)
	Lower inventory cost via postponement — hold generic sub-assemblies, not finished variants
	But: modularity is more expensive per unit at the component level — interfaces, connectors and fasteners add mass, space and cost that an integral design avoids. It also reduces peak performance . This is the genuine trade-off to acknowledge.
Customisation	The product can be adapted by adding or deleting modules — trim levels, infotainment tiers, powertrain options, drive configurations
	Late point differentiation becomes possible: base models are mass produced and audio systems or accessory packs are fitted at the dealership ; paint is applied late
	But: heavy module sharing may make devices look similar — the brand-dilution risk when several models are visibly the same car. Manufacturers counter this by keeping styling chunks (skin panels, lamps, interior trim) deliberately unique while sharing the hidden chunks
Manufacturing efficiency	Assembly modules / sub-assemblies can be built offline in parallel and married to the line, shortening line time
	OEM modularity — suppliers deliver complete modules (seats, cockpit, front-end carrier) more cheaply than in-house, and take on their own sub-tier management
	Decouples development and manufacturing tasks ; mixed-model assembly on one line becomes feasible; serviceability and repair improve because a faulty module is replaced rather than diagnosed to component level

Step 4 — Give the concrete automotive illustration. Modern platform strategies (a shared skateboard/platform with common floorpan, wheelbase-variable front and rear modules, and shared electrical architecture) let one platform produce hatchback, sedan, SUV and EV variants. The **differentiation plan** keeps body panels, interior trim and software unique; the **commonality plan** shares floorpan, suspension modules, HVAC and electrical architecture — exactly the printer-family logic in Fig 8.6.

Step 5 — Balanced conclusion (write this; it separates a good answer from an average one). The real architecture of a modern vehicle is **mixed, not purely modular**. Modularity is applied where variety, change and supplier leverage matter (interiors, infotainment, powertrain mounting, accessories); integral design is retained where performance, cost and copy-protection matter (body-in-white structure, crash load paths, engine block). So the correct answer is not "modularity is better" but **"architecture is chosen per chunk, against the six implications, and wide variety at acceptable cost requires modularity in the differentiating chunks."**

Q2. Explain the architecture design process with a neat schematic. Illustrate the four steps using a product of your choice, and distinguish between fundamental and incidental interactions. MUST DRAW HIGH [15]

- Draw the four-step flow (Fig 8.4).
- **Step 1 — schematic.** Draw it for your chosen product. Using an **inkjet printer** as in the slides: elements are supply paper, position paper in x-axis, position print cartridge in y-axis, store output, deposit ink on paper, store ink supply, supply DC power, communicate with host, control printer, enclose printer, display status.
- **Step 2 — cluster.** Group into chunks: print mechanism chunk, paper tray chunk, print cartridge chunk, power supply chunk, chassis/enclosure chunk, logic board chunk. State **which clustering consideration** drove each grouping — this is where the marks

are:

- Print mechanism + cartridge positioning → **geometric integration and precision**
- All electronic functions on one board → **similarity of design/production technology**
- DC power supply as its own chunk → **accommodating variety** (different regional electrical standards)
- Cartridge as its own chunk → **localisation of change** and **enabling standardisation**
- Logic board bought from a specialist → **capabilities of vendors**
- **Step 3 — rough geometric layout.** Draw the chunks in approximate physical position and note any packaging conflicts.
- **Step 4 — interactions.** Give both definitions and both examples from the table in section 8.6. Draw the interaction graph with **solid lines for fundamental** and **dashed for incidental**.

	Fundamental	Incidental
Origin	Planned — the lines on the schematic	Emerges from the physical implementation and geometric arrangement
Visibility	Known from the earliest schematic	Only discovered once chunks are embodied and laid out
Printer example	Paper flows from the paper tray to the print mechanism	Vibration from the paper tray actuators interferes with x-axis location of the print cartridge
Typical categories	Material, energy and signal flows between chunks	Thermal, vibration, EMI, acoustic, geometric interference / tolerance stack-up
How managed	Interface specification at the chunk boundary	System engineering methods , tolerance analysis, CAE, integration testing

Close with the practical concerns from section 8.7 and the team-chunk mirroring point.

Q3. Classify the types of modularity with a neat tree diagram. Explain each type with a suitable example. [15]

MUST DRAW

- **Draw the taxonomy tree** (Fig 8.2): Modular architecture splitting into manufacturing-based (assembly modules, OEM modules, sizable modules, conceptual modules) and function-based (bus, slot, sectional, mix).
- Use the two tables in section 8.4, with the class example against each type.
- Use **one single product family** to illustrate all of them — this is what turns a list into a good answer. The **modular table** works well: sectional (storage/support/top sections joined), bus (components built onto a bottom bus rail), OEM (bought-in castors and fasteners), assembly (pre-assembled drawer unit), sizable (the same leg profile cut to three heights), conceptual (three different foot designs all solving "stand on uneven floor").
- Close by noting that the concepts of integral and modular **apply at several levels** — system, sub-system and component (Fig 8.2) — so a product can be modular at system level and integral within a chunk.

Q4. What is delayed differentiation? Explain its principles and advantages with suitable industrial examples. How is it enabled by product architecture? [15]

- Give the three definitions verbatim from the box in section 8.8 — point of product differentiation, late point product differentiation, postponement.
- **Draw the two process diagrams side by side** (Fig 8.2): early differentiation branching near the start vs delayed differentiation with a long common trunk and a late fan-out. Mark the **point of differentiation** on each.
- Give the **Benetton** example in full, then the four industry examples (paint, automotive, fresh packaged foods, fashion).
- State the **two principles**: differentiating elements concentrated in one or a few chunks; the process designed so those chunks are added near the end of the supply chain.
- List the advantages, and explain the mechanism behind the headline one: **risk pooling** — a forecast of total demand across variants is far more accurate than forecasts of each variant, so holding generic inventory requires far less safety stock for the same service level. This is why postponement works "even if forecasts cannot be improved."
- **Link to architecture (the last part of the question)**: postponement is only possible if the architecture is **modular** in the differentiating features. If the variant-defining features are distributed across an integral design, the SKU is determined early and cannot be deferred. Hence the slide's statement that **modular design allows late point differentiation**, giving easier control, faster reaction to customer requirements, lower inventory costs and fewer interfaces. Note also that products usually must be **redesigned specifically** for delayed differentiation, and manufacturing steps **re-sequenced**.

Q5. What is platform planning? Explain the tension it resolves, and illustrate a differentiation plan and a commonality plan for a product family. [15]

- Definition: deciding what should be **shared** across products and what should be **unique**. Quote the Ford F150 driveshaft question.
- **The tension**: differentiating the product for various customers vs the economic benefits of common components. Neither extreme works — full commonality means products that customers cannot tell apart (the "may make devices look similar" con of modularity), full uniqueness means no volume leverage and unaffordable development cost.
- Build the two tables for a family of three printers (Fig 8.6), or for a vehicle family:

Chunk	Commonality plan (shared)	Differentiation plan (unique)
Platform / floorpan	Shared across all three models	—
Suspension modules	Shared, with spring/damper tuning variation	Tuning only
Powertrain	Common mounting interface	Engine/motor variant per model
Electrical architecture	Shared harness backbone and ECUs	—
Body panels and lamps	—	Unique to each model — carries the brand identity
Interior trim	Common substrate	Unique finish and colourway per trim level
Infotainment	Common hardware slot	Software tier per model

- **The key statement to close with:** product architecture determines **what trade-offs are available** — and if no good options exist, the answer is to **change the architecture** rather than to accept a bad compromise. Concentrating the differentiating content in a few visible chunks is what lets a firm be simultaneously distinctive to the customer and common in the factory.

Appendix A — Past Question Papers

A.1 Mid-Semester Test (EC-2 Regular), 15 March 2026

Closed Book • Weightage 30% • Duration 2 Hours • Six questions of 15 marks each

Note to students (printed on the paper)

1. Write all the answers precisely and to the point.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. **Assumptions made, if any, should be stated clearly at the beginning of your answer.**

Q	Question	Marks	Session
1	Explain the key stages of the product development with a neat block diagram. "Iterative product development has become increasingly dominant in technology firms over the past decade." — True or False? Justify your answer with examples.	15	Session 1
2	Explain the Subtract and Operate (SOP) analysis for identification of weaknesses in product design. What are its shortcomings in terms of innovation and the emergence of new functions? Illustrate with a practical example.	5+5+5	Session 5
3	Prepare a detailed House of Quality (HOQ) to prioritise the engineering characteristics for the development of the automotive front bumper for the passenger vehicle. The customer requirements have already been identified and listed in Table 1 along with their importance ratings. Also discuss any two possible outcomes from this HOQ study.	15	Session 4
4	An organisation is working on selecting a marketing strategy for a new product. They have considered Option A: Social Media Ads, Option B: Influencer Marketing and Option C: Trade Shows for promoting the product. Use a weighted decision matrix to evaluate and choose a right option. Consider criteria along with its weight as given in Table 2.	15	Session 7
5	Classify the various methods of concept generation. Discuss the limitations of traditional brainstorming techniques. How can structured approaches overcome these limitations?	15	Session 5
6	Can a company offer a wide range of automobile variants without adopting modular product design? Why or why not? Discuss how modularity impacts cost, customisation, and manufacturing efficiency.	15	Session 8

Table 1 — Customer requirements for the automotive front bumper (supplied with Q3):

SI No	Customer requirement	Importance (1-5)
CR1	High impact resistance	5
CR2	Lightweight	4
CR3	Low manufacturing cost	4
CR4	Corrosion resistance	3
CR5	Aesthetic surface finish	3
CR6	Easy assembly	4

Table 2 (criteria vs weight for Q4) was supplied as an image in the original paper. The worked solution in Session 7 uses a stated assumed set of criteria and weights — which is exactly what note 3 above permits.

What this paper tells you about the exam's shape

The six questions map **one to one onto the five content blocks** the instructor named, with two questions drawn from the concept generation block:

Q1 → Block 1 Planning • Q3 → Block 2 Specification • Q2, Q5 → Block 3 Concept Generation • Q4 → Block 4 Concept Evaluation • Q6 → Block 5 Architecture

Three of the six questions require a **matrix or block diagram** as the main deliverable (Q1, Q3, Q4). Two require **classification plus critique** (Q2, Q5). One is a **reasoned argument** (Q6). Practise all three answer shapes.

A.2 End-Semester Test (EC-3 Regular), 24 May 2026 — for reference only

Open Book • Weightage 40% (120 marks) • Duration 2½ Hours • Eight questions. **Most of this paper is beyond the mid-semester scope** (FMEA, reliability, make/buy, manufacturing process selection). It is reproduced because two questions fall squarely within Sessions 1–8 and are good practice.

Q	Question	In mid-sem scope?
1	Some companies have reportedly abandoned the practice of doing a customer test with early prototypes, preferring instead to go directly and quickly to market in order to observe the actual customer response. For what types of products and markets might this practice make sense? Justify.	Yes — Session 7, concept testing
2	Design a collaborative workspace for remote teams that reduces digital eye strain and repetitive hand movements. What ergonomic and human-centred strategies would you employ to balance engagement, efficiency, and health?	Partly — industrial design, Session 1
3	For a metal soda can, develop a Material-Energy-Toxicity Matrix.	No — later sessions
4	List the various factors that influence manufacturing process selection. How to quantify component shape complexity.	No
5	List down major factors in a Make/Buy decision, and decide make-vs-buy from the given cost data.	No
6	OEMs aim to reduce vehicle mass to improve fuel economy / EV range, but lighter structures often reduce crash energy absorption and occupant protection. Suggest possible solutions using the TRIZ approach. Discuss the benefits of the TRIZ method.	Yes — Session 6, worked in full in these notes
7	What is Failure Modes and Effects Analysis? Develop an FMEA for mobile phones.	No
8	Reliability of a ten-component series module; MTTF; availability.	No

Appendix B — The Instructor's Own Answer Key (EC-2 Make-up)

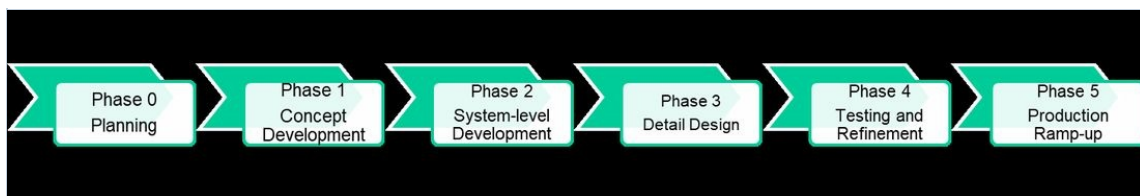
This is the make-up paper for the same mid-semester test, released **with the instructor's answers and diagrams**. It is the single best guide to what a full-mark answer looks like. All six questions are worked in the session notes above; this appendix reproduces her diagrams together so you can revise them as a set.

Q	Question	Worked in
1	Explain the difference between traditional product development and the concurrent engineering approach. Why is a concurrent engineering team best suited for the modern product development process in today's competitive world?	\$2.5 Q1
2	In recent days, designers have been paying more attention to ethnographic studies. Why is this trend significant? In which ways can ethnographic studies enhance creativity in product development?	\$3.11 Q2
3	What are the common issues with traditional customer surveys? List down the various quality attributes of the Kano model. Discuss possible methods for prioritising customer requirements.	\$3.11 Q1
4	What is functional decomposition? Why is it essential for concept generation? The designer would like to modify the electric kettle to be more compact, lightweight and aesthetically pleasing, while minimising heat loss. Draw a generic black-box function structure for an electric kettle.	\$5.9 Q3
5	Perform concept screening for four concepts of a portable laptop stand... Attack the negatives to formulate improvisation.	\$7.5 Q3
6	Select the best material for the car body panel. Apply AHP. (Note: students can skip the consistency check process.)	\$7.5 Q2

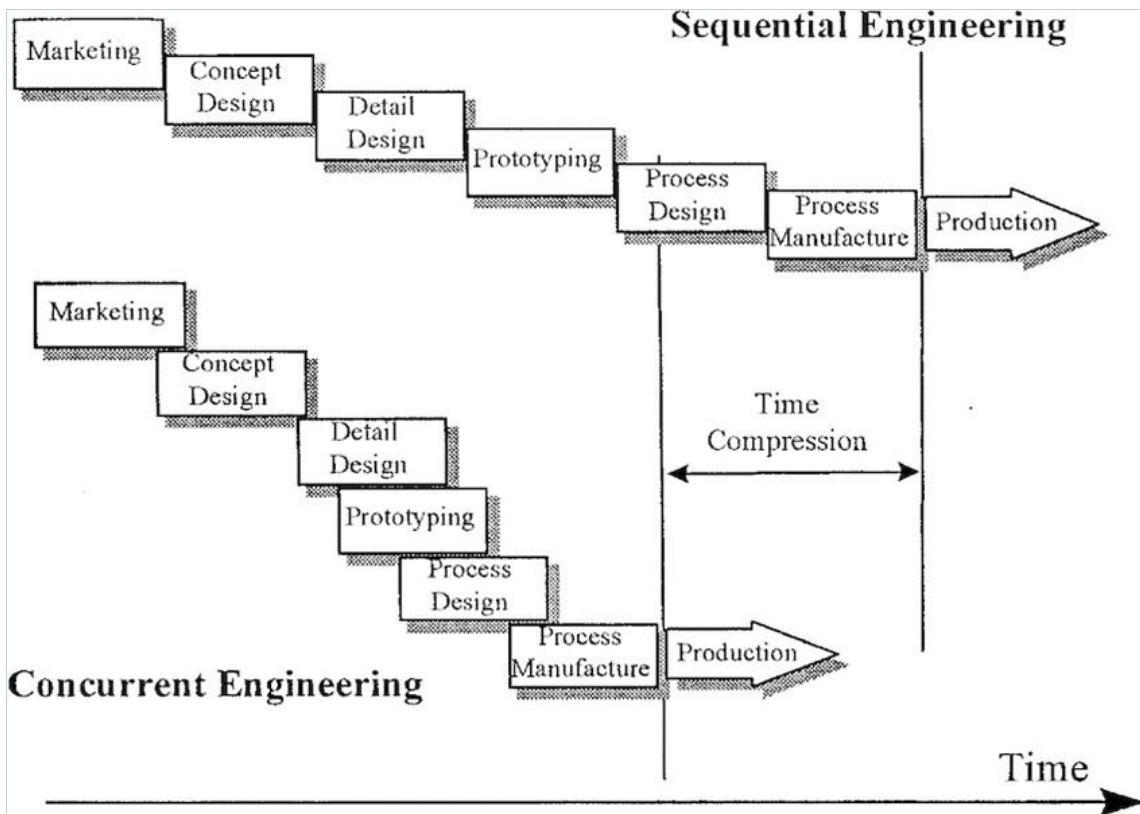
Two things her key says explicitly — remember both

- On the Pugh question: **"Student answer may be different and acceptable."** The method and the attack-the-negatives reasoning are what is graded, not which concept wins. (Her own text summary reads Concept B as "3 positives, 1 neutral, 1 negative" while her matrix shows four positives — a reminder that the tally is not the point.)
- On the AHP question: **"Students can skip the consistency check process while performing the AHP analysis."**

B.1 Her diagrams, collected



B-1 — The six-phase generic product development process



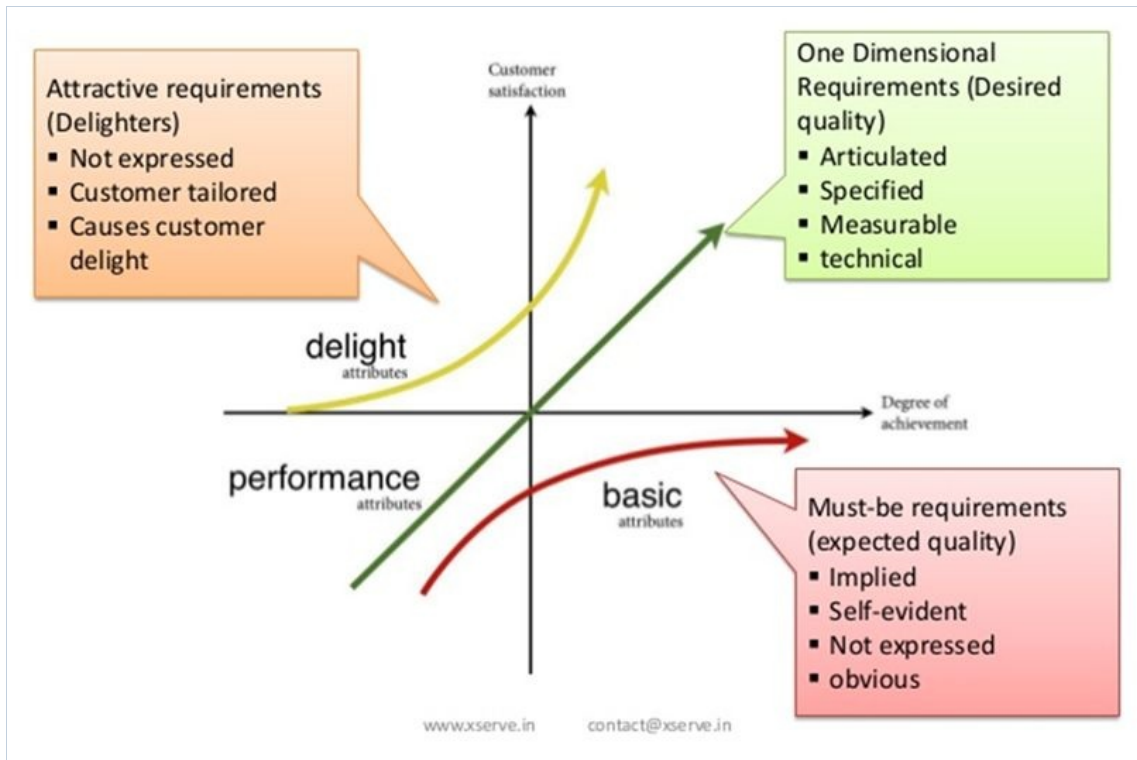
B-2 — Sequential vs concurrent engineering, with time compression marked

Dimension	Contribution of Ethnography
Hidden Needs Discovery	Reveals unspoken behaviors, frustrations, and desires that inspire breakthrough ideas.
Contextual Understanding	Shows how products fit into daily routines, social interactions, and cultural practices.
Empathy Building	Helps designers and engineers see the world through the user's eyes, sparking human-centered solutions.
Idea Generation	Provides rich narratives and scenarios that fuel brainstorming beyond technical constraints.
Innovation Validation	Ensures creative concepts resonate with real users, not just theoretical models.
Cross-Disciplinary Inspiration	Anthropological insights encourage collaboration between design, engineering, and marketing teams.

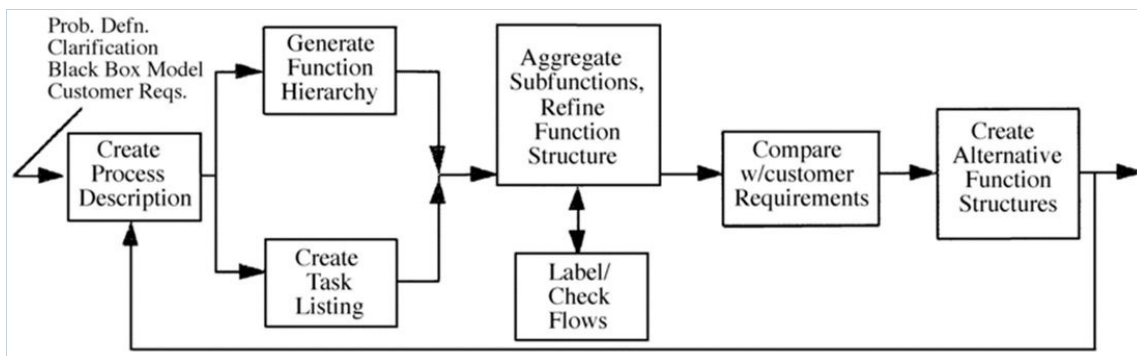
B-3 — Contribution of ethnography to product development, six dimensions

Issue	Why It's a Problem
Surface-Level Insights	Surveys usually capture what customers say they want, not what they <i>actually</i> need or do in real life.
Response Bias	Customers may give socially desirable answers or what they think the company wants to hear.
Limited Context	Surveys rarely capture the environment, emotions, or cultural factors influencing customer behavior.
Low Engagement	Many respondents rush through surveys, leading to incomplete or inaccurate data.
Static Questions	Predefined questions restrict exploration of unexpected insights; they don't adapt to evolving responses.
Sampling Issues	Surveys often rely on small or non-representative samples, skewing results.
Delayed Feedback	By the time survey results are analyzed, market conditions or customer preferences may have shifted.
Quantitative Overemphasis	Surveys focus on numbers and ratings, missing qualitative richness like stories, motivations, and subtle frustrations.

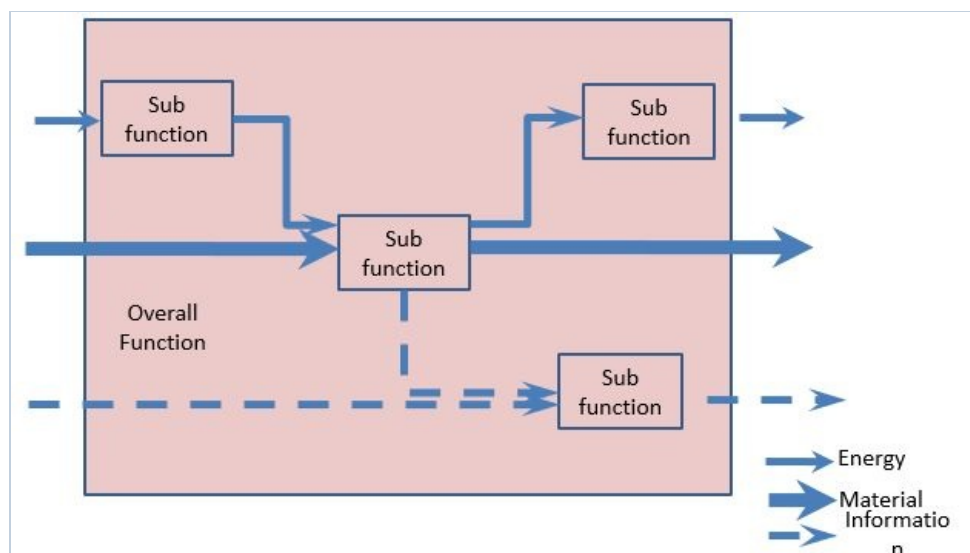
B-4 — Eight common issues with traditional customer surveys



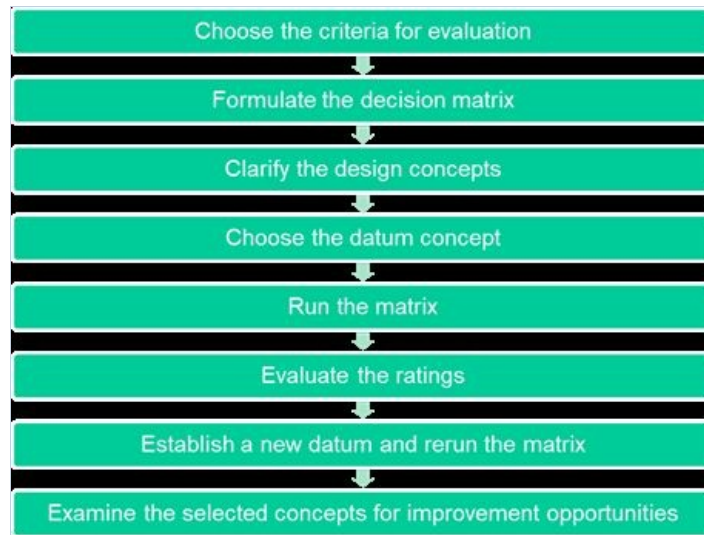
B-5 — The Kano model — **mandatory diagram** for any Kano question



B-6 — The functional decomposition process



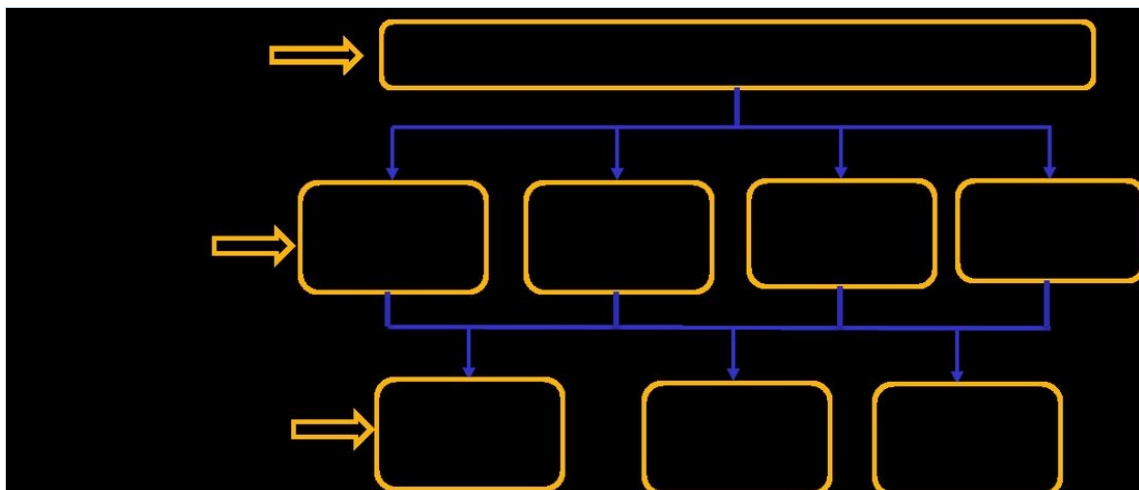
B-7 — Overall function decomposed into sub-functions, with energy / material / information flow conventions



B-8 — The eight steps of Pugh concept selection

Criteria	Concept A (Baseline)	Concept B	Concept C
Ease of Use	0 (simple foldable)	+ (compact, quick setup)	– (multi-joint, more complex to adjust)
Stability	0 (basic stability)	+ (solid base, angled support)	+ (adjustable joints allow stable positioning)
Durability	0 (light frame, moderate)	+ (solid construction, fewer moving parts)	– (multiple joints prone to wear)
Aesthetic Appeal	0 (functional, minimal)	+ (sleek integrated look)	0 (functional but bulky)
Manufacturability	0 (simple design, easy to produce)	– (complex base, higher cost)	– (multi-joint, precision manufacturing needed)

B-9 — Her worked Pugh screening matrix for the portable laptop stand



B-10 — The AHP hierarchy: Focus → Criteria → Alternatives

AHP's Ratings for Pairwise Comparison of Selection Criteria

Rating Factor	Relative Rating of Importance of Two Selection Criteria A and B	Explanation of Rating
1	A and B have equal importance.	A and B both contribute equally to the product's overall success.
3	A is thought to be moderately more important than B.	A is slightly more important to product success than B.
5	A is thought to be strongly more important than B.	A is strongly more important to product success than B.
7	A is thought to be very much more important than B, or is demonstrated to be more important than B.	A's dominance over B has been demonstrated.
9	A is demonstrated to have much more importance than B.	There is the highest possible degree of evidence that proves A is more important to product success than B.

The ratings of even numbers 2, 4, 6, and 8 are used when the decision maker needs to compromise between two positions in the table.

B-11 — AHP's 1-9 pairwise rating scale with explanations

Appendix C — The Instructor's Handwritten AHP Solution

Worked end to end for the mobile phone selection problem. This is the exact method and layout to reproduce in the exam — note how compact it is, and how each step is labelled A1, A2, A3, A4, A5. The typed version with all arithmetic checked is in §7.3.

2) AHP

①

Step 1: developing a model.

Focus (goal) - purchasing a smart phone

Criteria - cost, Battery backup, screen size, Ease of use

Alternatives - phone 1, phone 2, phone 3

Step 2 Derive priorities (weights) for the criteria

1 - Equally important, 3 - more important

5 - strong import, 7 - very strong

9 - extreme important

2, 4, 6, 8 - intermediate values betⁿ two adjacent judgement.

Pair wise comparison matrix,

(A1)

	cost	Battery Backup	Screen size	Ease of use
cost	1	1/7	1/5	1/3
Battery Backup	7	1	2	3
Screen size	5	1/2	1	3
Ease to use	3	1/3	1/3	1

$$\text{cost} = \left(1 \times \frac{1}{7} \times \frac{1}{5} \times \frac{1}{3}\right)^{1/4} = 0.31239$$

$$\text{Battery Backup} = \left(7 \times 1 \times 2 \times 3\right)^{1/4} = 0.25457$$

$$\text{Screen size} = \left(5 \times \frac{1}{2} \times 1 \times 3\right)^{1/4} = 1.0549$$

$$\text{Ease to use} = \left(3 \times \frac{1}{3} \times \frac{1}{3} \times 1\right)^{1/4} = 0.75984$$

C-1 - Step 1 developing the model; Step 2 deriving priorities; the pairwise comparison matrix [A1] and the geometric means

$$\begin{bmatrix} 0.31239 \\ 2.5457 \\ 1.6549 \\ 0.7598 \end{bmatrix}$$

$$\begin{bmatrix} 0.31239 / 5.2728 \\ 2.5457 / 5.2728 \\ 1.6549 / 5.2728 \\ 0.7598 / 5.2728 \end{bmatrix}$$

$$\text{sum} = 5.2728$$

weights

A2

$$\begin{bmatrix} 0.059246 \\ 0.4828 \\ 0.31385 \\ 0.1441 \end{bmatrix}$$

$$\text{sum} = 1.0$$

steps

consistency

check

(weight assigned correct or not)

$$A3 = A1 \times A2$$

$$\begin{bmatrix} 1 & 1/7 & 1/5 & 1/3 \\ 7 & 1 & 2 & 3 \\ 5 & 1/2 & 1 & 3 \\ 3 & 1/3 & 1/3 & 1 \end{bmatrix} \times$$

$$\begin{bmatrix} 0.059246 \\ 0.4828 \\ 0.31385 \\ 0.1441 \end{bmatrix}$$

$$\begin{bmatrix} 0.29902 \\ 1.9575 \\ 1.2438 \\ 0.58731 \end{bmatrix}$$

$$A4 = \frac{A3}{A2}$$

$$\begin{bmatrix} 4.0344 \\ 4.0545 \\ 4.0465 \\ 4.0782 \end{bmatrix}$$

$$\text{consistency index} = \frac{\lambda_{\max} - n}{n - 1}$$

$$CI = \frac{4.0631 - 4}{4 - 1}$$

$$CI = 0.0213$$

C-2 — Normalising to weights [A2]; Step 3 consistency check, $A3 = A1 \times A2$, $A4 = A3/A2$, $CI = 0.0213$

$$\text{consistency ratio} = \frac{\text{consistency index}}{\text{Random index}} \quad (2)$$

Table: Random index

Attributes	3	4	5	6	7	8	9	10
RI	0.52	0.89	1.11	1.21	1.35	1.4	1.47	1.49

$$RI = 0.89$$

$$\text{consistency ratio} = \frac{0.0213}{0.89} = 0.02386 \times < 0.1$$

(CR)

The proposed weights are consistent.

$$\begin{bmatrix} \text{cost} \\ \text{Battery Backup} \\ \text{screen size} \\ \text{Ease of use} \end{bmatrix} = \begin{bmatrix} 0.259246 \\ 0.4828 \\ 0.31385 \\ 0.1441 \end{bmatrix}$$

Battery Backup > screen size > Ease of use
> cost

Now, consider local priorities (or preferences) of the alternatives with respect to each criterion.

AS

	cost (Rs) (↓)	Battery Backup (mAh) (↑)	screen size (inches) (↓)	Ease of use (↑)
phone 1	15000	4000	5.6	Better (5)
phone 2	11000	3000	6	poor (3)
phone 3	18000	2500	5.2	Excellent (6)

Normalize AS matrix with least number

	cost (Rs) ($\downarrow$)	Battery Backup (MAh) ($\uparrow$)	Screen size (inches) ($\downarrow$)	Ease of use ($\uparrow$)
phone 1	$\frac{11000}{15000} = 0.7333$	$\frac{4000}{4000} = 1.0$	$\frac{5.2}{5.6} = 0.9285$	$\frac{5}{8} = 0.625$
phone 2	$\frac{11000}{11000} = 1.0$	$\frac{3000}{4000} = 0.75$	$\frac{5.2}{6} = 0.86667$	$\frac{3}{8} = 0.375$
phone 3	$\frac{11000}{16000} = 0.6875$	$\frac{2500}{4000} = 0.625$	$\frac{5.2}{5.2} = 1.0$	$\frac{8}{8} = 1.0$

Now, synthesis the model

	cost	Battery Backup	Screen size	Ease of use
weights	0.059246	0.4828	0.31385	0.1441
phone 1	0.7333	1.0	0.9285	0.625
phone 2	1.0	0.75	0.86667	0.375
phone 3	0.6875	0.625	1.0	1.0

$$\text{phone 1} = (0.7333 \times 0.059246) + (1.0 \times 0.4828) + (0.9285 \times 0.31385) + (0.625 \times 0.1441)$$

$$\boxed{\text{phone 1} = 0.90774}$$

$$\text{phone 2} = 0.7474$$

$$\text{phone 3} = 0.50043$$

$$\text{phone 4} =$$

Appendix D — Last-Minute Revision Sheet

D.1 The diagram checklist — can you draw each of these from memory?

Session	Diagram	Done
1	Six-phase generic PD process (chevrons, Phase 0 to Phase 5)	☐
1	Knowledge vs design freedom curve	☐
1	Asimow's seven phases	☐
1	Stage-gate / spiral / hybrid	☐
1	Technology push vs market pull sequences; technology S-curve	☐
2	Sequential vs concurrent engineering staircase, with time compression	☐
2	Functional / project / matrix organisation charts	☐
2	Five-step product planning process; opportunity funnel	☐
3	Five-step customer need identification	☐
3	Kano model graph — three curves, both axes labelled (MANDATORY)	☐
3	Kano evaluation table (functional × dysfunctional)	☐
3	Need hierarchy (primary / secondary / tertiary)	☐
4	Needs-metrics matrix	☐
4	House of Quality — all eight rooms labelled	☐
4	Benchmarking charts 1 and 2; trade-off map; flow-down	☐
5	Black box with material / energy / signal flows (MANDATORY convention)	☐
5	FAST diagram with HOW / WHY arrows and critical path	☐
5	Function tree; concept combination table; 6-3-5 layout	☐
6	Morphological chart with a concept traced through it	☐
6	TRIZ contradiction resolution flow (4 steps)	☐
6	Axiomatic design four domains; uncoupled / decoupled / coupled matrices	☐
7	Pugh matrix with $\Sigma+$, $\Sigma 0$, $\Sigma-$, net, rank	☐
7	Objective tree with branch weights multiplying to leaf weights	☐
7	AHP hierarchy: Focus → Criteria → Alternatives	☐
7	Weighted decision matrix layout	☐
7	Seven-step concept testing flow	☐
8	Modular vs integral function-component mapping (the crossing lines)	☐
8	Modularity taxonomy tree	☐
8	Architecture design process, four steps; interaction graph (solid = fundamental, dashed = incidental)	☐
8	Early vs delayed differentiation process diagrams	☐

D.2 Formula sheet

Topic	Formula
Kano — satisfaction	Extent of Satisfaction = $(A + O) / (A + O + M + I)$
Kano — dissatisfaction	Extent of Dissatisfaction = $(O + M) / [(A + O + M + I) \times (-1)]$
Kano — evaluation rule	$M > 0 > A > I$
HOQ — absolute importance	$AI_j = \sum_i (\text{Importance of } CR_i \times \text{Relationship}_{ij})$, with 9 / 3 / 1 strengths
HOQ — relative importance	$RI_j (\%) = AI_j / \sum AI \times 100$
Objective tree	Leaf weight = product of all branch weights from root to leaf; $\sum$ leaf weights = 1.0
Weighted decision matrix	Total score = $\sum (\text{weight factor} \times \text{rating})$
Pugh	Net = $\sum + - \sum -$ (+ better than datum, 0/S same, - worse)
AHP — normalise	$\text{NormC}_{ij} = C_{ij} / (\text{column sum } j)$ then $W_i = \text{row average of NormC}$
AHP — geometric mean shortcut	$w_{i \text{ raw}} = (a_{i1} \times \dots \times a_{in})^{1/n}$; $W_i = w_{i \text{ raw}} / \sum w_{i \text{ raw}}$
AHP — consistency	$A3 = [C]\{W\}$; $A4 = A3/\{W\}$; $\lambda_{\max} = \text{mean}(A4)$ $CI = (\lambda_{\max} - n)/(n - 1)$; $CR = CI/RI$; accept if $CR < 0.1$
AHP — RI table	$n = 3 \rightarrow 0.52 \cdot 4 \rightarrow 0.89 \cdot 5 \rightarrow 1.11 \cdot 6 \rightarrow 1.25 \cdot 7 \rightarrow 1.35 \cdot 8 \rightarrow 1.40 \cdot 9 \rightarrow 1.45 \cdot 10 \rightarrow 1.49$
AHP — normalise alternatives	Lower is better: $\min / \text{value}$ • Higher is better: $\text{value} / \max$
AHP — synthesis	Alternative value = $\sum (\text{normalised rating} \times \text{criterion weight})$ or $[F\text{Rating}]T\{W\}$
6-3-5	6 people $\times$ 3 ideas $\times$ 6 rounds = 108 ideas in 30 minutes (5 min per round)
Morphological / combination table	Number of concepts = product of the option counts in each row/column
Spiral PD model	$\{(1\text{st stage-gate}) + v1\} + \{(2\text{nd stage-gate}) + v2\} + \dots + \{(n\text{th stage-gate}) + vn\}$
Profit relation	Profit = Product Price (↑) – Product Cost (↓)

D.3 The numbered lists — rapid recall

Session 1

- 3 abilities for economic success
- 5 PD characteristics: quality, cost, dev time, dev cost, capability
- 5 PD challenges: trade-offs, dynamics, details, time pressure, economics
- 4 C's: creativity, complexity, choice, compromise
- 5 types of design: original, adaptive, redesign, selection, industrial
- 5 problem-solving steps
- 7 Asimow phases
- 6 generic PD phases (0-5)
- 8 process variations

Session 2

- 4 factors for choosing an organisation
- 4 key elements of concurrent engineering
- 5 product planning steps
- 4 types of PD projects
- 3 opportunity horizons
- 6 opportunity identification steps
- 4 prioritisation perspectives
- 4 competitive strategies
- 6 mission statement contents

Session 3

- 5 need identification steps
- 4 ethnographic study methods
- 6 contributions of ethnography
- 4 customer types: buyer, end user, lead user, extreme user
- 5 guidelines for need statements
- 6 Kano categories (M, O, A, I, R, Q)
- 4 Kano levels: expecters, spoken, unspoken, excitors
- 4 sources of resistance to benchmarking
- 6 reverse engineering steps

Session 4

- 4 target specification steps
- 5 final specification steps
- 5 metric guidelines
- 4 unit types
- 5 value expressions
- 8 HOQ rooms
- 9 QFD advantages

Session 5

- 5 mental block types
- 6 aids to creative thinking
- 5 concept generation steps
- 4 Osborn brainstorming rules
- 7 SCAMPER letters
- 3 flow types: material, energy, signal
- 3 secondary function types: required, aesthetic, unwanted
- 6 SOP steps

Session 6

- 5 morphological analysis steps
- 2 TRIZ basic principles
- 39 engineering parameters, 40 inventive principles
- 4 TRIZ procedure steps
- 4 axiomatic design domains (CA, FR, DP, PV)
- 2 axioms: independence, information
- 3 matrix forms: uncoupled, decoupled, coupled
- 3 feasibility categories
- 3 absolute evaluation filters
- 6 benefits of a structured method

Session 7

- 8 Pugh steps
- 5 Pugh guidelines
- 3 weighting factor methods
- 9 AHP rating levels (1,3,5,7,9 + evens)
- 6 AHP procedure steps
- 7 concept testing steps
- 5 sample size factors
- 5 survey formats
- 8 ways to communicate a concept

Session 8

- 3 preliminary design activities
- 6 implications of architecture
- 4 manufacturing-based modularity types
- 4 function-based modularity types (bus, slot, sectional, mix)
- 4 considerations at architecting
- 4 architecture design steps
- 7 clustering considerations
- 2 interaction types: fundamental, incidental
- 5 fundamental decisions
- 2 late point differentiation principles

D.4 Named people, places and figures worth quoting

Morris Asimow	Morphology of design — the seven phases
Karl Ulrich	Course textbook; the trailer architecture example; Research Policy 24 (1995) 419-441
Noriaki Kano	Kano model; four levels of customer requirements
Alex Osborn	Brainstorming — defer judgment, reach for quantity; four rules
Jiro Kawakita	Affinity diagram / KJ Method, 1960s
Genrich Altshuller	TRIZ / TIPS — 39 parameters, 40 principles, contradiction matrix
Fritz Zwicky	Morphological box, 1948
Nam P. Suh	Axiomatic design — independence and information axioms
Stuart Pugh	Concept screening matrix
Thomas Saaty	Analytic Hierarchy Process
Sun Tzu	500 BC — "know your enemy before you know yourself"; benchmarking
Xerox / Canon, 1979	Origin of modern competitive benchmarking
Toyota / QFD	Cost of a new model reduced by over 60%; development time cut by one-third
Benetton	Delayed differentiation — sweaters knitted white, dyed late
HP DeskJet	Architecture design process case study; separate DC power chunk for regional variety
GKN Aerospace / Melrose	RSP and RRSP — 50-year life-of-programme contracts, 19 RRSP contracts
Steve Jobs	Criticism of focus groups — quotable against the Kano/survey approach

D.5 The five answer-writing rules — read this on the morning of the exam

1. **Never write an essay.** Definition → diagram → bullets keyed to the diagram → named example. That is the shape of every answer.
2. **Draw first, write second.** If the diagram is on the page, the marks are already partly secured. The Kano graph and the black-box M-E-S structure are explicitly mandatory.
3. **State your assumptions at the start of the answer.** The paper's own instructions require it, and for every matrix question the assumed ratings and weights are accepted provided they are stated.
4. **Use a real, everyday product.** The questions will use general products; answer with one too, and name it.
5. **Watch the clock at 20 minutes per question.** An incomplete sixth answer costs more than a perfect first answer gains. If a calculation stalls, write the method and move on — method carries the marks.

All the best for the exam on 20 September.